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Article

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Fruquintinib plus paclitaxel versus placebo plus paclitaxel for gastric or gastroesophageal junction adenocarcinoma: the randomized phase 3 FRUTIGA trial

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Supplementary Table 1. Treatment-emergent adverse events (Safety analysis set)

	Fruquintinib plus paclitaxel group (N=350) n (%)		**Placebo plus paclitaxel group (N=349)** n (%)	
Preferred term	**Total n (%)**	**⩾3 n (%)**	**Total n (%)**	**⩾3 n (%)**
Number of subjects with at least one teae	348(99.4)	304(86.9)	347(99.4)	221(63.3)
Neutrophil count decreased	226(64.6)	163(46.6)	199(57.0)	102(29.2)
White blood cell count decreased	222(63.4)	117(33.4)	189(54.2)	65(18.6)
Anaemia	211(60.3)	41(11.7)	209(59.9)	37(10.6)
Alopecia	142(40.6)	0	152(43.6)	2(0.6)
Decreased appetite	104(29.7)	3(0.9)	65(18.6)	4(1.1)
Platelet count decreased	104(29.7)	13(3.7)	51(14.6)	4(1.1)
Leukopenia	101(28.9)	38(10.9)	79(22.6)	21(6.0)
Diarrhoea	98(28.0)	11(3.1)	58(16.6)	4(1.1)
Proteinuria	95(27.1)	4(1.1)	53(15.2)	2(0.6)
Constipation	91(26.0)	1(0.3)	65(18.6)	0
Hypoalbuminaemia	88(25.1)	0	46(13.2)	1(0.3)
Occult blood positive	86(24.6)	0	80(22.9)	0
Neutropenia	85(24.3)	52(14.9)	67(19.2)	28(8.0)
Palmar-plantar erythrodysaesthesia syndrome	84(24.0)	31(8.9)	17(4.9)	0
Weight decreased	84(24.0)	7(2.0)	45(12.9)	0
Aspartate aminotransferase increased	78(22.3)	6(1.7)	62(17.8)	6(1.7)
Blood bilirubin increased	74(21.1)	8(2.3)	43(12.3)	5(1.4)
Malaise	66(18.9)	2(0.6)	46(13.2)	2(0.6)
Pyrexia	65(18.6)	1(0.3)	28(8.0)	1(0.3)
Alanine aminotransferase increased	63(18.0)	5(1.4)	49(14.0)	5(1.4)
Dysphonia	61(17.4)	1(0.3)	8(2.3)	0
Vomiting	56(16.0)	9(2.6)	47(13.5)	3(0.9)
Epistaxis	55(15.7)	0	14(4.0)	0
Abdominal pain	52(14.9)	4(1.1)	37(10.6)	3(0.9)
Asthenia	52(14.9)	7(2.0)	34(9.7)	2(0.6)
Back pain	46(13.1)	4(1.1)	30(8.6)	1(0.3)
Hypokalaemia	46(13.1)	9(2.6)	16(4.6)	1(0.3)
Rash	46(13.1)	0	31(8.9)	0
Hypertension	45(12.9)	18(5.1)	14(4.0)	4(1.1)

Stomatitis	45(12.9)	10(2.9)	15(4.3)	3(0.9)
Abdominal distension	40(11.4)	1(0.3)	38(10.9)	1(0.3)
Cough	40(11.4)	0	21(6.0)	1(0.3)
Fatigue	39(11.1)	7(2.0)	29(8.3)	8(2.3)
Hypoaesthesia	39(11.1)	4(1.1)	49(14.0)	1(0.3)
Hyponatraemia	39(11.1)	6(1.7)	30(8.6)	5(1.4)
Myelosuppression	39(11.1)	32(9.1)	21(6.0)	18(5.2)
Nausea	39(11.1)	1(0.3)	35(10.0)	1(0.3)
Abdominal pain upper	37(10.6)	1(0.3)	27(7.7)	1(0.3)
Hypothyroidism	37(10.6)	0	14(4.0)	0
Insomnia	32(9.1)	0	19(5.4)	1(0.3)
Blood thyroid stimulating hormone increased	31(8.9)	0	12(3.4)	0
Hypocalcaemia	30(8.6)	3(0.9)	24(6.9)	1(0.3)
Arthralgia	29(8.3)	1(0.3)	21(6.0)	0
Dizziness	29(8.3)	2(0.6)	13(3.7)	0
Hypertriglyceridaemia	29(8.3)	6(1.7)	12(3.4)	2(0.6)
Mouth ulceration	29(8.3)	4(1.1)	4(1.1)	0
Pain in extremity	28(8.0)	2(0.6)	18(5.2)	0
Upper respiratory tract infection	28(8.0)	2(0.6)	19(5.4)	1(0.3)
Hepatic function abnormal	27(7.7)	10(2.9)	15(4.3)	11(3.2)
Thrombocytopenia	27(7.7)	2(0.6)	14(4.0)	0
Gingival pain	26(7.4)	1(0.3)	0	0
Bilirubin conjugated increased	24(6.9)	3(0.9)	19(5.4)	1(0.3)
Haemoglobin decreased	24(6.9)	1(0.3)	24(6.9)	4(1.1)
Pneumonia	23(6.6)	7(2.0)	14(4.0)	5(1.4)
Protein urine present	23(6.6)	0	23(6.6)	0
Blood alkaline phosphatase increased	22(6.3)	2(0.6)	29(8.3)	3(0.9)
Oropharyngeal pain	22(6.3)	0	2(0.6)	0
Toothache	22(6.3)	1(0.3)	3(0.9)	0
Headache	20(5.7)	2(0.6)	6(1.7)	0
Urinary tract infection	20(5.7)	2(0.6)	12(3.4)	0
Hypoproteinaemia	19(5.4)	1(0.3)	10(2.9)	0
Gamma-glutamyltransferase increased	18(5.1)	9(2.6)	26(7.4)	7(2.0)
Gingivitis	18(5.1)	5(1.4)	2(0.6)	0
Hyperuricaemia	18(5.1)	1(0.3)	20(5.7)	0
Lymphocyte count decreased	18(5.1)	7(2.0)	17(4.9)	6(1.7)
Neurotoxicity	18(5.1)	3(0.9)	14(4.0)	2(0.6)

Oedema peripheral	18(5.1)	0	8(2.3)	0
Tinnitus	17(4.9)	0	2(0.6)	0
Electrocardiogram t wave abnormal	16(4.6)	0	5(1.4)	0
Gastrooesophageal reflux disease	16(4.6)	0	9(2.6)	0
Hiccups	16(4.6)	0	10(2.9)	1(0.3)
Hyperglycaemia	16(4.6)	0	23(6.6)	2(0.6)
Pain	16(4.6)	1(0.3)	12(3.4)	0
Sinus tachycardia	16(4.6)	0	5(1.4)	0
Hypercholesterolaemia	15(4.3)	2(0.6)	9(2.6)	0
Hypophosphataemia	15(4.3)	3(0.9)	8(2.3)	0
Pruritus	15(4.3)	0	13(3.7)	0
Sinus bradycardia	15(4.3)	0	10(2.9)	0
Blood glucose increased	14(4.0)	1(0.3)	8(2.3)	0
Blood pressure increased	14(4.0)	5(1.4)	2(0.6)	0
Intestinal obstruction	14(4.0)	11(3.1)	8(2.3)	6(1.7)
Peripheral sensory neuropathy	14(4.0)	0	10(2.9)	0
Blood cholesterol increased	13(3.7)	0	7(2.0)	0
Haematuria	13(3.7)	0	7(2.0)	0
Haemorrhoids	13(3.7)	0	3(0.9)	0
Urinary occult blood positive	13(3.7)	0	5(1.4)	0
Urobilinogen urine increased	13(3.7)	0	5(1.4)	0
Weight increased	13(3.7)	0	14(4.0)	0
Hyperlipidaemia	12(3.4)	0	16(4.6)	1(0.3)
Productive cough	12(3.4)	0	7(2.0)	0
Amylase increased	11(3.1)	5(1.4)	6(1.7)	2(0.6)
Blood lactate dehydrogenase increased	11(3.1)	0	12(3.4)	0
Dyspnoea	11(3.1)	1(0.3)	8(2.3)	1(0.3)
Tri-iodothyronine free decreased	11(3.1)	0	9(2.6)	0
Upper gastrointestinal haemorrhage	11(3.1)	8(2.3)	6(1.7)	4(1.1)
Blood fibrinogen increased	10(2.9)	0	9(2.6)	0
Death	10(2.9)	10(2.9)	6(1.7)	6(1.7)
Febrile neutropenia	10(2.9)	10(2.9)	1(0.3)	1(0.3)
Gingival bleeding	10(2.9)	1(0.3)	2(0.6)	0
Blood albumin decreased	9(2.6)	0	4(1.1)	0
Blood creatinine increased	9(2.6)	0	5(1.4)	0
Myalgia	9(2.6)	0	7(2.0)	1(0.3)
Abdominal pain lower	8(2.3)	1(0.3)	8(2.3)	0

Aphthous ulcer	8(2.3)	1(0.3)	1(0.3)	0
Blood urea increased	8(2.3)	0	4(1.1)	0
Blood uric acid increased	8(2.3)	0	6(1.7)	0
Dry mouth	8(2.3)	0	3(0.9)	0
Fibrin d dimer increased	8(2.3)	0	5(1.4)	1(0.3)
Flatulence	8(2.3)	0	6(1.7)	0
Hypochloraemia	8(2.3)	0	1(0.3)	0
Supraventricular extrasystoles	8(2.3)	0	7(2.0)	0
White blood cells urine positive	8(2.3)	0	12(3.4)	0
Activated partial thromboplastin time prolonged	7(2.0)	0	5(1.4)	0
Chest discomfort	7(2.0)	1(0.3)	2(0.6)	0
Disease progression	7(2.0)	7(2.0)	1(0.3)	1(0.3)
Dysphagia	7(2.0)	2(0.6)	8(2.3)	0
Gingival swelling	7(2.0)	1(0.3)	0	0
Haematochezia	7(2.0)	0	2(0.6)	0
Nasopharyngitis	7(2.0)	0	5(1.4)	0
Neutrophil count increased	7(2.0)	1(0.3)	3(0.9)	0
Thyroid disorder	7(2.0)	0	0	0
Thyroxine free decreased	7(2.0)	0	0	0
Ventricular extrasystoles	7(2.0)	0	4(1.1)	0
Blood chloride decreased	6(1.7)	0	9(2.6)	1(0.3)
Bone pain	6(1.7)	1(0.3)	6(1.7)	0
Dyspepsia	6(1.7)	0	4(1.1)	0
Electrocardiogram qt prolonged	6(1.7)	0	2(0.6)	0
Gastrointestinal haemorrhage	6(1.7)	2(0.6)	3(0.9)	0
Haemorrhoidal haemorrhage	6(1.7)	0	5(1.4)	0
Hypoaesthesia oral	6(1.7)	0	0	0
Hypomagnesaemia	6(1.7)	1(0.3)	2(0.6)	0
Hypotension	6(1.7)	1(0.3)	7(2.0)	0
Neuropathy peripheral	6(1.7)	1(0.3)	5(1.4)	0
Urine ketone body present	6(1.7)	0	4(1.1)	1(0.3)
Vision blurred	6(1.7)	1(0.3)	1(0.3)	0
Abdominal discomfort	5(1.4)	0	13(3.7)	0
Anxiety	5(1.4)	0	1(0.3)	0
Ascites	5(1.4)	5(1.4)	5(1.4)	3(0.9)
Coagulopathy	5(1.4)	0	3(0.9)	0
Dysgeusia	5(1.4)	0	1(0.3)	0

Hyperbilirubinaemia	5(1.4)	0	1(0.3)	0
Hyperkalaemia	5(1.4)	2(0.6)	8(2.3)	1(0.3)
Hyperthyroidism	5(1.4)	0	1(0.3)	0
Influenza like illness	5(1.4)	0	10(2.9)	0
Leukocytosis	5(1.4)	2(0.6)	4(1.1)	2(0.6)
Low density lipoprotein increased	5(1.4)	0	2(0.6)	0
Non-cardiac chest pain	5(1.4)	0	4(1.1)	1(0.3)
Pharyngitis	5(1.4)	0	3(0.9)	1(0.3)
Proctalgia	5(1.4)	0	0	0
Protein total decreased	5(1.4)	0	6(1.7)	0
Rhinorrhoea	5(1.4)	0	1(0.3)	0
Skin exfoliation	5(1.4)	0	1(0.3)	0
Abdominal infection	4(1.1)	1(0.3)	1(0.3)	0
Anaesthesia	4(1.1)	1(0.3)	5(1.4)	0
Anal fissure	4(1.1)	1(0.3)	0	0
Bilirubin urine present	4(1.1)	0	1(0.3)	0
Blood triglycerides increased	4(1.1)	0	0	0
Cancer pain	4(1.1)	1(0.3)	7(2.0)	1(0.3)
Eczema	4(1.1)	0	1(0.3)	0
Electrocardiogram abnormal	4(1.1)	0	2(0.6)	0
Electrolyte imbalance	4(1.1)	1(0.3)	3(0.9)	1(0.3)
Eyelid oedema	4(1.1)	0	1(0.3)	0
Glucose urine present	4(1.1)	0	6(1.7)	0
Laryngeal pain	4(1.1)	0	3(0.9)	0
Liver injury	4(1.1)	3(0.9)	0	0
Muscular weakness	4(1.1)	0	7(2.0)	1(0.3)
Palpitations	4(1.1)	0	2(0.6)	0
Paraesthesia	4(1.1)	0	4(1.1)	0
Platelet count increased	4(1.1)	0	0	0
Respiratory failure	4(1.1)	4(1.1)	0	0
Small intestinal obstruction	4(1.1)	3(0.9)	3(0.9)	1(0.3)
Total bile acids increased	4(1.1)	0	1(0.3)	0
White blood cell count increased	4(1.1)	0	2(0.6)	0
Acne	3(0.9)	0	0	0
Anal abscess	3(0.9)	2(0.6)	1(0.3)	1(0.3)
Anal haemorrhage	3(0.9)	0	2(0.6)	0
Blood bilirubin unconjugated increased	3(0.9)	0	8(2.3)	1(0.3)

Dermatitis	3(0.9)	0	1(0.3)	0
Eructation	3(0.9)	0	4(1.1)	0
Gastrointestinal disorder	3(0.9)	0	1(0.3)	0
Hypoglycaemia	3(0.9)	0	1(0.3)	0
Ileus	3(0.9)	3(0.9)	1(0.3)	1(0.3)
Influenza	3(0.9)	0	3(0.9)	0
Infusion site extravasation	3(0.9)	0	3(0.9)	0
Jaundice cholestatic	3(0.9)	3(0.9)	5(1.4)	5(1.4)
Oral pain	3(0.9)	0	1(0.3)	0
Paronychia	3(0.9)	1(0.3)	2(0.6)	0
Pharyngeal haemorrhage	3(0.9)	0	0	0
Pneumonitis	3(0.9)	0	2(0.6)	1(0.3)
Prealbumin decreased	3(0.9)	0	0	0
Prothrombin time prolonged	3(0.9)	0	3(0.9)	0
Red blood cells urine positive	3(0.9)	0	3(0.9)	0
Urine bilirubin increased	3(0.9)	0	1(0.3)	0
Urine output decreased	3(0.9)	0	2(0.6)	0
Urticaria	3(0.9)	0	0	0
Vertigo	3(0.9)	0	2(0.6)	0
Arrhythmia	2(0.6)	0	2(0.6)	0
Atrial fibrillation	2(0.6)	1(0.3)	0	0
Bacteraemia	2(0.6)	1(0.3)	1(0.3)	0
Bacterial infection	2(0.6)	1(0.3)	0	0
Blood cholinesterase decreased	2(0.6)	0	0	0
Blood creatine phosphokinase increased	2(0.6)	0	1(0.3)	0
Blood phosphorus decreased	2(0.6)	0	0	0
Blood phosphorus increased	2(0.6)	0	3(0.9)	0
Blood thyroid stimulating hormone decreased	2(0.6)	0	1(0.3)	0
Dermatitis acneiform	2(0.6)	0	0	0
Dermatosis	2(0.6)	0	2(0.6)	0
Drug-induced liver injury	2(0.6)	1(0.3)	0	0
Dyschezia	2(0.6)	0	0	0
Dysuria	2(0.6)	0	0	0
Electrocardiogram low voltage	2(0.6)	0	0	0
Electrocardiogram p wave abnormal	2(0.6)	0	0	0
Electrocardiogram q wave abnormal	2(0.6)	0	0	0
Electrocardiogram st-t change	2(0.6)	0	1(0.3)	0

Febrile infection	2(0.6)	0	1(0.3)	0
Gastric perforation	2(0.6)	1(0.3)	0	0
Glucose tolerance impaired	2(0.6)	0	0	0
Haemoptysis	2(0.6)	1(0.3)	4(1.1)	0
Haemorrhage urinary tract	2(0.6)	0	0	0
Hepatic pain	2(0.6)	0	1(0.3)	0
Hypermagnesaemia	2(0.6)	0	1(0.3)	0
Hypoparathyroidism	2(0.6)	0	0	0
Inguinal hernia	2(0.6)	0	0	0
Interstitial lung disease	2(0.6)	1(0.3)	0	0
Keratitis	2(0.6)	0	0	0
Lipids increased	2(0.6)	0	0	0
Lower gastrointestinal haemorrhage	2(0.6)	0	1(0.3)	0
Lymphopenia	2(0.6)	0	0	0
Malnutrition	2(0.6)	0	1(0.3)	0
Melaena	2(0.6)	0	2(0.6)	0
Monocyte count decreased	2(0.6)	0	3(0.9)	0
Muscle spasms	2(0.6)	0	1(0.3)	0
Musculoskeletal discomfort	2(0.6)	0	1(0.3)	0
Musculoskeletal stiffness	2(0.6)	0	0	0
Myocardial ischaemia	2(0.6)	0	2(0.6)	0
Myringitis	2(0.6)	0	0	0
Nail discolouration	2(0.6)	0	2(0.6)	0
Nasal inflammation	2(0.6)	0	0	0
Neuralgia	2(0.6)	0	0	0
Neutropenic infection	2(0.6)	2(0.6)	0	0
Neutrophil percentage decreased	2(0.6)	2(0.6)	1(0.3)	1(0.3)
Obstruction gastric	2(0.6)	2(0.6)	2(0.6)	2(0.6)
Pain of skin	2(0.6)	0	0	0
Pancreatitis	2(0.6)	2(0.6)	0	0
Periarthritis	2(0.6)	0	0	0
Pericoronitis	2(0.6)	1(0.3)	0	0
Peripheral swelling	2(0.6)	0	6(1.7)	1(0.3)
Phlebitis	2(0.6)	0	4(1.1)	0
Pollakiuria	2(0.6)	0	1(0.3)	0
Poor quality sleep	2(0.6)	0	0	0
Pulmonary embolism	2(0.6)	2(0.6)	2(0.6)	1(0.3)

Pulpitis dental	2(0.6)	0	0	0
Rash pustular	2(0.6)	0	0	0
Red blood cell count decreased	2(0.6)	0	2(0.6)	0
Renal impairment	2(0.6)	1(0.3)	2(0.6)	0
Seizure	2(0.6)	0	0	0
Septic shock	2(0.6)	1(0.3)	0	0
Skin fissures	2(0.6)	0	0	0
Soft tissue infection	2(0.6)	1(0.3)	0	0
Spinal pain	2(0.6)	0	1(0.3)	0
Subileus	2(0.6)	0	0	0
Superficial vein thrombosis	2(0.6)	1(0.3)	0	0
Supraventricular tachycardia	2(0.6)	0	3(0.9)	0
Syncope	2(0.6)	1(0.3)	1(0.3)	0
Temperature intolerance	2(0.6)	0	0	0
Tongue ulceration	2(0.6)	0	0	0
Tri-iodothyronine free increased	2(0.6)	0	1(0.3)	0
Urethritis	2(0.6)	1(0.3)	0	0
Urinary incontinence	2(0.6)	0	0	0
Vaginal haemorrhage	2(0.6)	0	2(0.6)	0
Abdominal tenderness	1(0.3)	0	0	0
Acute kidney injury	1(0.3)	1(0.3)	1(0.3)	1(0.3)
Administration site extravasation	1(0.3)	0	1(0.3)	0
Agitation	1(0.3)	0	0	0
Albumin urine present	1(0.3)	0	0	0
Albuminuria	1(0.3)	0	0	0
Alkalosis	1(0.3)	1(0.3)	0	0
Alpha hydroxybutyrate dehydrogenase increased	1(0.3)	0	1(0.3)	0
Anal dilatation	1(0.3)	0	1(0.3)	0
Anal fistula	1(0.3)	0	0	0
Anal pruritus	1(0.3)	0	0	0
Anal ulcer	1(0.3)	0	0	0
Angina pectoris	1(0.3)	0	0	0
Angiotensin ii increased	1(0.3)	0	0	0
Angular cheilitis	1(0.3)	0	0	0
Anorectal infection	1(0.3)	1(0.3)	0	0
Anoxia	1(0.3)	0	0	0
Aortic aneurysm	1(0.3)	0	0	0

Appendicitis	1(0.3)	1(0.3)	1(0.3)	1(0.3)
Appendicitis perforated	1(0.3)	1(0.3)	0	0
Arrhythmia supraventricular	1(0.3)	0	1(0.3)	0
Aspartate aminotransferase decreased	1(0.3)	0	0	0
Asthma	1(0.3)	0	0	0
Asthma exercise induced	1(0.3)	0	0	0
Asymptomatic bacteriuria	1(0.3)	0	3(0.9)	0
Atrioventricular block	1(0.3)	0	1(0.3)	0
Benign prostatic hyperplasia	1(0.3)	0	0	0
Blood aldosterone decreased	1(0.3)	0	0	0
Blood calcium decreased	1(0.3)	0	0	0
Blood creatine phosphokinase mb increased	1(0.3)	0	0	0
Blood creatinine decreased	1(0.3)	0	0	0
Blood fibrinogen decreased	1(0.3)	0	1(0.3)	0
Blood iron decreased	1(0.3)	0	0	0
Blood potassium decreased	1(0.3)	0	1(0.3)	0
Blood sodium decreased	1(0.3)	0	2(0.6)	0
Blood thyroid stimulating hormone abnormal	1(0.3)	0	0	0
Blood urine present	1(0.3)	0	0	0
Bradycardia	1(0.3)	0	0	0
Bronchitis	1(0.3)	0	0	0
Bronchopulmonary aspergillosis	1(0.3)	1(0.3)	0	0
Bundle branch block left	1(0.3)	0	0	0
Burns first degree	1(0.3)	0	0	0
C-reactive protein increased	1(0.3)	0	1(0.3)	0
Carbon dioxide combining power decreased	1(0.3)	0	0	0
Catheter site infection	1(0.3)	0	0	0
Cerebral infarction	1(0.3)	1(0.3)	1(0.3)	1(0.3)
Cerebrovascular insufficiency	1(0.3)	1(0.3)	0	0
Chest pain	1(0.3)	0	2(0.6)	1(0.3)
Chills	1(0.3)	0	0	0
Choking	1(0.3)	0	0	0
Cholecystitis	1(0.3)	1(0.3)	1(0.3)	0
Cholecystitis acute	1(0.3)	1(0.3)	0	0
Cholelithiasis	1(0.3)	0	1(0.3)	0
Cholestasis	1(0.3)	0	0	0
Chromaturia	1(0.3)	0	0	0

Coagulation test abnormal	1(0.3)	0	0	0
Cognitive disorder	1(0.3)	1(0.3)	1(0.3)	0
Coma	1(0.3)	0	0	0
Conduction disorder	1(0.3)	0	0	0
Contusion	1(0.3)	0	0	0
Deafness unilateral	1(0.3)	0	0	0
Decubitus ulcer	1(0.3)	0	1(0.3)	0
Degenerative bone disease	1(0.3)	0	0	0
Depressed level of consciousness	1(0.3)	1(0.3)	1(0.3)	1(0.3)
Depressed mood	1(0.3)	0	0	0
Dermatitis allergic	1(0.3)	0	1(0.3)	0
Dermatophytosis of nail	1(0.3)	0	0	0
Device related infection	1(0.3)	1(0.3)	0	0
Dry eye	1(0.3)	0	1(0.3)	0
Dry skin	1(0.3)	0	0	0
Dyslipidaemia	1(0.3)	0	0	0
Dysmenorrhoea	1(0.3)	0	0	0
Dysphoria	1(0.3)	0	0	0
Ear pain	1(0.3)	0	0	0
Eating disorder	1(0.3)	0	0	0
Electrocardiogram qrs complex abnormal	1(0.3)	0	0	0
Electrocardiogram rr interval prolonged	1(0.3)	0	0	0
Electrocardiogram st segment depression	1(0.3)	0	2(0.6)	0
Enteritis infectious	1(0.3)	1(0.3)	0	0
Entropion	1(0.3)	1(0.3)	1(0.3)	1(0.3)
Epididymitis	1(0.3)	0	0	0
Epilepsy	1(0.3)	0	0	0
Escherichia infection	1(0.3)	0	0	0
Extrasystoles	1(0.3)	0	1(0.3)	0
Eye pain	1(0.3)	0	0	0
Facial pain	1(0.3)	0	0	0
Faeces hard	1(0.3)	0	0	0
Feeling hot	1(0.3)	0	0	0
Fibrin d dimer decreased	1(0.3)	0	0	0
Fistula	1(0.3)	0	0	0
Flank pain	1(0.3)	0	1(0.3)	0
Fournier's gangrene	1(0.3)	1(0.3)	0	0

Gastric disorder	1(0.3)	0	1(0.3)	0
Gastric haemorrhage	1(0.3)	1(0.3)	0	0
Gastric ulcer	1(0.3)	0	0	0
Gastrointestinal anastomotic stenosis	1(0.3)	0	0	0
Gastrointestinal infection	1(0.3)	0	0	0
Gingival disorder	1(0.3)	0	0	0
Glomerular filtration rate decreased	1(0.3)	0	1(0.3)	0
Glycosuria	1(0.3)	0	1(0.3)	0
Haematocrit decreased	1(0.3)	0	0	0
Herpes simplex	1(0.3)	0	0	0
Herpes virus infection	1(0.3)	0	0	0
Herpes zoster	1(0.3)	0	0	0
High density lipoprotein decreased	1(0.3)	0	0	0
Hypercalcaemia	1(0.3)	0	1(0.3)	0
Hyperhidrosis	1(0.3)	0	1(0.3)	0
Hypernatraemia	1(0.3)	1(0.3)	2(0.6)	0
Hyperparathyroidism	1(0.3)	0	0	0
Hyperphosphataemia	1(0.3)	0	0	0
Hypersensitivity	1(0.3)	0	1(0.3)	0
Hyperthermia	1(0.3)	0	0	0
Hypophagia	1(0.3)	1(0.3)	0	0
Hypovolaemic shock	1(0.3)	1(0.3)	0	0
Immune thrombocytopenia	1(0.3)	1(0.3)	0	0
Immunodeficiency	1(0.3)	0	0	0
Impaired gastric emptying	1(0.3)	0	0	0
Incision site pain	1(0.3)	0	0	0
Infected neoplasm	1(0.3)	1(0.3)	0	0
Infection	1(0.3)	0	1(0.3)	0
Infusion related reaction	1(0.3)	0	1(0.3)	0
Infusion site inflammation	1(0.3)	0	0	0
Intermenstrual bleeding	1(0.3)	0	0	0
International normalised ratio increased	1(0.3)	0	1(0.3)	0
Intervertebral disc protrusion	1(0.3)	0	0	0
Jaundice	1(0.3)	1(0.3)	1(0.3)	1(0.3)
Jugular vein thrombosis	1(0.3)	0	0	0
Lacrimation increased	1(0.3)	0	0	0
Lacunar infarction	1(0.3)	0	0	0

Large intestinal obstruction	1(0.3)	1(0.3)	0	0
Laryngitis	1(0.3)	0	0	0
Laryngopharyngitis	1(0.3)	0	0	0
Ligament sprain	1(0.3)	0	0	0
Limb injury	1(0.3)	0	1(0.3)	0
Lipase increased	1(0.3)	1(0.3)	0	0
Lipoprotein increased	1(0.3)	0	1(0.3)	0
Localised oedema	1(0.3)	0	1(0.3)	0
Loss of consciousness	1(0.3)	1(0.3)	0	0
Malignant neoplasm progression	1(0.3)	1(0.3)	2(0.6)	2(0.6)
Mean cell haemoglobin concentration decreased	1(0.3)	0	0	0
Meibomian gland dysfunction	1(0.3)	0	0	0
Memory impairment	1(0.3)	0	2(0.6)	0
Menstrual disorder	1(0.3)	0	6(1.7)	0
Metabolic acidosis	1(0.3)	0	0	0
Metastases to lymph nodes	1(0.3)	0	0	0
Micturition urgency	1(0.3)	0	0	0
Middle ear inflammation	1(0.3)	0	0	0
Mouth haemorrhage	1(0.3)	0	0	0
Mouth injury	1(0.3)	0	0	0
Mucosal inflammation	1(0.3)	0	1(0.3)	1(0.3)
Multiple organ dysfunction syndrome	1(0.3)	1(0.3)	0	0
Muscle twitching	1(0.3)	0	0	0
Musculoskeletal pain	1(0.3)	0	0	0
Myalgia intercostal	1(0.3)	0	0	0
Nail bed bleeding	1(0.3)	0	0	0
Nail hypertrophy	1(0.3)	0	0	0
Narrow anterior chamber angle	1(0.3)	0	0	0
Nasal dryness	1(0.3)	0	0	0
Nasal ulcer	1(0.3)	0	0	0
Nephrolithiasis	1(0.3)	0	0	0
Neutrophil function disorder	1(0.3)	1(0.3)	0	0
Neutrophil percentage increased	1(0.3)	0	1(0.3)	0
Neutrophilia	1(0.3)	0	1(0.3)	0
Night sweats	1(0.3)	0	0	0
Ocular discomfort	1(0.3)	0	0	0
Ocular icterus	1(0.3)	0	0	0

Oedema	1(0.3)	0	0	0
Oesophageal obstruction	1(0.3)	0	0	0
Oliguria	1(0.3)	0	0	0
Onychomadesis	1(0.3)	0	0	0
Oral infection	1(0.3)	0	0	0
Pancreatitis acute	1(0.3)	1(0.3)	0	0
Papule	1(0.3)	0	1(0.3)	0
Paralysis	1(0.3)	0	0	0
Pelvic fluid collection	1(0.3)	0	1(0.3)	1(0.3)
Pericardial effusion	1(0.3)	1(0.3)	0	0
Periodontal disease	1(0.3)	0	0	0
Periodontal inflammation	1(0.3)	1(0.3)	0	0
Periodontitis	1(0.3)	0	0	0
Peripheral nerve paresis	1(0.3)	0	0	0
Peritonitis	1(0.3)	1(0.3)	0	0
Pharyngeal swelling	1(0.3)	0	0	0
Pneumothorax	1(0.3)	1(0.3)	0	0
Post procedural haemorrhage	1(0.3)	1(0.3)	0	0
Prostatitis	1(0.3)	0	0	0
Protein total increased	1(0.3)	0	1(0.3)	0
Psoriasis	1(0.3)	0	0	0
Rash maculo-papular	1(0.3)	0	0	0
Regurgitation	1(0.3)	0	0	0
Renal vein embolism	1(0.3)	0	0	0
Renin increased	1(0.3)	0	0	0
Respiratory tract infection	1(0.3)	0	0	0
Retching	1(0.3)	0	0	0
Retinol binding protein decreased	1(0.3)	0	0	0
Rheumatoid arthritis	1(0.3)	0	0	0
Rhinitis atrophic	1(0.3)	0	0	0
Rib fracture	1(0.3)	0	0	0
Rotator cuff syndrome	1(0.3)	0	0	0
Salivary hypersecretion	1(0.3)	0	0	0
Sciatica	1(0.3)	0	0	0
Scrotal dermatitis	1(0.3)	0	0	0
Sensation of foreign body	1(0.3)	0	0	0
Sepsis	1(0.3)	0	0	0

Shock	1(0.3)	1(0.3)	0	0
Shock haemorrhagic	1(0.3)	1(0.3)	1(0.3)	1(0.3)
Sinus arrhythmia	1(0.3)	0	2(0.6)	0
Sinusitis	1(0.3)	0	0	0
Sitophobia	1(0.3)	0	1(0.3)	0
Skin infection	1(0.3)	0	0	0
Skin irritation	1(0.3)	0	0	0
Skin ulcer	1(0.3)	1(0.3)	0	0
Sleep deficit	1(0.3)	0	0	0
Sleep disorder	1(0.3)	0	1(0.3)	0
Somnolence	1(0.3)	0	1(0.3)	0
Spinal stenosis	1(0.3)	0	0	0
Sputum increased	1(0.3)	0	0	0
Swelling	1(0.3)	0	0	0
Tachycardia	1(0.3)	0	1(0.3)	1(0.3)
Taste disorder	1(0.3)	0	0	0
Thermal burn	1(0.3)	0	0	0
Thirst	1(0.3)	0	1(0.3)	0
Throat irritation	1(0.3)	0	0	0
Thrombosis in device	1(0.3)	1(0.3)	0	0
Thyroid function test abnormal	1(0.3)	0	0	0
Thyroid mass	1(0.3)	0	0	0
Thyroiditis subacute	1(0.3)	0	0	0
Thyroxine free increased	1(0.3)	0	1(0.3)	0
Tinea manuum	1(0.3)	0	0	0
Tongue exfoliation	1(0.3)	0	0	0
Tonsillar inflammation	1(0.3)	0	0	0
Trichiasis	1(0.3)	1(0.3)	0	0
Trigeminal nerve disorder	1(0.3)	0	0	0
Tuberculosis	1(0.3)	0	0	0
Tumour pain	1(0.3)	0	0	0
Urethral haemorrhage	1(0.3)	0	0	0
Urethral pain	1(0.3)	0	1(0.3)	0
Urinary casts present	1(0.3)	0	0	0
Urinary retention	1(0.3)	0	1(0.3)	0
Urine leukocyte esterase positive	1(0.3)	0	0	0
Urine output increased	1(0.3)	0	1(0.3)	0

Vasculitis	1(0.3)	0	0	0
Vena cava embolism	1(0.3)	0	0	0
Venous thrombosis limb	1(0.3)	0	1(0.3)	0
Ventricular arrhythmia	1(0.3)	0	1(0.3)	0
Visual impairment	1(0.3)	0	1(0.3)	0
Vith nerve disorder	1(0.3)	0	0	0
Wound complication	1(0.3)	0	0	0
Yellow skin	1(0.3)	0	0	0
Abnormal faeces	0	0	2(0.6)	0
Activated partial thromboplastin time shortened	0	0	1(0.3)	0
Anal spasm	0	0	1(0.3)	0
Anaphylactic reaction	0	0	3(0.9)	1(0.3)
Anorectal swelling	0	0	1(0.3)	0
Aphasia	0	0	2(0.6)	0
Apolipoprotein a-i increased	0	0	1(0.3)	0
Arterial thrombosis	0	0	1(0.3)	0
Arthropathy	0	0	1(0.3)	0
Bacterial test positive	0	0	2(0.6)	0
Bladder spasm	0	0	1(0.3)	0
Blindness	0	0	1(0.3)	0
Blood magnesium decreased	0	0	1(0.3)	0
Blood potassium increased	0	0	1(0.3)	0
Breath odour	0	0	1(0.3)	0
Bronchitis chronic	0	0	1(0.3)	0
Bundle branch block right	0	0	1(0.3)	0
Carbohydrate antigen 72-4 increased	0	0	1(0.3)	0
Carbon dioxide decreased	0	0	1(0.3)	1(0.3)
Cardiac aneurysm	0	0	1(0.3)	0
Cardiac failure	0	0	1(0.3)	1(0.3)
Catheter site thrombosis	0	0	1(0.3)	0
Choking sensation	0	0	1(0.3)	0
Conjunctival hyperaemia	0	0	1(0.3)	0
Conjunctival ulcer	0	0	1(0.3)	0
Conjunctivitis	0	0	1(0.3)	0
Deep vein thrombosis	0	0	2(0.6)	0
Depression	0	0	1(0.3)	0
Dermatitis bullous	0	0	1(0.3)	1(0.3)

Dermo-hypodermitis	0	0	1(0.3)	0
Diabetes mellitus	0	0	1(0.3)	1(0.3)
Diplegia	0	0	1(0.3)	0
Disseminated intravascular coagulation	0	0	1(0.3)	0
Dyspnoea exertional	0	0	1(0.3)	0
Eczema asteatotic	0	0	1(0.3)	0
Electrocardiogram st segment abnormal	0	0	2(0.6)	0
Electrocardiogram st segment elevation	0	0	1(0.3)	0
Electrocardiogram t wave alternans	0	0	1(0.3)	0
Electrocardiogram t wave amplitude decreased	0	0	1(0.3)	0
Erythema	0	0	1(0.3)	0
Erythropenia	0	0	1(0.3)	0
Eyelid pain	0	0	1(0.3)	0
Face oedema	0	0	1(0.3)	0
Facial bones fracture	0	0	1(0.3)	0
Faecal volume increased	0	0	1(0.3)	0
Fibrin degradation products increased	0	0	1(0.3)	0
Fracture	0	0	1(0.3)	0
Frequent bowel movements	0	0	1(0.3)	0
Gastric dilatation	0	0	1(0.3)	0
Gastric stenosis	0	0	1(0.3)	1(0.3)
Gastroenteritis	0	0	1(0.3)	0
Gastrointestinal tract adenoma	0	0	1(0.3)	0
Generalised oedema	0	0	1(0.3)	0
Glomerular filtration rate increased	0	0	1(0.3)	0
Haemoglobinuria	0	0	1(0.3)	0
Haemorrhagic ascites	0	0	1(0.3)	1(0.3)
Head discomfort	0	0	1(0.3)	0
Hepatic encephalopathy	0	0	1(0.3)	1(0.3)
Hepatic failure	0	0	1(0.3)	1(0.3)
Hepatic steatosis	0	0	1(0.3)	0
Hot flush	0	0	1(0.3)	0
Infusion site pain	0	0	1(0.3)	0
Interleukin level increased	0	0	1(0.3)	0
Intestinal fistula	0	0	1(0.3)	1(0.3)
Intestinal perforation	0	0	1(0.3)	1(0.3)
Iron deficiency anaemia	0	0	1(0.3)	0

Joint swelling	0	0	2(0.6)	0
Left ventricular hypertrophy	0	0	2(0.6)	0
Leukaemoid reaction	0	0	1(0.3)	0
Limb discomfort	0	0	1(0.3)	0
Lymph node pain	0	0	1(0.3)	0
Lymphadenopathy	0	0	1(0.3)	0
Lymphocyte percentage decreased	0	0	2(0.6)	0
Lymphocyte percentage increased	0	0	1(0.3)	0
Menstruation irregular	0	0	1(0.3)	0
Monocyte percentage decreased	0	0	1(0.3)	0
Muscle strain	0	0	1(0.3)	0
Musculoskeletal chest pain	0	0	1(0.3)	0
Nail pigmentation	0	0	2(0.6)	0
Nasal congestion	0	0	1(0.3)	0
Neck pain	0	0	2(0.6)	1(0.3)
Oropharyngeal discomfort	0	0	1(0.3)	0
Pain in jaw	0	0	1(0.3)	0
Patella fracture	0	0	1(0.3)	0
Pharyngeal oedema	0	0	1(0.3)	0
Pleural effusion	0	0	4(1.1)	0
Pneumomediastinum	0	0	1(0.3)	0
Procalcitonin increased	0	0	1(0.3)	0
Pulmonary toxicity	0	0	1(0.3)	0
Qrs axis abnormal	0	0	1(0.3)	0
Rash pruritic	0	0	1(0.3)	0
Red blood cell sedimentation rate increased	0	0	1(0.3)	0
Red cell distribution width increased	0	0	1(0.3)	0
Rhinitis	0	0	1(0.3)	0
Sacral pain	0	0	1(0.3)	0
Serum amyloid a protein increased	0	0	1(0.3)	0
Skin injury	0	0	1(0.3)	0
Soft tissue swelling	0	0	1(0.3)	0
Spinal osteoarthritis	0	0	1(0.3)	0
Subclavian vein thrombosis	0	0	1(0.3)	0
Swelling face	0	0	1(0.3)	0
Thermal burns of eye	0	0	1(0.3)	0
Thrombocytosis	0	0	1(0.3)	0

Tongue dry	0	0	1(0.3)	0
Transfusion reaction	0	0	1(0.3)	0
Troponin i increased	0	0	1(0.3)	0
Troponin t increased	0	0	1(0.3)	0
Tumour haemorrhage	0	0	1(0.3)	1(0.3)
Upper respiratory tract inflammation	0	0	1(0.3)	0
Ureteric obstruction	0	0	1(0.3)	0
Volvulus	0	0	1(0.3)	0
Vulvitis	0	0	1(0.3)	0
Wound infection	0	0	2(0.6)	0
Xerophthalmia	0	0	1(0.3)	0

Supplementary Table 2. Treatment-emergent adverse events leading to death (Safety analysis set)

System Organ Class (SOC) Preferred Term (PT)	Fruquintinib plus paclitaxel group (N=350) n (%)		Placebo plus paclitaxel group (N=349) n (%)	
	All	Related	All	Related
Any TEAE leading to death	34 (9.7)	18 (5.1)	16 (4.6)	6 (1.7)
General disorders and administration site conditions	17 (4.9)	4 (1.1)	7 (2.0)	2 (0.6)
Death	10 (2.9)	4 (1.1)	6 (1.7)	2 (0.6)
Disease progression	7 (2.0)	0	1 (0.3)	0
Gastrointestinal disorders	7 (2.0)	6 (1.7)	3 (0.9)	2 (0.6)
Upper gastrointestinal hemorrhage	3 (0.9)	3 (0.9)	2 (0.6)	2 (0.6)
Intestinal obstruction	2 (0.6)	1 (0.3)	1 (0.3)	0
Large intestinal obstruction	1 (0.3)	1 (0.3)	0	0
Small intestinal obstruction	1 (0.3)	1 (0.3)	0	0
Infections and infestations	3 (0.9)	3 (0.9)	0	0
Neutropenic infection	1 (0.3)	1 (0.3)	0	0
Pneumonia	1 (0.3)	1 (0.3)	0	0
Septic shock	1 (0.3)	1 (0.3)	0	0
Vascular disorders	3 (0.9)	2 (0.6)	0	0
Hypovolemic shock	1 (0.3)	1 (0.3)	0	0
Shock	1 (0.3)	0	0	0
Shock hemorrhagic	1 (0.3)	1 (0.3)	0	0
Blood and lymphatic system disorders	2 (0.6)	2 (0.6)	0	0
Immune thrombocytopenia	1 (0.3)	1 (0.3)	0	0
Myelosuppression	1 (0.3)	1 (0.3)	0	0

Respiratory, thoracic and mediastinal disorders	2 (0.6)	2 (0.6)	0	0
Pulmonary embolism	1 (0.3)	1 (0.3)	0	0
Respiratory failure	1 (0.3)	1 (0.3)	0	0
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	1 (0.3)	0	3 (0.9)	1 (0.3)
Malignant neoplasm progression	1 (0.3)	0	2 (0.6)	0
Tumor hemorrhage	0	0	1 (0.3)	1 (0.3)
Nervous system disorders	1 (0.3)	1 (0.3)	1 (0.3)	0
Cerebral infarction	1 (0.3)	1 (0.3)	0	0
Hepatic encephalopathy	0	0	1 (0.3)	0
Cardiac disorders	0	0	1 (0.3)	0
Cardiac failure	0	0	1 (0.3)	0
Hepatobiliary disorders	0	0	1 (0.3)	1 (0.3)
Hepatic failure	0	0	1 (0.3)	1 (0.3)

Fruquintinib Clinical Study Protocol Cover Page

Study Protocol

Title: A randomized, double-blind, phase III trial to evaluate the efficacy and safety of fruquintinib in combination with paclitaxel for the treatment of patients with advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma

Abbreviated Title: A phase III trial of fruquintinib in combination with paclitaxel for the treatment of advanced gastric or GEJ adenocarcinoma

Protocol No.:	2017-013-00CH1
Study Drug:	Fruquintinib capsules (HMPL-013)
Sponsor:	HUTCHMED Limited
Version No.:	6.0
Version Date:	26 Aug 2021

Statement of ownership: This Protocol is owned by the sponsor, and it shall not be distributed, photocopied or published by individuals or organizations without authorization.

Note: This is a translated copy of the Chinese protocol_version 6.0 of 2017-013-00CH1. In the event of any discrepancy between this translation and the Chinese verison, please refer to the Chinese version.

Clinical Trial Protocol Signature Page - 01

Signature of Sponsor Representative

The signature below indicates the agreement to the clinical trial protocol that is currently provided.

Full name: [redacted] Title: [redacted]

Signature: Date:

Clinical Trial Protocol Signature Page - 02

Signatures of Principal Investigators of study sites

The signature below indicates agreement to the clinical trial protocol that is currently provided.

Full name:

Signature: Date:

Version History

Version No.	Version Date
1.0	Apr 28, 2017
2.0	Jun 28, 2017
3.0	Feb 9, 2018
4.0	Oct 16, 2019
5.0	Nov 03, 2020
6.0	Aug 26, 2021

Contact Information of the Sponsor, HUTCHMED Limited

Should medical emergencies occur, the Case Officer of the clinical study team can be contacted. If the Case Officer cannot be contacted, the Safety Reporting Contact listed in the following table can be contacted.

	Full name	**Contact number**
Clinical Project Lead	[redacted]	[redacted] [redacted] [redacted]
Safety Reporting Contact	[redacted]	[redacted] [redacted] [redacted]

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1 Abbreviations and Terminologies

Abbreviations and Terminologies	Definition
AE	Adverse event
AESI	Adverse event of special interest
A/G	Albumin/globulin
ALB	Albumin
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
ANC	Absolute neutrophil count
APTT	Activated partial thromboplastin time
AST	Aspartate aminotransferase
AUC	Area under the concentration-time curve
BCRP	Breast cancer resistance protein
BSC	Best supportive care
BUN	Blood urea nitrogen
CFDA	China Food and Drug Administration
CRC	Colorectal cancer
CrCl	Creatinine clearance rate
CHOL	Cholesterol
CL/F	Apparent clearance
C_{max}	Peak concentration
c-MET	Mesenchymal-epithelial transition factor
CNS	Central nervous system
CR	Complete remission
eCRF	Electronic case report form
CT	Computed tomography
DBiL	Direct bilirubin
DCR	Disease control rate
DLT	Dose-limiting toxicity
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
EGFR	Epidermal growth factor receptor
FDG-PET	Fluorodeoxyglucose positron emission tomography
FIB	Fibrinogen
Glu	Glucose

Abbreviations and Terminologies	Definition
GEJ	Gastro-esophageal junction
HFS	Hand-foot syndrome
HR	Hazard ratio
HUVEC	Human umbilical vein endothelial cells
IB	Investigator's Brochure
IBiL	Indirect bilirubin
IC_{50}	Half maximal inhibitory concentration
ICH	International Council for Harmonization
IEC/IRB	Independent Ethics Committee/Institutional Review Board
INR	International normalized ratio
ISF	Investigator Site File
ITT	Intention-to-treat analysis
LDH	Lactate dehydrogenase
MedDRA	Medical Dictionary for Regulatory Activities
MRI	Magnetic resonance imaging
MMRM	Mixed model for repeated measures
MRT	Mean residence time
NCI-CTC	National Cancer Institute Common Toxicity Criteria
NCI CTCAE	National Cancer Institute Common Terminology Criteria for Adverse Events
NOAEL	No observed adverse effect level
NSCLC	non-small cell lung cancer
NYHA	New York Heart Association Functional Classification
ORR	Objective response rate
OS	Overall survival
PD	Progressive disease
PDGFR	Platelet-derived growth factor receptor
PFS	Progression-free survival
PK	Pharmacokinetics
PLT	Platelet
POC	Proof of Concept
PPS	Per protocol set
P-gp	P-glycolprotein
PR	Partial response
PT	Prothrombin time
QD	quaque die (once a day)

Abbreviations and Terminologies	Definition
RECIST	Response Evaluation Criteria In Solid Tumors
REML	Restricted maximum likelihood
SAE	Serious adverse event
Scr	Serum creatinine
SS	Safety analysis set
SD	Stable disease
$t_{1/2}$	Terminal elimination half life
TBiL	Total bilirubin
TC	Total cholesterol
TG	Triglyceride
TGI	Tumor growth inhibition
TKI	Tyrosine kinase inhibitor
T_{max}	Time of peak concentration
TNM	TNM classification of malignant tumors
TP	Total protein
TT	Thrombin time
TTP	Time to progression
TSH	Thyroid-stimulating hormone
ULN	Upper limit of normal
VEGF	Vascular endothelial growth factor
VEGFR	Vascular endothelial growth factor receptor
V_z/F	Apparent volume of distribution during terminal phase
WHO	World Health Organization

2 Synopsis

Investigational drug name	Fruquintinib capsule (laboratory code: HMPL-013)
Control drug	Placebo capsule
Protocol no.	2017-013-00CH1
Study title	A randomized, double-blind, phase III trial to evaluate the efficacy and safety of fruquintinib in combination with paclitaxel for the treatment of patients with advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma
Sponsor	HUTCHMED Limited
Principal investigator	Professor Xu Ruihua
Study site	Sun Yat-Sen University Cancer Center and several other clinical research institutions
Number of patients to be enrolled	About 700
Study duration	The estimated enrollment duration is approximately 54 months. Estimated study period: September 2017 to March 2023
Indications	Advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma
Study population	Patients with advanced gastric or GEJ adenocarcinoma confirmed through histology or cytology who did not respond to first-line standard chemotherapy (containing platinum and fluorouracil)
Study objectives	To compare the efficacy and safety of fruquintinib combined with paclitaxel with those of paclitaxel monotherapy for second-line treatment of advanced gastric or GEJ adenocarcinoma
Study design	This is a randomized double-blind placebo-controlled multicenter phase III clinical study to compare the efficacy of fruquintinib combined with paclitaxel with that of paclitaxel monotherapy for second-line treatment in patients with advanced gastric or GEJ adenocarcinoma. Approximately 700 patients will be randomly (at a 1:1 ratio) assigned to treatment groups below: ● Fruquintinib+paclitaxel

	● Placebo + paclitaxel Patients will be randomly stratified according to following factors: ● The gastroesophageal junction vs the stomach ● Presence of peritoneal metastasis vs absence of peritoneal metastasis ● ECOG PS 0 vs 1 All patients will receive fruquintinib (or placebo) combined with paclitaxel. One treatment cycle consists of 4 weeks. Tumor assessment will be performed through imaging method once every 4 weeks in the first 3 cycles and once every 8 weeks thereafter till the presence of disease progression. Safety observations include: AEs, changes in laboratory values, vital signs and ECG changes, etc. The status of tumor treatment and the follow-up of survival after progression will be recorded. During the study, all patients are not allowed to cross over to the other treatment, but they are allowed to receive the best supportive care. An Independent Data Monitoring Committee (IDMC) will be established to review safety and efficacy data.
Treatment regimen	● Investigational drug: Fruquintinib Capsules Fruquintinib Capsules (HMPL-013) 4 mg, PO, once a day (QD) for 3 consecutive weeks followed by 1-week break (a cycle=4 weeks); ● Control: Placebo Capsules Placebo Capsules 4 mg, PO QD for 3 consecutive weeks followed by 1-week break (a cycle=4 weeks); ● Paclitaxel injection: Paclitaxel injection 80 mg/m^2 iv once weekly in the first 3 weeks (i.e. at days 1, 8 and 15, respectively) (a cycle=4 weeks); ○ Remark: Paclitaxel will be administered for at most 6 cycles of treatment and then, if no disease progression occurs, replaced with fruquintinib (or fruquintinib placebo) monotherapy.
Treatment termination for patient	Patients should receive treatment until they present with any of following conditions: ● Criteria for completion of treatment: – Disease progression determined according to the Response

	Evaluation Criteria in Solid Tumors (RECIST 1.1); – Death; – End of the entire study • Criteria for premature termination of treatment: – Presence of toxicity to the patients that requires drug withdrawal; – Patient is pregnant – Patients or their legal representatives request to withdraw from the study – Loss to follow-up – The patient has poor compliance and cannot comply with the study protocol – The investigator decides to terminate the treatment for the patient's best interests
Evaluation Criteria	
Efficacy Endpoints	**Dual primary efficacy endpoints:** Overall survival (OS), progression-free survival (PFS) (defined according to RECIST 1.1) **Secondary efficacy endpoints:** Objective response rate (ORR), disease control rate (DCR), duration of disease response and quality-of-life score (EORTC QLQ-C30, version 3.0) **Exploratory objective:** To explore biomarkers related to the antitumor activity of fruquintinib.
Safety Endpoints	Adverse events (AE), severe adverse event (SAE), ECOG score, vital signs, physical examination, laboratory test, 12-lead electrocardiogram (ECG) and cardiac ultrasound findings.
Inclusion criteria	In accordance with the Guidelines for Clinical Trials of Anti-tumor Drugs, patients must meet all of the following criteria before inculsion: 1. The Informed Consent Form has been signed; 2. Aged 18-75 years (inclusive); 3. Patients with metastatic or locally advanced gastric or GEJ adenocarcinoma definitively diagnosed by histological and/or cytological examinations; all other histological types are excluded;

	4. At least one lesion outside the stomach that can be evaluated according to RECIST1.1; a local mass in the stomach or gastroesophageal junction is classified as "unmeasurable"; 5. Patients with advanced gastric or GEJ adenocarcinoma who failed* to respond to the standard first-line chemotherapy regimen (which shall contain platinum and fluorouracil for treatment for no less than 2 weeks) *Remark: Failure to respond to standard first-line chemotherapy is defined as disease progression (that occurs during treatment or within 4 weeks after final administration of drugs (including a single first-line drug for maintenance treatment) at the end of treatment). Neoadjuvant or adjuvant therapy is permitted, in which paclitaxel-containing drug can be included. But if relapse or metastasis occurs during the neoadjuvant or adjuvant therapy period or within 6 months after completion of the above treatment, that adjuvant therapy is considered as the failure of the first-line systemic chemotherapy for progressive disease; 6. No history of systemic antitumor treatment within 4 weeks before the first dose of study drug, which involves systemic chemotherapy (washout period of oral fluorouracil is 2 weeks), radiotherapy, immunotherapy, biological therapy and/or hormone therapy (except for traditional Chinese medicine treatment with anti-tumor indications not verified by randomized trial); 7. Body weight ≥40 kg; or BMI >18.5; 8. Physical status score (ECOG score) 0-1; 9. Expected survival >12 weeks.
Exclusion criteria	Patients meeting any of the following criteria must be excluded from this study: 1. Presence of squamous cell carcinoma (adenosquamous carcinoma that mainly contains squamous cell carcinoma); 2. Known Her-2 positive gastric or GEJ cancer that has not undergone this targeted (e.g. Herceptin) treatment (patients who experienced disease progression after treatment with Herceptin can be enrolled); 3. Patients who have not recovered from toxicity caused by previous anti-cancer treatment (CTCAE>grade 1), or not completely recovered from previous surgery;

	4. Patients with brain metastasis at baseline; 5. Patients who have had other malignancies over the past 5 years (except those with squamous-cell or basal cell carcinoma, and cervical carcinoma in situ, who received radical resection); 6. Presence of active infection that is not controlled clinically, such as acute pneumonia and hepatitis B or C in the active stage (patients with a past history of HBV infection whose HBV DNA is $\geq 1 \times 10^4$ copies or $\geq$2000 IU/mL, regardless of whether medication is taken or not); 7. Cannot take oral medications due to dysphagia, or intractable vomiting or known drug malabsorption; 8. Present active gastric/duodenal ulcer, ulcerative colitis, intestinal obstruction and other gastrointestinal diseases or other conditions that may lead to gastrointestinal bleeding or perforation according to the investigators' judgment; or those who had intestinal perforation or intestinal fistula within 6 months; or those having had surgeries for intestinal perforation or intestinal fistula and did not recover; 9. Patients with a history of arterial thrombosis or deep venous thrombosis within the past 6 months; 10. Patients who had an evident hemorrhagic tendency or a history of hemorrhage within 2 months before randomization, such as tarry stools, hematemesis, hemoptysis, and fecal occult blood (+ + or more) (patients tested 1+ for fecal occult blood and have primary lesions must undergo gastroscopy to rule out bleeding or active ulcers before they can enrolled); 11. Stroke and/or transient ischemic attack within 12 months prior to randomization; 12. Appropriate organ function. Patients with any of the following conditions will be excluded: ● Absolute neutrophil count (ANC) $<2\times10^9$/L, or platelet (PLT) $<100\times10^9$/L or hemoglobin (Hb) <90g/L within 1 week prior to randomization (blood transfusion or receiving G-CSF treatment within 1 week before blood sampling are not acceptable); ● Serum total bilirubin (TBiL) >1.5 times upper limit of normal (ULN), or alanine aminotransferase (ALT) or aspartate aminotransferase (AST) >2.5 times ULN (based on normal reference in the clinical study site); for patients with liver

	metastasis, ALT or AST >5 times ULN; ● Electrolyte abnormality of clinical significance; ● Serum creatinine (Scr) >ULN and creatinine clearance rate (CrCl) <50mL/min; ● Urine protein 2+ or more, and 24 h urine protein quantification ≥1.0 g/24 h; ● Activated partial thromboplastin time (APTT) and prothrombin time (PT) >1.5×ULN (according to normal reference in each clinical study site); 13. Hypertension that still cannot be controlled even after routine treatment is given (systolic blood pressure≥140 mmHg or diastolic blood pressure ≥90 mmHg); 14. Heart function evaluation: left ventricular ejection fraction <50% (echocardiography); 15. Acute myocardial infarction, severe/unstable angina or coronary bypass surgery within 6 months prior to randomization; or patients with ≥NYHA grade 2 cardiac insufficiency; 16. Skin wound, surgical site, wound site, severe mucosal ulcer or fracture without complete healing; 17. Macrovascular invasion by tumor, such as pulmonary artery, superior vena cava, and inferior caval vein; 18. Patients who have undergone major surgery or major invasive procedure within 60 days prior to randomization; 19. Female patients who are pregnant or lactating or of child bearing potential with positive pregnancy test result before the first dose; 20. Patients of child bearing potential who or whose sexual partners are not willing to take contraceptive measures; 21. The patient is unfit to participate in this clinical trial according to the investigator's judgment; 22. With severe psychological or mental abnormality; 23. Patients who have undergone brachytherapy (such as radioactive seeds implantation) within 60 days prior to randomization; 24. Patients having participated in other clinical trials on other oral investigational drugs that have not been approved in China within 4

	weeks before randomization, except for patients who have participated in observational studies; 25. Patients known to have a past history of serious allergic reactions to the investigational drug including its excipients; 26. Presence of relapse or metastasis during and 6 months after the enoadjuvant/adjuvant therapy in which the patients received administration of paclitaxel-containing drug; 27. Patients having received paclitaxel-containing drug for first-line treatment; 28. Patients having a past history of systemic treatment targeting the vascular endothelial growth factor receptor (VEGFR) signaling pathway.
Statistical analysis	**Analytical methods:** The objective of this study is to evaluate overall survival (OS) and PFS after treatment with fruquintinib plus paclitaxel vs. fruquintinib placebo plus paclitaxel in patients with advanced gastric cancer who did not respond to first-line chemotherapy. OS and PFS are the dual primary endpoints. OS is defined as the time from patient randomization to patient death because of any cause. PFS is defined as the time from randomization date to either the first observation of PD or death, whichever occurs first. The investigators will evaluate PD based on RECIST v1.1. Patients who have not progressed or died at the time of analysis will be censored at the time of the latest evaluable tumor assessment. If a subject has no evaluable tumor assessment after baseline, the subject will be censored on the date of randomization. Intergroup comparison in OS and PFS will be performed using the stratified log-rank test based on the intention to treat analysis (ITT) set. Stratification factors include the primary tumor sites (the gastroesophageal junction vs the stomach), EOCG PS (0 vs 1) and peritoneal metastasis (presence of peritoneal metastasis vs absence of peritoneal metastasis). The statistical test is a two-tailed test where 0.05 denotes a total significance level. In addition, estimate the median survival and PFS and corresponding 95% confidence interval of the two treatment groups using Kaplan-Meier method, and draw the Kaplan-Meier curve that can provide visual description on the differences between both treatment groups. Estimation of treatment effects of OS and PFS shall be expressed by the hazard ratio (HR) estimated by Cox proportional hazards model and its 95%

	confidential interval. **Statistical hypothesis and sample size determination:** The primary endpoint is to verify if the efficacy (OS or PFS) of fruquintinib combined with paclitaxel is superior to placebo combined with paclitaxel in patients with advanced gastric or GEJ adenocarcinoma as the second line therapy. Null hypothesis (H_0) of OS: OS HR = 1.0 (fruquintinib combined with paclitaxel vs placebo combined with paclitaxel) Alternative hypothesis (H_1) of OS: OS HR < 1.0 (fruquintinib combined with paclitaxel vs placebo combined with paclitaxel) The number of primary endpoint of OS required for the evaluation of efficacy is calculated based on the following assumptions: ● The significance level is one-sided 0.025 (which is equal to two-tailed 0.05) ● 80% power of test ● The OS hazard ratio (HR) of treatment group/control group is 0.75, in other words, the median overall survival is extended from 7.0 months to 9.33 months ● Randomization at an 1:1 ratio As of the date of protocol version 4.0 amendment, the first interim analysis planned in protocol version 3.0 has been completed, which occurred when 100 patients were enrolled and the last enrolled patient completed the 6-month follow-up. The objective of the interim analys is to early terminate the study if negative result is obtained. The criteria of early termination were provided in the Appendix C in IDMC charter (Version 1.1, issued on 11 Mar 2019). Based on the results from the first interim analysis and pre-supposed criteria for early termination, IDMC suggested to continue the study, and an additional interim analysis could be added when 50% of the OS events have been observed. The sponsor accepted the suggestion. The second interim analysis will use Lan-DeMets spending function to construct the valid or invalid O'Brien-Fleming type stopping boundary for study termination. Then a total of 401 OS events need to be observed at the final analysis to achieve the 80% power of test. Assuming that the non-homogeneous enrollment will be completed within 29 to 30 months and the last patient need to be followed up for 7 months, a total of 544 patients are

required to be enrolled to observe 401 OS events.

The purpose of the second interim analysis is to early terminate the study by superiority or inferiority results, or continue the study by adjusting the sample size, and evaluate the safety profile of fruquintinib combined with paclitaxel. If the results of the interim analysis meet the criteria for early termination by superiority results, the sponsor may accept the suggestion from IDMC and terminate the study. The second interim analysis will occurred when about 201 OS events achieved (approximately 50% of toal OS events).

- If the statistic exceed the efficacy critical value (z≥2.958, two-tailed $p \leq 0.003$), the study will be terminated by superior efficacy. Under the hypothesis of exponential distribution, the critical value is equal to HR ≤ 0.659.
- If the statistic exceed the futility critical value (z≤0.564, two-tailed $p \geq 0.573$), the study will be terminated by futility. Under the hypothesis of exponential distribution, the critical value is equal to HR ≥ 0.924.
- If the results of the interim analysis does not meet the criteria for study termination but fall in the promising zone, the sample size will be increased by CHW method (Cui, Hung& Wang, 1999). In the final analysis, the statistic will be calculated by CHW method. For the reasons of confidentiality, the rules of re-estimation of sample size will be provided in an independent document, and will inform all members of IDMC before IDMC meeting.
- If the results of the interim analysis does not meet the criteria for study termination and the sample size does not need to be adjusted, the final analysis will be conducted when around 401 OS events occurred. The corresponding efficacy critical value: z≥1.969, two-tailed $p \leq 0.049$. Under the hypothesis of exponential distribution, the critical value is equal to HR ≤ 0.821.
- If the results of the interim analysis indicate serious safety issues, the study will be early terminated.

O's Brien-Fleming boundary values

OS Ratio number of events	Boundary value for statistic z		Boundary value for nominal α (two-tailed)		Cumulative α consumption	Cumulative β consumption
	Valid	Invalid	Valid	Invalid		

201	0.501	2.958	0.564	0.003	0.573	0.003	0.070
401	1.0	1.969	1.969	0.049	0.049	0.05	0.20

After the second interim analysis, Sponsor determined to go on the study with increasing the sample size as suggested by Study Scientific Monitor Committee and Center for Drug Evaluation (CDE). The final analysis will be performed when approximately 700 patients are enrolled and approximately 525 OS events are observed.

Considerations for an increase in OS event count to 525:

- ✓ Fuquinitinib placebo plus paclitaxel (paclitaxel monotherapy), median OS = 8.3 months (results from Keynote-061)
- ✓ Fuquinitinib plus paclitaxel versus paclitaxel monotherapy, 2-month improvement of OS at 2 months (HR = 0.806)
- ✓ Final analysis significance level was two-sided 0.049 (accounting for consumption of one completed interim analysis for OS)
- ✓ Power of final analysis: approximately 70%

The protocol is currently revised to version 6.0. Based on further communication with the CDE, the Sponsor decided to change the single primary endpoint of OS to dual primary endpoints of PFS and OS. The final analysis of PFS and OS will be performed simultaneously.

- ✓ Graphical method will be used (alpha distribution and recycling) to control type I error for the final analysis of the dual primary endpoints at two-sided 0.049 level (considering the consumption of one completed OS interim analysis). The remaining total two-sided significance level 0.049 will be split into two parts, 0.01 for PFS test (two-sided) and 0.039 for OS test (two-sided).
- ✓ If it's statistically significant for PFS analysis, the two-sided level 0.01 of PFS will be recycled and transformed to OS, OS will be test on the two-sided 0.049 level.
- ✓ If it's statistically significant for OS analysis, the two-sided level 0.039 of OS will be recycled and transformed to PFS, PFS will be test on the two-sided 0.049 level.

The sample size for this study was determined by the final analysis of OS. The final analysis will be performed when approximately 700 patients are enrolled and approximately 525 OS events are observed. Based on the number of 656 PFS events and a two-sided 0.01 significance level,

	approximately 97% power would be provided to detect a statistical difference in PFS between the two groups (assumed HR=0.7). **Final analysis** The final analysis will be performed when approximately 700 patients are enrolled and approximately 525 OS events are observed.

Table 1 Study Flow Chart

Study period	Screening period[22]		Treatment period							Follow-up period	
			Cycle 1		Cycle 2		Cycle 3		Cycle 4 to end of treatment (EOT)[23]	Observation period after EOT[24]	Survival follow-up period
	Screening phase 1	Screening phase 2	Day 1 (first dose)	Day 15	Day 1	Day 15	Day 1	Day 15	Day 1	30 days after the last administration	Once every 2 months after EOT
Window period	**Within 21 days prior to administration**	**within 7 days prior to administration[23]**	**0 day**	**±2 days**	**±3 days**	**±2 days**	**±3 days**	**±2 days**	**±3 days**	**±3 days**	**±7 days**
Study protocol item											
Informed consent form[1]	X										
Medical history/tumor history[2]	X										
Demographics	X										
Present illness/baseline symptoms		X									
Physical examination[3]		X		X	X	X	X	X	X	X	
ECOG score		X		X	X	X	X	X	X	X	
QLQ-C30 score[4]		X			X		X		X	X	
Vital signs[5]		X		X	X	X	X	X	X	X	
Blood routine[6]		X		X	X	X	X	X	X	X	
Blood biochemistry[7]		X		X	X	X	X	X	X	X	
Coagulation[8]		X		X	X	X	X	X	X	X	
Urine analysis[9]		X		X	X	X	X	X	X	X	
Fecal occult blood		X		X	X	X	X	X	X	X	

Study period	Screening period[22]		Treatment period							Follow-up period	
			Cycle 1		Cycle 2		Cycle 3		Cycle 4 to end of treatment (EOT)[23]	Observation period after EOT[24]	Survival follow-up period
	Screening phase 1	Screening phase 2	Day 1 (first dose)	Day 15	Day 1	Day 15	Day 1	Day 15	Day 1	30 days after the last administration	Once every 2 months after EOT
Window period	**Within 21 days prior to administration**	**within 7 days prior to administration[23]**	**0 day**	**±2 days**	**±3 days**	**±2 days**	**±3 days**	**±2 days**	**±3 days**	**±3 days**	**±7 days**
Pregnancy test (if applicable)[10]		X								X	
12-lead ECG[11]	X			X	X		X		X	X	
Thyroid function test[12]	X				X		X		X	X	
Echocardiography[13]	X										
Tumor assessment[14]	X		Tumor assessment will be performed through imaging method once every 4 weeks in the first 3 cycles and once every 8 weeks thereafter till the presence of disease progression. It is recommended that, for patients who withdraw from the study early or are suspended for reasons unassociated with disease progression, evaluation should be continued until the presence of disease progression.							X[14]	
Inclusion/exclusion criteria		X									
Drug distribution/return[15]		X			X		X		X	X	

Study period	Screening period[22]		Treatment period							Follow-up period	
			Cycle 1		Cycle 2		Cycle 3		Cycle 4 to end of treatment (EOT)[23]	Observation period after EOT[24]	Survival follow-up period
	Screening phase 1	Screening phase 2	Day 1 (first dose)	Day 15	Day 1	Day 15	Day 1	Day 15	Day 1	30 days after the last administration	Once every 2 months after EOT
Window period	**Within 21 days prior to administration**	**within 7 days prior to administration[23]**	**0 day**	**±2 days**	**±3 days**	**±2 days**	**±3 days**	**±2 days**	**±3 days**	**±3 days**	**±7 days**
Investigational drug (fruquintinib or placebo)[16]			X In a 28-day treatment cycle, drug is orally administered qd in the first 21 days (3-week administration + 1-week break)								
Concomitant drug therapy[17] (Paclitaxel)			X In a 28-day treatment cycle, paclitaxel is administered iv gtt at D1, D8 and D15.								
Concomitant drugs/treatments[18]	X	X	X	X	X	X	X	X	X	X	
Collection of adverse events[19]	X	X	X	X	X	X	X	X	X	X	
Biomarker sample (blood)[20]			X			X					
Collection of subsequent anti-cancer treatment										X	X
Survival follow-up[21]											X

Notes:

1. Informed consent form must be signed before all study specific procedures, and it can be done as early as 21 days before the administration .
2. Past history/history of tumors: date of initial diagnosis of gastric adenocarcinoma and its site (the stomach or gastroesophageal junction), pathological type by Lauren classification (intestinal type, diffuse type, mixed type, other type, or unknown type), its grade (well differentiated, moderately differentiated, poorly differentiated, or unknown), and its TMN stage (according to AJCC criteria (version 7, 2010)); date of diagnosis of the initial distant metastasis (or local advance) and its site (the liver, peritoneum or abdomen, etc.); previous therapeutic regimens (including the first-line regimen), the dates of their initiation and termination, the best response to them, and the date of disease progression during their use; a past history of chemotherapy with targeted drugs (HER2, EFGR or others), which shall contain the dates of initiation and termination of chemotherapy and target sites for chemotherapy; as well as a past history of surgery (total or subtotal gastrectomy) for the primary lesion (tumor) and its name and time. Previous significant procedures (e.g. gastroscope, biopsy and other diagnostic or therapeutic non-invasive procedures) must be recorded in case report form, including start and end dates, name of procedure and procedure site.
3. Physical examination includes height (only in screening), body weight, head, eyes, ears, nose, larynx, neck, heart, chest (including lungs), abdomen, limbs, skin, lymph nodes, neurological condition and general status
4. QLQ-C30 is assessed before the start of each treatment cycle, at the end of treatment, and at follow-up conducted 30 days after end of treatment.
5. Vital signs include blood pressure, heart rate, respiratory rate and body temperature. If antihypertensive treatment is received at baseline or during study, blood pressure should be monitored 3 ±2 h after administration of antihypertensive drug on the day of visit.
6. Blood routine test: red blood cell, hemoglobin, neutrophil, platelet count and absolute leukocyte differential count. If neutrophil $\leqslant 1\times10^9$/L or platelet $\leqslant 50\times10^9$/L, frequency of re-examination should be increased (once every 2-3 days, the results of test conducted outside the hospital can be accepted); blood routine should be re-examined before every administration of paclitaxel.
7. Blood biochemistry: includes total protein (TP), albumin (ALB), fasting blood glucose, urea nitrogen (BUN) or urea, creatinine, alkaline phosphatase (ALP), lactate dehydrogenase (LDH), total bilirubin (TBiL), direct bilirubin (DBiL), indirect bilirubin (IBiL), aspartate aminotransferase (AST), alanine aminotransferase (ALT), calcium, phosphorus, potassium, sodium, pancreatic amylase or serum amylase, uric acid triglycerides (TG) and total cholesterol (TC). If ALT or AST increases by 3 times and above, and total bilirubin increases by 2 times and above, it is recommended that the frequency of examination should be increased (recommend 1-2 times/week, the results of test conducted outside the hospital can be accepted); at baseline, creatinine clearance should be calculated according to creatinine value CrCl (calculation formula: CrCl=(140-age)×body weight (kg)/[72×Scr(mg/dL)] or CrCl=[(140-age)×body weight (kg)]/[0.818×Scr (umol/L)]). Attention should be paid to unit of creatinine in calculation of creatinine clearance. For females, it is calculation result ×0.85).

8. Coagulation: include prothrombin time (PT) and activated partial thromboplastin time (APTT). For baseline, the international normalized ratio (INR) should be determined.
9. Routine urine test: protein, red blood cell count, white blood cell count and urine glucose. If urine routine indicates urine protein ++ or more, 24 h urine protein has to be determined (the results of test conducted outside the hospital can be accepted).
10. Pregnancy: female patients of child bearing age have to undergo blood pregnancy test in screening period and 30 days after drug discontinuation. If pregnancy is suspected, additional pregnancy test should be performed. It is not applicable to postmenopausal female patients. For postmenopausal patients, time of menopause should be recorded.
11. 12-lead ECG: includes heart rate, P-R interval, RR interval, QRS time limit, Q-T interval, QTc and ECG diagnosis.
12. Thyroid function test: includes serum free triiodothyronine (fT3), serum free tetraiodothyronine (fT4) and thyroid stimulating hormone (TSH).
13. Echocardiography: includes left ventricular ejection fraction and diagnosis result. Re-examination can be performed if there is clinical indication during the trial.
14. Tumor assessment: baseline tumor assessment should be completed within 3 weeks prior to administration (-7 days window period) in accordance with standards of RECIST 1.1. For the imaging inspection (CT or MRI) on the chest and abdomen, if clinically necessary, proper measures (CT or MRI) can be adopted to inspect any other known or suspected disease sites (such as pelvic cavity and brain). Record the presence or absence of a primary lesion and a measurable/unmeasurable lesion. For baseline and following assessments, the same imaging inspection should be adopted, and by the same investigator as far as possible. The visit window of tumor assessment may be advanced to day -5. Tumor assessment will be performed through imaging method once every 4 weeks in the first 3 cycles and once every 8 weeks thereafter till the presence of disease progression. Tumor assessment of every cycle should be carried out on the first day of the next cycle until disease progression happens. Scheduled visit time for tumor assessment is calculated based on Cycle 1 Day 1 (C1D1) and not affected by medication discontinuation. If patients have no tumor assessment within 28 days prior to end of treatment/withdrawal, tumor assessment should be performed at end of treatment/withdrawal. Patients with symptoms of history of bone metastasis at baseline should receive bone scan at baseline. If there is bone lesion at baseline, local bone lesion should be followed up. Meanwhile, brain imaging examination must be conducted to rule out brain metastasis at baseline. For patients with suspected disease progression before the next scheduled assessment, unscheduled tumor assessment should be performed.
15. Study drug distribution/ return: after administration, unused drugs and drug bottles from the last cycle will be returned on D1 visit of each cycle and drugs for the next cycle will be distributed. The safety visit must be completed before the distribution of drugs.
16. See Section 9 of the Protocol for administration of study drug. The dosing cycle of fruquintinib and its placebo is fixed, without window period and not affected by

medication discontinuation or dose reduction. The window period shown in the flow chart does not apply to fruquintinib and its placebo.

17. The dosing cycle of paclitaxel (every 28 days is one cycle) is fixed, starting from C1D1, and the window period of each injection is +3 days (except for C1D1). The first administration of paclitaxel must be on the same day as the first administration of fruquintinib (or fruquintinib placebo). The hematologic criteria in Section 9.4.3 of the protocol must be met before each administration of paclitaxel. The scheduled injection time for paclitaxel can be delayed depending on adverse events. If the administration of paclitaxel is delayed, the scheduled time for the subsequent administration in the same cycle will be delayed accordingly, but the last dose of this cycle must be completed no later than D22 (included); otherwise, an injection must be stopped. The administration of paclitaxel does not affect the scheduled time for safety evaluation. The window period shown in the flow chart does not apply to paclitaxel injection.
18. Concomitant drugs/treatments: treatments from 4 weeks prior to the first dose of investigational drug until the observation period after end of study treatment (EOT) or before initiating the new anti-tumor therapy (whichever occurs first) must be recorded, including generic name and daily dose of the drug; reason for taking such drug for treatment; start and end dates of drug treatment.
19. Duration for collecting adverse event (AE) will be from the first dose of the investigational drug to 30 days after last administration or start of new anti-tumor treatment (whichever occurs first). Collection of serious adverse event (SAE) will be from the date of signing the Informed Consent Form to 30 days after last administration or start of new anti-tumor treatment (whichever occurs first), afterwards, only the SAEs related to the investigational drug confirmed by the investigator will be collected.
20. Blood sample collection will be performed twice: at D1 in the first cycle, and at D15 in the second cycle before administration of paclitaxel and fruquintinib (or fruquintinib placebo).
21. Survival follow-up: Survival follow-up will be performed every 2 months after EOT to collect subsequent anti-tumor treatment and trial related SAEs. The frequency of telephone follow-up may be increased as required.
22. Screen failures have a chance to undergo re-screening.
23. The collection of safety data must be completed within ETO + 7 days. If safety evaluation has been conducted within 7 days before EOT, the safety evaluation at EOT can be waived.
24. The collection of safety and other data must be completed during the observation period visit at 30 days after last administration or before the start of new anti-tumor treatment (whichever occurs first). It may be waived if the new anti-tumor treatment starts within 7 days after the end of treatment.

3 Amendments to the Protocol

The protocol may not be amended without consent from both the Principal Investigators and Sponsor.

If amendments to the protocol are necessary, the amendment contents or the new version of the protocol must be submitted to the Ethics Committee for review, and written approval must be obtained, unless the amendments are to the trial personnel or contact information, or if emergency protocol amendments are implemented out of consideration for the safety of patients.

4 Background of the Study

4.1 Gastric Cancer

Gastric cancer is a common malignant tumor of the digestive tract. According to GLOBOCAN[1], in 2012, there were more than 950,000 new cases of gastric cancer worldwide, with the highest incidence rates in East Asia; more than 72,000 deaths resulted from gastric cancer, making it the third leading cause of cancer death after lung cancer and liver cancer. In China, there were more than 400,000 new cases of gastric cancer in 2012, accounting for the second highest incidence rate after lung cancer and 42.6% of the world's new cases. Approximately 330,000 deaths resulted from gastric cancer in 2012, making it the third leading cause of cancer death after lung cancer and liver cancer, and accounting for approximately 45% of the world's deaths. According to statistics published in CA in 2016, in China, approximately 680,000 new cases of gastric cancer were estimated, and close to 500,000 deaths resulted from gastric cancer in 2015[2]. These statistics indicate that gastric cancer has posed a serious threat to the health and lives of the population.

In particular, advanced gastric cancer accounts for a large proportion of new cases. As surgical indications are usually not applicable for such patients, combination therapy is used for the treatment of advanced gastric cancer. Even if surgical resection is possible, most patients will develop metastases or local progression. For patients with distant metastases or local progression, flucytosine (5-FU or capecitabine) in combination with cisplatin or oxaliplatin is usually the regimen of choice for first-line therapy. Other alternative regimens include paclitaxel in combination with cisplatin or oxaliplatin, single-agent flucytosine, single-agent paclitaxel, flucytosine in combination with irinotecan, etc. If paclitaxel or irinotecan was not used during first-line therapy, such drugs will be the drugs of choice for second-line therapy[3]. Currently, research is focused on the use of small-molecule targeted drugs in combination with chemotherapy drugs in clinical research. In particular, angiogenesis inhibitors are one class of drugs being studied[4]. In 2014, the FDA approved ramucirumab (made by Eli Lilly and Company) for use in combination with paclitaxel as second-line therapy for advanced gastric or gastroesophageal junction adenocarcinoma[5]. This regimen has already been included as one of the regimens of choice for second-line therapy in the NCCN Guidelines for Gastric Cancer[3]. The RAINBOW trial[6] has demonstrated that ramucirumab in combination with paclitaxel as second-line therapy for advanced gastric adenocarcinoma can significantly prolong overall survival (mOS 9.6m vs 7.4m, HR=0.807, P=0.017), and significantly prolong progression-free survival (mPFS 4.4m vs 2.9m, HR=0.635, $p<0.001$). In 2014, the CFDA approved apatinib, a small-molecule targeted drug which specifically targets and inhibits VEGFR-2, as a third-line and beyond therapy for advanced gastric adenocarcinoma[7]. The Phase III trial of apatinib in patients with gastric cancer indicated that, compared with the placebo group, apatinib can

significantly prolong overall survival (mOS 6.5m vs 4.7m, HR=0.709, p= 0.0149) and significantly prolong progression-free survival (mPFS 2.6m vs 1.8m, HR=0.44, p<0.001) in patients receiving third-line and beyond therapy for advanced gastric adenocarcinoma[8].

Angiogenesis is the most critical step in the occurrence and progression of malignant tumors. According to a variety of studies, angiogenesis in the tumor region provides nutrients for tumors and facilitates the removal of metabolites, and tumor cells can be transferred to the rest of the body through new blood vessels. Therefore, effective inhibition of angiogenesis in the tumor region can inhibit the growth of tumor cells, and reduce the incidence of metastasis as well. The development of tumor angiogenesis is associated with a variety of vascular factors. In the absence of oxygen, rapidly growing tumor cells can secrete a variety of angiogenic growth factors that stimulate angiogenesis. One important factor is the vascular endothelial cell growth factor (VEGF). The vascular endothelial cell growth factor/vascular endothelial cell growth factor receptor (VEGFR) has been found to be one of the main inducing factors related to tumor angiogenesis, therefore the VEGF/VEGFR signaling pathway is considered to be one of the most promising targets in molecular targeted therapies[4].

Fruquintinib is a drug developed by HUTCHMED Limited, which owns the completely independent intellectual property rights to the drug. It is a small-molecule compound that targets the VEGF receptor kinase family, and is closely linked to angiogenesis. Fruquintinib mainly acts on the VEGFR transmembrane receptor family (VEGFR 1, 2 and 3) in vascular endothelial cells, and has high kinase selectivity. The half maximal inhibitory concentration (IC_{50}) of fruquintinib against VEGFR2, VEGFR1 and VEGFR3 are 35nM, 33nM and 0.5nM respectively. In addition, it has no inhibitory activity (IC_{50}>3μM) towards a variety of kinases associated with the cell cycle or cell proliferation, including cyclin-dependent kinases (CDK1, 2, 5), epidermal growth factor receptors (EGFR) and mesenchymal-epithelial transition factors (c-Met). Fruquintinib also has no significant inhibitory activity towards beta-type platelet-derived growth factor receptors (PDGFRβ) (IC_{50}>1μM). Therefore, with high kinase selectivity, such a compound has a good selectivity at the molecular level[9].

4.2 Fruquintinib

4.2.1 Preclinical Evidence of Fruquintinib

The direct killing effect of fruquintinib is very weak in 13 kinds of cells, including primary human umbilical vein endothelial cells (HUVEC) (IC_{50}⩾30 μM, with IC_{50} of 18.7μM for primary HUVEC). Compared with the VEGF-dependent proliferation of HUVEC (with IC_{50} of only 1.7 nM), the difference in inhibitory activity is more than 10,000 times. Fruquintinib has high effectiveness and low toxicity at enzymatic and cellular levels.

In nude mice models with subcutaneous transplantations of human colon cancer HT-29 cells, human non-small cell lung cancer NCI-H460 cells and human renal cancer Caki-1 cells, fruquintinib was administered orally once a day for 3 consecutive weeks with doses of 0.77, 1.92, 4.8 and 12 mg/kg (increments of 2.5 times), and fruquintinib demonstrated a dose-dependent tumor growth inhibitory effect. In refractory tumor models such as malignant melanoma A375, pancreatic BXPC-3, pancreatic Miapaca and hepatoma Bel-7402, the daily doses of 1.5, 5 and 15 mg/kg of fruquintinib had significant inhibitory effects on tumor growth. The human gastric carcinoma BGC-823 model had the highest sensitivity towards fruquintinib, as tumor growth was almost completely inhibited with a daily dose of 2 mg/kg.

In addition, we conducted a preliminary study on the anti-tumor effects of fruquintinib in

combination with chemotherapy drugs commonly used clinically. Using nude mice models with subcutaneous transplantations of primary tumor tissue from cancer patients, we found that for the primary colon cancer model COL1T0117P4, the respective tumor growth inhibition (TGI) for oral administration of 1 mg/kg of fruquintinib and single-agent oxaliplatin (10 mg/kg, i.v., once a week) were 36.3% and 23.9%, and the TGI for the fruquintinib/oxaliplatin combination group was 68.1%. For the primary human poorly differentiated gastric adenocarcinoma transplantation model GAS1T0113P5, the TGI for oral administration of 1 mg/kg of fruquintinib was 44.0%, the TGI for docetaxel (5 mg/kg, i.v., once a week) was 46.3%, and the TGI was improved to 73.4% when the combination of both drugs was used. For the primary pancreatic cancer model PAN1T0108P7, the TGI for 2 mg/kg of fruquintinib and 25 mg/kg of gemcitabine (i.p., once a week) were 52.8% and 46.9% respectively, and the TGI was improved to 71.9% when the combination of both drugs was used. In the aforementioned models, statistically significant differences ($P<0.05$ or $P<0.01$) were observed in the tumor volumes of the combination groups and single-agent groups, which indicate that the combination of fruquintinib and the aforementioned chemotherapy drugs enhances drug effectiveness. On the other hand, symptoms of increased animal toxicity were not observed in the use of fruquintinib in combination with the chemotherapy drugs. The body weight changes of animals in the combination groups and the single-agent groups were largely similar.

For detailed results of the acute toxicity, chronic toxicity and genotoxicity studies of fruquintinib, please refer to the Investigator's Brochure (IB, v10.0).

4.2.2 Clinical Experience with Fruquintinib

Results of Clinical Pharmacokinetic Studies

In the study 2009-013-00CH1, the pharmacokinetics of fruquintinib in patients with advanced malignant solid tumors were evaluated. After oral administration of single doses of 1, 2, 4, 5 and 6 mg of fruquintinib, extremely low clearance was observed. The once-daily dosage regimen was established based on the long half-life of more than 35 hours. Within the dose range of 1 to 6 mg, area under the concentration-time curve (AUC) and peak concentration (C_{max}) essentially increased in proportion to the dose level. After administration of multiple-daily doses of fruquintinib, equilibrium of the exposure level was achieved 14 days after administration. Based on the AUC for a 24-hour interval, the exposure level of fruquintinib increased by approximately 3 times from the first administration to the equilibrium state. To enable comparison with administration for 4 consecutive weeks, at dose levels of 5 mg and 6 mg, a dosage regimen of suspended administration for 1 week after administration for 3 consecutive weeks (once daily) was investigated. The pharmacokinetic results indicated that during identical dates of administration, the pharmacokinetic status of fruquintinib for both dosage regimens were similar.

In the study 2012-013-00CH3, two dosage regimens, i.e. "4 mg QD, 4 consecutive weeks" and "5 mg QD, 3 weeks on/1 week off" for patients with advanced colorectal cancer were compared and evaluated. In general, the pharmacokinetic properties of fruquintinib derived from this study were similar to the results of the previous study (2009-013-00CH1).

In the study 2012-013-00CH2, the impact of food consumption after oral administration of a single dose (4 mg) in healthy male volunteers on the pharmacokinetics of fruquintinib was studied. The impact of food consumption was evaluated through the study of bioequivalence under fasting conditions and fed conditions. The exposure levels of fruquintinib were expressed

as $AUC_{0-\infty}$ and C_{max}. The criteria for bioequivalence was met for the geometric mean (90% confidence interval) under fasting/fed conditions, i.e. 80%-125% for AUC, and 70%-143% for C_{max}. Absorption was slower under fed conditions as compared to fasting conditions. The main metabolites of fruquintinib in blood plasma are M9 (a carboxylic acid metabolite) and M11 (a desmethyl metabolite). In the study 2014-013-00CH5, the concentrations of fruquintinib and its metabolites, M9 (HM5024093) and M11 (HM5025423) in the blood plasma of healthy male volunteers were tested. The $AUC_{0-\infty}$ of the metabolites M9 and M11 were, respectively, 8.1% and 35.6% of the $AUC_{0-\infty}$ of the parent drug, fruquintinib.

In the study 2014-013-00CH3, the preliminary PK properties of fruquintinib after combination with paclitaxel were as follows:

- In 28 patients whose PK properties were evaluable, a good linear relationship was observed between the PK properties of fruquintinib at steady state and dose;
- PK properties of paclitaxel were evaluated in 11 patients, the exposure level for fruquintinib in combination with paclitaxel at steady state was increased by 30% compared to that of single-agent paclitaxel; in addition, the half-life of paclitaxel during combination therapy was slightly prolonged compared to that of single-agent paclitaxel.

Efficacy Overview of Fruquintinib in the Treatment of Advanced Tumors

Currently, the efficacy data derived from the Phase I clinical study (2009-013-00CH1) indicates that fruquintinib has demonstrated good clinical anti-tumor activity, as partial response (PR) and stable disease (SD) were observed in most patients with advanced cancer who had undergone repeated treatments.

In addition, the results of two Proof of Concept (POC) studies for fruquintinib as third-line or beyond therapy for advanced colorectal cancer (CRC) and third-line therapy for advanced non-small cell lung cancer (NSCLC) indicated that compared to placebo, fruquintinib had better clinical efficacy in patients with metastatic CRC and NSCLC, and can significantly prolong PFS. The PFS results indicated that the outcomes of both POC studies were positive, and the primary efficacy endpoints were successfully met.

Studies of third-line or beyond therapy for CRC patient population (including 2009-013-00CH1, 2012-013-00CH3 and 2012-013-00CH1, n=121)

In 121 patients with colorectal cancer, the objective response rate and disease control rate were 8.26% (10/121) and 71.1% (86/121) respectively.

In the study 2012-013-00CH3: median progression-free survival (mPFS)=5.3 months, median overall survival=8.8 months

In the study 2012-013-00CH1: fruquintinib versus placebo: mPFS=4.7 months vs 1.0 month, stratified HR=0.31, P<0.001; mOS=7.7 months vs 5.5 months, HR=0.71, P=0.27.

Studies of third-line therapy for NSCLC patient population (including 2009-013-00CH1 and 2014-013-00CH1, n=68)

In 68 patients with advanced lung cancer, the objective response rate and disease control rate were 20.6% (14/68) and 72.1% (48/68) respectively.

In the study 2014-013-00CH1: fruquintinib versus placebo: mPFS=3.81 months vs 1.15 months, stratified HR=0.27, P<0.001

Phase Ib/II study of fruquintinib in combination with paclitaxel as second-line therapy for advanced gastric cancer (n=34)

For the Phase II clinical trial of the recommended dose of fruquintinib (4 mg, QD, 3/1 regimen) in combination with paclitaxel (80 mg/m^2) as second-line therapy for advanced gastric cancer, ORR=25.9%, DCR=66.7%, mPFS=4 months and OS=8.5 months.

Safety Overview of Fruquintinib in the Treatment of Advanced Tumors

Based on the safety issues of same class of drugs as fruquintinib, the results of non-clinical toxicological studies of fruquintinib and the evaluation of data from six phase I-III clinical studies completed, the important risks of fruquintinib are determined to be bleeding, abnormal liver function, dermatological toxicity, hypertension and proteinuria. Important potential risks of fruquintinib include gastrointestinal perforation, surgical and wound healing complications, and embryotoxicity.

In the phase Ib/II trial of fruquintinib in combination with paclitaxel for the treatment of advanced gastric cancer, during the dose escalation stage, dose-limiting toxicity (DLT) was observed in 2 patients from "4 mg fruquintinib+80 mg/m^2 paclitaxel" dose group (one patient had Grade 3 febrile neutropenia and one patient had Grade 4 neutropenia). In the said dose group, the most common CTC AE Grade ⩾ 3 drug-related adverse events with an incidence of > 5% was reduced neutrophil count, reduced white blood cell count, hypertension, hand-foot syndrome and anemia. Fruquintinib-related toxicities were similar to those observed in previous studies.

For detailed clinical trial data, please refer to version 10.0 of the Investigator's Brochure for fruquintinib.

4.3 Rationale for Study Design

The recommended dose for fruquintinib in combination with paclitaxel (80mg/m^2) for Phase III trials is 4 mg, QD, 3 weeks on/1 week off.

In the dose escalation stage of the study 2014-013-00CH3, DLT did not occur in the 2 mg and 3 mg dose groups; 2 cases of DLT occurred in the 4 mg dose group (febrile neutropenia for both cases). The facts that such toxicity is related to paclitaxel, is commonly observed clinically, and easily controlled and managed using G-CSF, have been taken into consideration. In addition, the overall incidence of fruquintinib-related toxicities in the 4 mg dose group is low, and is similar to the safety results of previous studies. Adverse events of Grade 3 and above did not occur within the DLT observation period. Moreover, during the dose escalation study, further validation and observation of safety and tolerability in the particular dose group were also performed.

5 Study Objectives

5.1 Primary Objective

To compare the OS and PFS for fruquintinib plus paclitaxel versus placebo plus paclitaxel as

second-line therapy in patients with advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma.

5.2 Secondary Objectives

- Objective response rate (ORR), disease control rate (DCR) and duration of response (DoR);
- To evaluate the improvement to the quality of life of patients based on EORTC QLQ-C30 version 3.0 (Appendix 3);
- Safety and tolerability.

5.3 Exploratory Objective

To explore the biomarkers related to the anti-tumor activity of fruquintinib and the possible mechanisms of drug resistance.

6 Study Design

6.1 Overview of Study Design

This is a randomized, double-blind, placebo-controlled, multicenter Phase III clinical trial comparing the efficacy and safety of fruquintinib plus paclitaxel versus placebo plus paclitaxel as second-line therapy for patients with advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma. The type of comparison used in this study is a superiority test.

After screening, patients who satisfy the eligibility criteria will be randomly assigned to the fruquintinib plus paclitaxel group or the placebo plus paclitaxel group in a 1: 1 ratio.

Approximately 700 patients will be randomly (in a 1: 1 ratio) assigned to the following treatment groups:

- fruquintinib plus paclitaxel
- placebo plus paclitaxel

Stratified randomization of the randomized patients will be performed based on the following factors

- gastroesophageal junction vs stomach
- peritoneal metastasis, present vs absent
- ECOG PS 0 vs 1

Remarks: Paclitaxel will be administered for at most 6 cycles of treatment and then, if no disease progression occurs after 6 cycles, the patient is switched to fruquintinib (or fruquintinib placebo) monotherapy for maintenance.

All patients will receive fruquintinib (or placebo) treatment with 3 weeks on/1 week off regimen as one treatment cycle, and will concurrently receive paclitaxel treatment for 4 continuous weeks to constitute one treatment cycle. Imaging methods will be used to assess tumor status once every 4 weeks for the first 3 cycles and once every 8 weeks for subsequent cycles, until disease progression. Safety indicators include: AE, changes in laboratory values, vital signs and ECG changes, etc. In addition, the tumor treatment status and survival follow-up after disease

progression will be recorded.

Crossover treatment is not allowed during the trial; however, all patients are allowed to receive best supportive care.

The study is divided into 3 periods: baseline period, treatment period, and follow-up period. For the time definitions of the study periods, please refer to **Table 2**.

Table 2 Time Definitions of the 3 Periods of the Study

Screening/baseline	Treatment	Follow-up
Day -21 to Day 1 (Before the 1st administration of the study drug on Day 1)	Day 1 of the 1st cycle to end of treatment	End of treatment to end of study*

*The criteria for end of study for patients is death, loss of follow-up, refusal of survival follow-up, or the end of the entire study.

When any of the following criteria is met, the entire clinical study will end:

- The sponsor decides to terminate the trial early.
- Unblinding and analysis have been conducted after achieving at least 525 OS events during final analysis.

After the end of the entire study, if there are patients in the fruquintinib group without disease progression, upon assessment by the investigator(s), fruquintinib, but not paclitaxel, will continue to be provided after approval of the sponsor.

6.2 Randomization Method and Inclusion Process

The judgment of the investigator(s) on previous chemotherapy and item 5 of the enrollment criteria are patient to confirmation by the sponsor before randomization. If patients satisfy all eligibility criteria, randomization can be performed. Investigators must login to the Interactive Web Response System (IWRS) and complete the IWRS worksheet to register for screening. Patients who satisfy the eligibility criteria shall be randomly assigned in accordance with blinding principles to the fruquintinib plus paclitaxel group or the placebo plus paclitaxel group in a 1: 1 ratio. Factors for stratified randomization are the primary site (gastroesophageal junction vs stomach); EOCG PS (0 vs 1); peritoneal metastasis (present vs absent). The randomization number of each patient will be provided to the investigators by the IWRS.

Patients must begin the study treatment within 2 days after randomization.

Patients who end the study after signing the Informed Consent Form and before randomization shall be considered as "screen failures". Patients who failed screening previously can be re-screened. However, a new Patient ID number must be assigned by the IWRS. Such situations must be reviewed by the sponsor on a case-by-case basis, and patients can only be re-screened once.

6.3 Number of Patients

The planned enrollment for this study is 700, with 350 patients each in the experimental group and the control group.

6.4 Study Sites

There will be approximately 37 study sites, including the Sun Yat-sen University Cancer Center. The Sun Yat-sen University Cancer Center is the leading unit for this study.

7 Target Population

Patients histologically or cytologically diagnosed with advanced gastric or GEJ adenocarcinoma who failed first-line standard chemotherapy (use of platinum-based drugs or fluorouracil drugs for at least 2 treatment cycles must be included).

7.1 Inclusion Criteria

In accordance with the Guidelines for Clinical Trials of Anti-tumor Drugs, patients must satisfy all of the following criteria before inclusion:

1. Signed the Informed Consent Form;
2. 18-75 years of age (inclusive);
3. Histologically or cytologically diagnosed with metastated or locally progressed gastric or GEJ adenocarcinoma; all other histological types are excluded
4. At least one lesion beyond the stomach that is evaluable according to the standard of RECIST1.1; local tumors limited to the stomach or the gastroesophageal junction are classified as "unmeasurable";
5. Patients with advanced gastric or GEJ adenocarcinoma who failed first-line standard chemotherapy* (use of platinum-based drugs or fluorouracil drugs for at least 2 treatment cycles must be included);

   *Note: Failed first-line standard chemotherapy is defined as disease progression during the treatment process, disease progression within 4 months after the last administration of the drug at the end of treatment (including single-agent maintenance therapy for first-line therapy). Neoadjuvant or adjuvant therapy is allowed, and paclitaxel can be used for neoadjuvant or adjuvant therapy. However, if recurrence or metastasis occurs during adjuvant therapy or within 6 months after completion of the adjuvant therapy, it will be considered as failed first-line systemic chemotherapy for progressive disease;
6. Patients must not have received anti-tumor therapy, such as systematic chemotherapy (washout period of oral fluorouracil is 2 weeks) or radiotherapy, immunotherapy, biological or hormonal therapy within 4 weeks before the first dose of study drug (except for traditional Chinese medicine treatment with anti-tumor indications not verified by randomized trial);
7. Body weight ⩾ 40 kg; or BMI >18.5
8. ECOG performance status: 0-1;
9. Expected survival rate of more than 12 weeks.

7.2 Exclusion Criteria

If any of the following criteria is satisfied, patients must be excluded from this study:

1. Squamous carcinoma (including adenosquamous carcinoma predominantly composed of squamous carcinoma);

2. Known Her-2 positive gastric or GEJ cancer that has not undergone this targeted (e.g. Herceptin) treatment (patients who experienced disease progression after treatment with Herceptin can be enrolled);

3. Patients with unrestored toxicity from previous anti-cancer therapy (CTCAE >level 1), not fully recovered from previous surgery;

4. Patients with brain metastases at baseline;

5. Malignant tumors within the past 5 years (excluding squamous cell carcinoma or basal cell carcinoma, or patients with cervical carcinoma in situ who had undergone radical surgery);

6. Clinically uncontrolled active infections such as acute pneumonia, active phase of hepatitis B or hepatitis C (history of hepatitis B virus infection regardless of whether medications have been used to control the infection, HBV DNA $\geqslant 1\times10^4$ copies/mL or $\geqslant$ 2000 IU/mL);

7. Patients who cannot take oral medications due to dysphagia, patients with intractable vomiting or known drug absorption disorders;

8. Present active gastric/duodenal ulcer, ulcerative colitis, intestinal obstruction and other gastrointestinal diseases or other conditions that may lead to gastrointestinal bleeding or perforation according to the investigators' judgment; or those who had intestinal perforation or intestinal fistula within 6 months; or those having had surgeries for intestinal perforation or intestinal fistula and did not recover;

9. Patients who have a history of arterial thrombosis or deep vein thrombosis within the past 6 months;

10. Patients who show obvious evidence or history of bleeding within 2 months before randomization e.g. black stool, hematemesis, hemoptysis, fecal occult blood ++ or above, (patients tested 1+ for fecal occult blood and have primary lesions must undergo gastroscopy to rule out bleeding or active ulcers before they can be enrolled),etc.;

11. Patients who have had stroke and/or transient cerebral ischemia within 12 months prior to randomization;

12. Regarding adequate organ function, the patient should be excluded if any of the following conditions is found:

- Absolute neutrophil count (ANC) $<2\times10^9$/L, or platelet (PLT) $<100\times10^9$/L or hemoglobin (Hb) <90g/L within 1 week prior to randomization (blood transfusion or receiving G-CSF treatment within 1 week before blood sampling are not acceptable);
- serum total bilirubin (TBiL) exceeds 1.5 times the upper limit of normal (ULN), or alanine aminotransferase (ALT) and aspartate aminotransferase (AST) exceed 2.5 times ULN (based on the normal value of clinical trial research center); the levels of ALT or AST are higher than 5 times ULN in patients with liver metastases;
- clinically significant electrolyte abnormality;
- serum creatinine (Scr) exceeds the normal upper limit and creatinine clearance

(CrCl) is below 50mL/min;

- urine protein test 2+ or more, and 24 hours urine protein quantification ⩾ 1.0g/24h;
- partial prothrombin time (APTT) and prothrombin time (PT) exceeds 1.5 × ULN (based on the normal value of clinical trial research center);

13. High blood pressure that cannot be controlled (systolic blood pressure ⩾ 140mmHg or diastolic blood pressure ⩾ 90mmHg) even when the normative treatment is given;

14. If the evaluation of cardiac function shows that left ventricular ejection fraction <50% (echocardiography);

15. Patient has suffered acute myocardial infarction, or undergone severe/unstable angina or coronary artery bypass surgery within 6 months prior to randomization; or patients with NYHA Class 2 cardiac insufficiency or above;

16. Patients whose skin wounds, surgical site, traumatic site, mucosal ulcer or fracture is not completely healed;

17. Macrovascular invasion by tumor, such as pulmonary artery, superior vena cava, and inferior caval vein;

18. Patients who have undergone major surgery or major invasive procedure within 60 days prior to randomization;

19. Pregnant or lactating women or women who may be pregnant and have a positive result of the pregnancy test before first medication administration;

20. Patients with bearing potential but are not willing to receive contraceptive measures or their sex partners are unwilling to receive contraceptive measures;

21. The investigators believe that the patient is unfit to participate in this clinical study;

22. There is a serious psychological or mental abnormality;

23. Patients who have undergone brachytherapy (such as radioactive seeds implantation) within 60 days prior to randomization;

24. Patients who have participated in other clinical trials and have taken other investigational drug(s) that have not been approved in China within 4 weeks before randomization, except for patients who have participated in observational studies;

25. Patients who are known to have a serious drug allergic reaction to the study drug, including its excipients;

26. Presence of relapse or metastasis during and 6 months after the enoadjuvant/adjuvant therapy in which the patients received administration of paclitaxel-containing drug;

27. Patients who were treated with paclitaxel as the first-line treatment;

28. Patients who have previously received the systemic therapy that targets the VEGFR signaling pathway.

7.3 Criteria for Termination of the Treatment

Patients should receive the treatment unless any of the following conditions occurs:

- The criteria for completion of the study treatment:
  - confirmation of progressive disease (PD) according to the Response Evaluation Criteria in Solid Tumors (RECIST1.1 version);
  - death;
  - end of the entire study.
- Criteria for early termination of the treatment:
  - the patient shows toxicities and needs to stop taking the drug;
  - the patient is pregnant;
  - request of the patient or his/her legally authorized representative for withdrawal;
  - the patient is lost to follow-up;
  - the patient shows poor compliance and cannot follow the instructions of the trial protocol;
  - the investigator determines that termination of the treatment provides the maximum benefit to the patient.

7.4 Patient Withdrawal Criteria

The reason(s) that the patient withdraws from the study must be recorded. The follow-up results of all patients who terminate their treatment must be recorded and unused study drug must be counted and returned.

All patients who terminate their treatment, regardless of the reason, will be patiented to survival follow-up until death, except for those who withdraw their informed consent and state clearly they reject the follow-ups. The survival status will be evaluated every 2 months after completion of the treatment. Before the end of the study, all surviving patients will be patiented to one survival follow-up. After the end of the study, follow-ups will not be conducted on the patients. In addition, tumor assessment information will be collected from those patients who do not show progressive disease at the end of the treatment until progressive disease is identified.

Criteria for Patients to Complete the Study:

1. Death;
2. End of the study.

Criteria for Patient's Early Withdrawal:

1. The patient refuses to undergo survival follow-up;
2. The patient is loss to follow-up.

7.5 Replenishment of Patients

The patients who withdraw from the study will not be replaced or replenished.

7.6 Dropout Criteria

All the patients who signed the Informed Consent Form, passed the screening and are

randomized into the study are entitled to withdraw from this study at any time.

1. For patients who only signed the Informed Consent Form and passed the screening, but are not randomized, they will not be considered dropout cases.
2. After randomization, as long as the patient cannot undergo safety evaluation because he/she did not take the study drug at all, and/or cannot undergo effectiveness evaluation due to failure to complete one cycle of trial, he/she will be considered a dropout case, except for the following reasons:
    1) Patients who experienced disease progression after randomization and have clear medical evidence will not be considered dropout cases, but imaging evidence must be provided;
    2) Patients who withdrew due to pregnancy or intolerable toxicity will also not be considered dropout cases;
    3) The investigator judges that the treatment should be terminated for the patient's best interest;
    4) The patient dies after randomization.

For dropout cases, the investigator must record the reason(s) for dropout, make every effort to complete the relevant tumor assessment items and record the situation of the last visit.

8 Evaluation Plan and Steps

8.1 Evaluation Indicators for Screening Phase

The Informed Consent Form (ICF) must be signed before implementing any procedure of this study. The trial site should save all signed ICFs, including the ICFs of the patients who have failed the screening procedure.

8.1.1 Documentation of Complete Medical History during the Screening Period

1. Demographic information: date of birth, gender, ethnic group/race, weight, height; body surface area; history of drinking, and history of smoking.
2. Past history (except for the indication):
    a. past/existing disease(s) upon enrollment into this study;
    b. history of surgery: start and end date, name of surgery and site of surgery, etc.
    c. past relevant operations: start and end date, name of operation and site of operation, etc.
    d. other baseline features.
3. History of gastric or GEJ adenocarcinoma
    a. date of first diagnosis, site of primary lesion (stomach or GEJ) and TNM stage;
    b. histological or cytological diagnosis, pathologic Lauren classification (intestinal, diffuse, mixed type or other), tumor grade (well differentiated, moderately differentiated, poorly differentiated, unknown);
    c. presence or absence of the primary lesion, presence or absence of extra-abdominal measurable lesion;

d. the diagnosis date of first distal metastasis (or local progression), the site of metastasis (presence or absence of liver metastasis, presence or absence of peritoneal metastasis, presence or absence of ascites, etc.) at baseline;

e. past anti-cancer treatment(s): drug class (chemotherapy, immunotherapy, target therapy, others), start/end date, best efficacy, treatment failure/date of progressive disease;

f. history of radiotherapy: start/end date, site of radiotherapy, dosage and frequency;

g. history of surgery: name of surgery (total gastrectomy, subtotal gastrectomy, or others [abdominal exploration, etc.]), time;

h. whether or not any other meaningful operation (diagnostic or non-invasive operation such as endoscopy, biopsy, etc.) is conducted: start/end date, name of operation and site;

4. Medication administration status within 4 weeks before the first dose of study drug

All drug treatments and meaningful non-drug treatments received within 4 weeks before the first dose of study drug must be recorded, including: the generic name of the drug, daily dose, reason of administration of said drug, the start/end date of administration of said drug or whether administration of said drug is still ongoing at the time of enrollment.

8.1.2 Evaluation Indicators for Screening Period

Patients who have failed the screening have a chance to undergo re-screening.

1. Comprehensive physical exam including height (only at the time of screening), weight, head, eyes, ears, nose, throat, neck, heart, chest (including the lungs), abdomen, limbs, skin, lymph nodes, nervous system status and general status.
2. Performance status (ECOG PS): the ECOG PS scale is used at the screening phase and patients' general status is monitored during the screening phase.
3. Vital signs including blood pressure, heart rate, breathing rate and temperature. If an anti-hypertensive drug is used at the baseline, patient's blood pressure must be monitored 3 hours (±2 hours) after taking the anti-hypertensive drug on the day of follow-up.
4. Hematology test: red blood cell (RBC), hemoglobin, neutrophils, platelet count and absolute white blood cell count.
5. Blood biochemistry test: alkaline phosphatase (ALP), total bilirubin (TBiL), direct bilirubin (DBiL), indirect bilirubin (IBiL), aspartate aminotransferase (AST), and alanine aminotransferase (ALT); including total protein (TP), albumin (ALB), blood glucose, urea or urea nitrogen (BUN), creatinine, lactate dehydrogenase (LDH), calcium, phosphorus, potassium, sodium, pancreatic amylase or serum amylase, uric acid, triglyceride (TG), and total cholesterol (TC).
6. The creatinine clearance rate is calculated based on the level of creatinine for baseline.
7. During the course of the study, increase the frequency of test (for evaluation conducted during the treatment period and follow-up period) if the level of ALT or AST increased $\geqslant$ 3 × ULN and the total bilirubin increased $\geqslant$ 2 × ULN.
8. Coagulation test: including PT, APTT and INR.
9. Urine routine: Including urine protein, red blood cells, white blood cells, urine sugar.
10. Fecal occult blood: If the patient is unable to give a stool sample at the follow-up for fecal occult blood test, the patient should be advised to closely monitor the stool condition.

A timely follow-up should be scheduled in the event of occurrence of black stool.

11. Pregnancy test: Women of childbearing potential are required to have a blood test during the screening period and within 30 days after they stop taking the drug. If a patient is suspected of being pregnant during the course of the study, additional pregnancy test must be performed. Pregnancy test does not apply to women who are not of childbearing potential. Women of non-childbearing potential are defined as women in natural menopause for more than 24 consecutive months, or after total hysterectomy or bilateral oophorectomy.
12. 12-lead ECG: including heart rate, P-R interval, RR interval, QRS duration, Q-T interval, QTc and ECG diagnosis results.
13. Thyroid function test: Including serum free triiodothyronine (fT3), serum free thyroxine (fT4) and thyroid stimulating hormone (TSH).
14. Cardiac ultrasonography: Including left ventricular ejection fractions (LVEF) and results of diagnosis, and the exam can be performed again if a clinical indicator is found during the study period.
15. Completed Health Questionnaire (QLQ-C30 version 3.0)
16. Baseline Tumor Assessment: See Section 8.2.1 for details.

8.2 Evaluation for Treatment Period

During the study period, unless otherwise specified, the visit(s) of each treatment cycle should be completed during the window period of the scheduled date for safety and efficacy evaluation. See Figure 1 for details.

8.2.1 Efficacy Evaluation

Efficacy Endpoint

- Dual primary efficacy endpoint: OS, PFS
- Secondary efficacy endpoint: ORR, DCR (where SD ⩾ 8 weeks), duration of disease response, quality of life score (EORTC QLQ-C30, Version 3.0)

Efficacy Evaluation

- Tumor Assessment: The investigator conducts tumor and efficacy evaluation according to RECIST 1.1 (The Response Evaluation Criteria in Solid Tumors) along with clinical condition. For specific evaluation criteria see Appendix 2. The method of imaging evaluation of tumor will be determined by the investigator, either CT or MRI may be used, but the method and technical parameters of imaging evaluation selected should be consistent for the same patient during the study period. A contrast agent should be used if there is no contraindication. If the tumor evaluation has been performed 28 days prior to the first administration and the same method is used in the same hospital, this tumor evaluation can be used as a baseline for tumor evaluation. Baseline tumor evaluation should include the chest, abdomen, brain (will only be carried out on patients with or suspected of having brain lesions in the subsequent visits) and any other sites with known or suspected tumor lesions (such as pelvic etc.). Patients who have bone metastases at baseline requires a bone scan at baseline, and if there is a bone lesion at baseline, local bone lesions should be followed up.

Records of target lesion: Number, location and description of lesions, maximum diameter of each lesion (except for lymph nodes), minimum diameter of lymph nodes and total diameter of all target lesions.

Records of non-target lesion: All lesions except the target lesion, including the pathological lymph nodes, may be considered non-target lesions and no measurements are required, but the lesion should be recorded at baseline evaluation. The treatment period is recorded as "present", "missing" or in extremely rare situations "clear progress". Target lesions that exist extensively may be recorded together with the target organ (e.g. extensively augmented lymph nodes of the pelvis or large scale liver metastasis).

Imaging evaluation of tumor should be conducted in patients once every four weeks in the first three weeks and then every 8 weeks until progressive disease is identified

If patients discontinuate the study medication by toxicity or other reasons, the investigators should conduct the scheduled tumor assessment, until PD, death, starting new anti-tumor therapy, lost to follow-up, withdrawn of informed consent, or study terminated by sponsor.

- Quality of life assessment: Completion of QLQ-C30 during the window period as indicated in the study process diagram 1.

8.2.2 Safety Evaluation

- Determination and classification of the adverse events according to NCI CTCAE 4.03, and conducting a combined evaluation to assess safety and tolerability via the incidence, severity and outcome of adverse events. Investigators should take appropriate measures in a timely manner to handle the adverse event (AE) or serious adverse event (SAE) and determine the relationship between the study drug and the AE/SAE. If there is a serious safety issue in the study, the sponsor and the investigator will decide whether to terminate the study or not. The sponsor may take the initiative to terminate this study depending on the situation. For the reporting procedure for AE/SAE please refer to Appendix 6.
- The changes of the vital signs (including blood pressure, heart rate, breathing rate, temperature), physical examination and laboratory tests (blood routine, blood biochemistry, coagulation, urine routine, etc.), 12-lead ECG and echocardiography.
- Performance Status Assessment: According to the ECOG PS Assessment (see Appendix 1).

8.3 Follow-up Period Assessment

- The follow-up after EOT should be completed within 30 ±3 days after administration of the last dose of the study drug: vital signs, physical examination, ECOG PS, QLQ-C30, laboratory tests, imaging evaluation of tumor, 12-lead electrocardiogram, collection of AE, SAE, concomitant medication/therapy and unused study drug.

- Survival follow-up should be conducted every 2 months ± 7 days after the EOT visit, and subsequent anti-tumor therapy and study-related SAEs should be collected.

8.4 Exploratory Study Assessment (Optional)

Samples collected for this part of the study will be used to explore the correlation between the anti-tumor activity of fruquintinib and the expression levels of biomarkers in the blood. For example, to investigate whether activation of the HGF / c-MET, FGF / FGFR signaling pathway has an effect on the efficacy of fruquintinib.

For patients who successful pass the screening, blood samples will be collected before the administration of fruquintinib and paclitaxel on C1D1 and C2D15, if they give the consent for biomarker testing (15 mL per blood collection). Patients may voluntarily choose whether or not

to participate in the blood sample collection.

These results will not be used during the study to guide the treatment, but after the end of the study, the test results of the center may be reported upon the investigator's request.

9 Study Drug and Method of Application

9.1 Fruquintinib/Fruquintinib Placebo

9.1.1 Description of Fruquintinib/Fruquintinib Placebo

Fruquintinib/Fruquintinib Placebo capsules are provided by HUTCHMED Limited.

Fruquintinib / Fruquintinib Placebo capsules are manufactured and packaged by a qualified manufacturer retained by HUTCHMED Limited, and HUTCHMED Limited provides technical guidance and quality control. See Table 9.1.1 for details. See **Table 3** for details.

Table 3 Description of Fruquintinib/Fruquintinib Placebo

Name of study drug	Dosage form	Specifications	Method of administration	Expiration dating period
Fruquintinib	Capsule	1mg	Oral administration	4 years
Fruquintinib placebo	Capsule	1mg	Oral administration	4 years

For packaging and sample labeling details, see the investigator clinical trial file.

The packaging of the study drug should have the unique drug number.

Re-supply of the study drug during the study period will be managed by the IWRS system. For details, see the IWRS User Manual.

9.1.2 Storage, Dispensing and Management of Fruquintinib/Fruquintinib Placebo

All fruquintinib/fruquintinib placebo must be stored as instructed in a safe and enclosed place at 10-30℃. The storage temperature must be recorded and saved in corresponding files. The study drug can only be used as specified in this protocol.

The investigator or research nurse will dispense the study drug based on IWRS.

The investigator/pharmacist/supervisor responsible for the safekeeping of the study drug must record the medications that are delivered to the research center, inventory and usage of each patient and return all unused drug to the applicant or handle these drugs as instructed. In addition, when returning the unused study drug to the sponsor, the investigator/pharmacist/supervisor responsible for the safekeeping of the study drug should confirm that the patient returns all unused (with intact packaging) or unfinished drug (opened) and the research center also has no remaining study drug.

The quantity of the fruquintinib/fruquintinib placebo given to each patient should be calculated by dispensed quantity-returned quantity-lost quantity.

9.1.3 Disposal of Fruquintinib/Fruquintinib Placebo

All remaining fruquintinib, fruquintinib placebo, and fruquintinib/fruquintinib placebo collected after the end of the study must be returned to and destroyed by the sponsor. If the study drug is destroyed by the research center, a certificate of disposal must be submitted, including the following documents:

- Identification (drug number or patient treatment assignment) of the destroyed study drug
- Destroyed quantity
- Method of disposal
- Signature of the staff (unit) who is responsible for disposal of the drug and date of disposal.

9.1.4 Administration and Dosage of Fruquintinib/Fruquintinib Placebo

Fruquintinib capsule (HMPL-013) or Fruquintinib placebo 4 mg, QD, oral administration for 3 consecutive weeks followed by a 1-week off period. Every 28 days (4 weeks) is defined as one cycle until progressive disease (PD) or the patient shows intolerable toxicity. The dosing cycle of fruquintinib or its placebo is fixed, without window period, and not affected by medication discontinuation or dose reduction. The safety evaluation must be completed before starting each dosing cycle of fruquintinib or fruquintinib placebo.

Take fruquintinib or fruquintinib placebo with 100-200 mL of drinking water before or after meal. Taking fruquintinib or fruquintinib placebo at the same time during the treatment cycle is recommended.

9.1.5 Rules for Making up for a Missed Dose of Fruquintinib (or Fruquintinib Placebo)

Missed doses must be reported to the investigator, do not increase the daily dose to make up for a missed dose. If vomiting occurs within 30 minutes after taking fruquintinib and the intact capsule of the study drug is visible in the vomits, the daily dose should be made up.

Strenuous exercise during the study period should be avoided; alcohol, tobacco as well as caffeine-containing drinks are not allowed. Avoid ingestion of grapefruit, citrus fruit, pomelo or drinks containing the abovementioned fruit juices (see Appendix 4).

9.2 Study Drug Paclitaxel

9.2.1 Description of Paclitaxel

Paclitaxel used in this project will be provided by the sponsor. In this study, 5 mL: 30mg is used and the specification of the packaging is 1 bottle/package. Store at 25°C and below and keep away from light. Paclitaxel must be diluted as instructed in the instruction for use before IV injection.

9.2.2 Dispensing and Management of Paclitaxel

Paclitaxel should be saved by each research center. The investigator or research nurse will

dispense paclitaxel according to IWRS, and the research center is responsible for its injection.

The investigator/pharmacist/supervisor responsible for the safekeeping of the study drug must record the medications that are delivered to the study center, inventory and usage of each patient and return all unused drug to the applicant or handle these drugs as instructed.

9.2.3 Disposal of Paclitaxel

Diluted paclitaxel and empty bottles must be destroyed by following the SOP for disposal of the research center. All unopened study drug, paclitaxel, and the packaging of used paclitaxel must be returned to the sponsor. If the study drug is destroyed by the research center, a certificate of disposal must be submitted, including the following documents:

- Identification (drug number) of the destroyed study drug
- Destroyed quantity
- Method of disposal
- Signature of the staff (unit) who is responsible for disposal of the drug and date of disposal.

9.2.4 Administration and Dosage of Paclitaxel

The patients of both groups all receive the paclitaxel treatment.

Starting from Day 1 of this study, patients will receive paclitaxel 80 mg/m^2, IV injection, 28 days per treatment cycle and the injection will be given on Day 1 of each week for the first 3 weeks of each treatment cycle (Days 1, 8 and 15), until PD or the presence of intolerable toxicity, or until 6 treatment cycles are completed. The situation that occurs first will be the date of discontinuous administration of paclitaxel. If the patient's disease shows no progression after completion of the 6 treatment cycles, monotherapy of fruquintinib (or fruquintinib placebo) may be given until identification of progressive disease or occurrence of intolerability.

The dose of paclitaxel for patients will be calculated and generated by IWRS based on the baseline height (Ht) and weight (Wt). During the study period, the weight must be measured in each cycle and entered into the IWRS to calculate the dose of paclitaxel.

When giving paclitaxel, the investigator should follow the guide of research center, package insert and past routine procedures for administration of paclitaxel. For more information on administration, preparation, storage and handling of paclitaxel, see the package insert.

9.3 Common Adverse Events/Adverse Reactions of Study Drug

9.3.1 Common Adverse Events of Fruquintinib

The trial results of the monotherapy of fruquintinib have indicated: Common adverse events relating to fruquintinib (≥ 10%) are palmar-plantar erythrodysaesthesia syndrome (mostly shown as hand-foot syndrome), hypertension, proteinuria, dysphonia (shown as hoarseness), blood thyroid stimulating hormone increased, diarrhea, decreased appetite, weight decreased, aspartate aminotransferase or alanine aminotransferase increased, stomatitis, cough, asthenia, fatigue, blood bilirubin increased, occult blood positive, constipation, rash, back pain, and hypothyroidism.

The phase Ib/II trial results of the combined therapy of fruquintinib and paclitaxel have indicated: Dose-limiting toxicity (DLT) is febrile neutropenia and neutropenia. The common adverse events (≥ 10%) include white blood cell count decreased, neutrophil count decreased, dizziness, rash, palmar-plantar erythrodysaesthesia, nausea, platelet count decreased, aspartate aminotransferase or alanine aminotransferase increased, mouth ulceration, asthenia, discomfort, fever, headache, alopecia, epistaxis, diarrhea, anemia, decreased appetite, dysphonia (shown as hoarseness), arthralgia, back pain, and hypertension.

9.3.2 Common Adverse Events of Paclitaxel

Allergic reaction: Paclitaxel may have a fatal allergic reaction with or without pre-medication. Suspension of the treatment is not necessary for mild symptoms such as flushing, rash, skin reactions, dyspnea, hypotension, tachycardia and other minor symptoms. However, the incidence of severe allergic reactions characterized by dyspnea and hypotension and hypotension, angioneurotic edema, and systemic urticaria, account for 2% -4% of patients treated with paclitaxel, which require medical attention and may be mediated by histamine. Discontinue the use of paclitaxel immediately upon identification of these symptoms and actively treat specific symptoms.

Bone marrow depression (mainly neutropenia) is the main dose-limiting toxicity (DLT) of paclitaxel. The median number of neutrophils in the trough was Day 11. Re-administration of this product is permitted only when the neutrophil count is at least 1500/mm^3 and platelets is at least 100,000/mm^3. See **Table 4** for the adverse effects of paclitaxel.

Table 4 Paclitaxel-Related Adverse Reactions (Extracted from the Instructions for Use-Paclitaxel)

Reported adverse drug events (MedDRA terminology)		
System organ classification	**Frequency of occurrence**	**MedDRA terminology**
Infection and invasion	Very common	Infection[1]
	Uncommon	Septic shock
	Unknown	Cryptococcal meningitis, sepsis, white matter disease, opportunistic infections, cytomegalovirus infections, Pneumocystis carinii infection, Mycobacterium tuberculosis complex infection, esophageal candidiasis, cryptosporidium gastroenteritis, pneumonia, simple herpes, urinary tract infection, upper respiratory tract infection, sinusitis and rhinitis
Hematopoietic and lymphatic system abnormalities	Very common	bone marrow failure, hemorrhage, neutropenia, anemia, thrombocytopenia, leukopenia
	Rare	neutropenia with fever
	Unknown	acute myeloid leukemia, myelodysplastic syndrome,

		blood toxicity, reduced platelet count
Immune system abnormalities	Very common	allergic reactions, hot flashes
	Uncommon	respiratory distress, vasoconstriction, whole body urticaria
	Unknown	anaphylactic shock, allergy shock-like reaction[2]
Metabolic and nutritional disorders	Unknown	anorexia
Mental disorders	Unknown	conscious fuzzy state
Nervous system abnormalities	Very common	neurotoxicity[3], peripheral neuropathy, visual evoked potential abnormalities
	Unknown	seizures, epilepsy, encephalopathy, peripheral neuropathy, dizziness, ataxia, high tension, paresthesia, headache, insomnia
Eye disorders	Unknown	optic nerve abnormalities, flash dark spots, flash illusion, floaters
Ear and inner ear disorders	Unknown	deafness, ototoxicity, dizziness, tinnitus
Heart abnormalities	Very common	ECG abnormalities
	Common	bradycardia
	Uncommon	myocardial infarction, cardiomyopathy, ventricular tachycardia, atrioventricular block, tachycardia
	Unknown	ventricular failure, heart failure[4], congestive heart failure, atrial fibrillation, decreased ejection fraction, supraventricular tachycardia, conduction abnormalities, contraction, sinus bradycardia, electrocardiographic repolarization abnormality
Blood vessel abnormalities	Very common	hypotension
	Uncommon	thrombosis, hypertension, thrombophlebitis
	Unknown	shock, phlebitis
Breathing, chest and mediastinal abnormalities	Unknown	respiratory failure, pulmonary embolism, pulmonary fibrosis, interstitial lung disease, radiation pneumonia, dyspnea, pleural effusion, epistaxis, cough
Gastrointestinal abnormalities	Very common	abdominal pain, diarrhea,

		vomiting, nausea
	Unknown	intestinal obstruction, intestinal perforation, mesenteric vascular thrombosis, ischemic colitis, pancreatitis, pseudomembranous colitis, neutropenic enterocolitis, ascites, esophagitis, mucosal inflammation, constipation
Liver and gallbladder abnormalities	Common	elevated aspartate aminotransferase, elevated blood alkaline phosphatase, abnormal liver function
	Uncommon	elevated blood bilirubin
	Unknown	liver necrosis[2], hepatic encephalopathy[2]
Skin and subcutaneous tissue abnormalities	Very common	hair loss
	Common	skin abnormalities, nail abnormalities
	Unknown	Stevens-Johnson syndrome, toxic epidermal necrolysis, erythema multiforme, exfoliative dermatitis, urticaria, skin necrosis, memory resuscitation, skin loss, skin fibrosis, cellulitis, itching, rash, erythema, onycholysis, acne
Skeletal muscle, connective tissue and bone abnormalities	Very common	joint pain, muscle pain
	Uncommon	backache
	Unknown	limb pain
Kidney and urinary tract abnormalities	Unknown	renal failure, renal toxicity
Body abnormalities and administration sites	Common	extravasation, injection site response, local edema, pain, induration, tenderness, skin discoloration, chills, injury
	Unknown	dehydration, fever, edema, chest pain, sweating, fatigue, discomfort
Test	Unknown	elevated serum creatinine

1. Infection, death accounts for 1% of all patients.
2. Reporting of patient death events.
3. Including the CNS toxicity (cases associated with death are rare) of pediatric patients reported by trial sites.
4. The incidence and severity of cardiac dysfunction were higher in patients who had previously been treated with anthracycline before receiving trastuzumab and paclitaxel when compared to patients treated with monotherapy of paclitaxel, but cases of death were rare.

9.4 Dosage Adjustment of the Study Drug

9.4.1 The Principles of Dosage Adjustment of the Study Drug

During the course of this study, if drug-related adverse events occur, the drug may be suspended or the dose may be reduced depending on the level of adverse events. The general rule for dosage adjustment is as follows:

1. Once an adverse event occurs, the investigator is required to provide active symptomatic treatment. The investigator will conduct dosage adjustment as per protocol, and is recommended to discuss the issue with the sponsor if necessary;
2. If no progressive disease occurs, suspension of one of the study drugs does not affect the continued administration of another study drug;
3. If the investigator determines that the adverse event is only associated with one of the study drugs, whether the adjustment for the other study drug is required will be determined by the investigator;
4. Dosage should be adjusted based on the most severe adverse event if several adverse events occur at the same time;
5. The suspension rules for fruquintinib:Patients who return to level 1 or baseline within 14 days after medication suspension (the 7-day off period of the fruquintinib treatment cycle is not included in the 14 days) can continue the treatment; those who cannot return to level 1 or baseline after medication suspension for more than 14 days are considered as intolerant and the study drug associated with the event should be discontinued permanently.
6. The suspension rules for paclitaxel: paclitaxel will be permanently discontinuated if any toxicity is not recovered after 28 days post the last dosing.
7. Any of the following conditions will permanently discontinue the use of the adverse event-associated study drug: NCI CTCAE Grade 4 liver dysfunction; NCI CTCAE Grade 3 or above bleeding; any degree of arterial thrombosis; and any other Grade 4 life-threatening adverse reactions;
8. The dosage should not be increased to the previous level after the dosage reduction;
9. A maximum of 2 dosage adjustments is allowed to be performed for each patient; dosage adjustment is not allowed after two dosage adjustments, but suspension of medication is permitted.

9.4.2 Dosage Adjustment of Fruquintinib

General rules

For adverse events that require reductions to the dose of fruquintinib, following the dosage levels indicated in **Table 5** is recommended.

In the event of suspension of medication or dose reduction, the dosing cycle will not be adjusted. The total number of days of medication administration cannot exceed 21 days, and the number of consecutive days of medication discontinuation cannot exceed 14 days throughout the medication administration period. If the duration of medication suspension ranged from 7 to 14 days in a dosing cycle (first 21 days), it is possible to consider making up no more than five days' doses within the first five days of the medication suspension period (Day 22 – 28 of each

cycle), but no medication can be taken on the last two days of medication suspension period.

Table 5 Pre-Set Dosage Adjustment Levels of Fruquintinib

Number of dose reduction of fruquintinib	Dosage	Number of capsules of fruquintinib/fruquintinib placebo
Dose level - 0 (initial dose)	4mg	Fruquintinib 1 mg capsule, 4 capsules or corresponding 4 placebo capsules
Dose level - 1 (first dose reduction)	3mg	Fruquintinib 1 mg capsule, 3 capsules or corresponding 3 placebo capsules
Dose level - 2 (second dose reduction)	2mg	Fruquintinib 1 mg capsule, 2 capsules or corresponding 2 placebo capsules

For dose reduction for blood and non-blood toxicities of fruquintinib, it is recommended to carry out with reference to **Table 6**. Since hand-foot syndrome, hypertension, proteinuria, platelet count reduction, bleeding, abnormal liver function do not belong in the scope of this guide, please refer to the corresponding dosage adjustment rules. In the event of the presence of any degree of arterial thrombosis, administration of the drug must be permanently terminated.

Table 6 Guide to Dosage Adjustment of Fruquintinib[a]

Level of adverse events (NCI CTCAE 4.03)	Administration	Dosage Adjustment[b]
level 1	Administration of medications as prescribed	No change
level 2	Administration of medications as prescribed	No change is recommended
level 3[c]	temporary suspension of administration until the level of toxicity restored to ≤ level 1 or baseline is recommended	Reducing the dose 1 level below is recommended
level 4[d]	Permanent termination of the treatment	Permanent termination of the treatment

a: Except for hand-foot syndrome, hypertension, proteinuria, platelet count reduction, bleeding and abnormal liver function; in the event of the presence of any degree of arterial thrombosis, permanent termination of the treatment is required

b: Asymptomatic laboratory test abnormal should not lead to drug interruption, adjustment, or discontinuation, unless it is evaluated as abnormal clinical significant or life-threatening by investigators

c: Including level 3 diarrhea and level 3 stomatitis that cannot be ameliorated after medications, except for level 3 prolonged menstrual cycle

d: Only refers to life-threatening AE

Skin and Mucosal toxicity

a. Oral Mucositis

If the patient shows level 1-2 stomatitis (including oral ulcers, mucositis, gingivitis, throat

discomfort, angular cheilitis, etc.), topical use of antibiotics (including anti-anaerobic antibiotics) and antifungal agents, mucosal protective agents and local anesthetic use, oral antacids, and the use of watermelon frost sprays and oral administration of silver yellow detox tablets are allowed and discontinuation of the treatment is not required. Avoid the usage of iodine and prolonged usage of hydrogen peroxide; a diet of soft and non-irritating food is recommended, as well as avoiding spicy, sour or irritating food. If the patient cannot swallow food or fluid, parenteral or nutritional support may be required.

If stomatitis after treatment cannot be restored and affects eating and weight, or if level 3 or above stomatitis is present, temporary suspension of the medication is required. If the patient's condition returns to level 1 or baseline within 14 days, you may consider continuing with the original dose or with dose reduction treatment.

b. Treatment of Hand-foot Syndrome

See **Table 7** for the dosage adjustment regimen for hand-foot syndrome.

If the patient shows < level 2 hand-foot syndrome, symptomatic treatment may be provided. Avoid rubbing or pressing hands and feet and avoid contact with objects with high temperature. Keep hands and feet moist or use urea cream when needed. This may help with alleviating the symptoms and curing the lesions. Patients with severe symptoms (especially those with pain) can use burn pain ointment or oral diclofenac sodium enteric-coated tablets (Voltaren) or other drugs to help alleviate the symptoms.

Patients who have level 3 or above hand-foot syndrome (severe skin reactions, including skin exfoliation, blisters, bleeding and edema with pain which can affect daily life), temporary suspension of medication administration is recommended. If the toxicity has restored to level 1 or below within 14 days, both resuming medication administration and reducing the dosage are permitted.

Table 7 Dosage Adjustment Regimen for Hand-Foot Syndrome

Classification criteria of AEs	Dosage adjustment regimen	Comments on handling
Level 1: numbness, dysesthesia, feeling dull, erythema, painless edema, molting, thickening of the skin or discomfort without affecting the normal activity; no pain	maintain the current dose level	Seek supporting treatment to relieve symptoms; use "urea cream" or other treatments to treat the symptoms;
Level 2: painful erythema, accompanied by hand or foot swelling and/or discomfort, affecting normal activity	temporary suspension of medication administration if necessary, maintain the current dose level if recovered to Level 1 within 14 days	Use of the burn pain ointment manufactured by Wuhan Jianmin or other treatments to treat the symptoms is recommended.
Level 3: wet peeling, ulcers, blisters or severe hand-foot pain, or serious discomfort leading to the inability of the patient to work or to complete daily activities	temporary suspension of medication administration is recommended, reduce the dose one level below if recovered to Level 1 within 14 days	Permanent termination of the treatment may be considered if intolerance still present after two dosage reductions

Hypertension

For the treatment in response to one of the adverse events, hypertension (or worsened high

blood pressure, or elevated blood pressure), it is recommended for the investigator to perform dosage adjustment of fruquintinib in accordance with **Table 8**. While conducting close monitoring of blood pressure, it is strongly recommended to use appropriate treatment measures to effectively control blood pressure. The choice of antihypertensive drugs should be based on the investigator's judgment by reference to the guidelines for treatment of hypertension. Please consult a physician of internal medicine if necessary. All drug treatment should be recorded.

Table 8 Dosage Adjustment Regimen for Hypertension

NCI CTCAE Grades and Definitions	Dosage adjustment regimen	Comments on handling
Grade 1: early phase of high blood pressure (systolic blood pressure 120-139 mmHg or diastolic blood pressure 80-89 mmHg)	maintain the current dose level	None
Grade 2: systolic blood pressure 140-159 mmHg or diastolic blood pressure 90-99 mmHg, or diastolic pressure increased >20 mmHg	maintain the current dose level	Treatment goal: Control the blood pressure to 140/90 mmHg and below. Start the treatment of antihypertensive drugs/or increase the dose of current antihypertensive drugs/or use of additional antihypertensive drugs, use of antihypertensive drugs or dosage adjustment should be in accordance with the guidelines for treatment of hypertension. Please consult a physician of internal medicine if necessary.
Grade 3: systolic blood pressure ⩾160 mmHg or diastolic blood pressure ⩾100 mmHg; or use of more than one antihypertensive drug	if the blood pressure remains higher than 160/100 mmHg for 7 consecutive days or longer after antihypertensive treatment or dosage adjustment of antihypertensive drugs, temporarily discontinue the use of fruquintinib, if recovered to level 1 or baseline within 14 days, reduce the dose to one level below.	Treatment goal: Control the blood pressure to 140/90 mmHg and below. Start the treatment of antihypertensive drugs/or increase the dose of current antihypertensive drugs/or use of additional antihypertensive drugs, use of antihypertensive drugs or dosage adjustment should be in accordance with the guidelines for treatment of hypertension. Please consult a physician of internal medicine if necessary.
Grade 4: Life-threatening (such as malignant hypertension, temporary or permanent neurological deficits and high blood pressure crisis)	Permanent termination of the treatment	Intervention of emergency medicine

Hepatoxicity

Patients with impaired liver function may have a higher risk of developing toxicity. See **Table**

9 for dosage adjustment of fruquintinib.

Investigator should report the patients with ALT or AST ⩾ 3 × ULN and TBiL ⩾ 2 × ULN in accordance with the adverse event of special interest (AESI) as specified in the protocol. It is recommended that four routine serum monitoring parameters (ALT, AST, ALP, and TBiL) to be reviewed over the next 48 to 72 hours to identify abnormalities and observe their development trends. Inquiry about the corresponding symptoms while performing closely monitoring according to Appendix 5 (clinical evaluation of liver injury) is recommended.

For patients who show the early symptoms of liver damage (such as loss of appetite, nausea, vomiting, right upper abdominal discomfort, fatigue, etc.) before discovery of blood biochemical abnormalities, blood biochemical test is recommended to be conducted immediately, and the frequency of monitoring should be increased.

Table 9 Dosage Adjustment for Abnormal Liver Function

AE Level	Dosage adjustment regimen	Comments on handling
Level 1	maintain the current dose level	Conduct visits as scheduled
Level 2	Temporary suspension of medication administration may be considered. Resume original dose if recovered to Level 1 within 14 days	Actively provide liver-protection treatment and perform close monitoring
Level 3	Temporary suspension of medication administration is recommended. It is possible to consider reducing the dose one level below if recovered to Level 1 or baseline within 14 days	Actively provide liver-protection treatment and perform close monitoring until the toxicity is recovered to Level 1 or baseline or a reason is identified
Level 4	Permanent termination of the treatment	Actively provide liver-protection treatment and perform close monitoring until the toxicity is recovered to Grade 1 or baseline or a reason is identified)

a: If Grade 2 -3 abnormal liver function meets the potential Hy's Law, the treatment must also be terminated immediately. Potential Hy's Law: ALT/AST ⩾ 3 × ULN coupled with TBil ⩾ 2 × ULN, and ALP < 2 × ULN, and cannot find other reasons that can explain the elevations in liver enzymes and bilirubin (such as bile duct obstruction indicated by elevated ALP, viral hepatitis or other suspected concomitant medications, etc.). See "Clinical Evaluation of Liver Function Impairment" provided in Appendix 5.

Proteinuria

It is recommended to perform the 24-hour urine protein quantification within 1 week if the urine test if a patient shows urine protein 2+ or above. Dosage adjustment can be performed on the patient with proteinuria according to **Table 10**.

Table 10 Dosage Adjustment Regimen for Proteinuria

NCI CTCAE Grades	Dosage adjustment regimen	Comments on handling
Grade 1: urine test shows urine protein +; or the 24-hour urine	maintain the current dose level	Conduct visits as scheduled

NCI CTCAE Grades	Dosage adjustment regimen	Comments on handling
protein quantification < 1.0 g		
Grade 2: the 24-hour urine protein quantification is 1.0 g to 2.0 g (2.0 g is excluded)	maintain the current dose level	Active treatment with urine routine monitoring (repeat the test every week is recommended) and consult with a physician of renal medicine if necessary.
Grade 2: the 24-hour urine protein quantification is 2.0 g to 3.5g (3.5 g is excluded)	Temporary suspension of medication administration is recommended. The dose can be reduced by one level below if recovered to Grade 1 or baseline within 14 days	Active treatment and consult with a physician of renal medicine if necessary.
Grade 3: the 24-hour urine protein quantification ⩾ 3.5g	Temporary suspension of medication administration and reduce the dose one level below if recovered to Level 1 or baseline within 14 days	Active treatment and consult with a physician of renal medicine if necessary. Permanent termination of the treatment is required upon third detection of proteinuria

Gastrointestinal Toxicities

Supportive care should be given promptly for dehydration caused by severe or persistent diarrhea, nausea, vomiting or anorexia.

a. Nausea, vomiting

- If severe nausea or vomiting occurs during the study period, symptomatic treatment should be provided. If vomiting occurs after taking the medication, see the section of making up for the missed doses for details regarding how to make up for the missed dose(s).

b. Diarrhea

- If the patient has Grade 1-2 diarrhea, close monitoring or use of the drugs that are capable of improving intestinal function is permitted.
- In the event of occurrence of Grade 3 or above diarrhea, symptomatic supportive care should be provided, and use of loperamide hydrochloride capsules (imudium) and/or other drugs to improve intestinal function is permitted. Generally, the symptom will be relieved within 3 days and use of the study drug can be continued. If the symptom lasts more than 3 days without improvement, temporary suspension of administration of the study drug is required.
- In the event of occurrence of Grade 3 or above diarrhea, administration of the study drug should be temporarily suspended if the investigator determines continued use of the study drug is not under control.
- If Grade 4 diarrhea that may cause serious hemodynamic collapse occurs, temporary suspension of the study drug is required and should be dealt with actively.

Reduced Platelet Count

If the patient has a reduced platelet count (<75 × 10^9/L), suspending the medication and performing routine blood monitoring in accordance with clinical practice is advised. If the patient condition is not recovered to Grade 1 or baseline within 7 days of discontinuation, injection of recombinant human interleukin-11 for platelet therapy is recommended. Transfusion of apheresis platelets may be conducted in the presence of Grade 4 thrombocytopenia. See **Table 11** for dosage adjustment regimen for platelet reduction.

Table 11 Dosage Adjustment Regimen for Platelet Reduction

Classification of AEs	Dosage adjustment regimen	Comments on handling
Level 1: platelet count is 100 to 75×10^9/L	maintain the current dose level	Conduct visits as scheduled
Level 2: platelet count is 75 to 50×10^9/L	Temporary suspension of medication is recommended and maintain the current dose level if recovered to Level 1 or baseline within 7 days	Re-test blood routine in accordance with clinical practice and use of interleukin-11 for active treatment is permitted if necessary. Close follow-up is required subsequently.
	Temporary suspension of medication and reduce the dose one level below if recovered to Level 1 or baseline within 7-14 days	Re-test blood routine in accordance with clinical practice and use of interleukin-11 for active treatment is permitted if necessary. Close follow-up is required subsequently.
Level 3: platelet count is 50 to 25×10^9/L	Temporary suspension of medication administration and reduce the dose one level below if recovered to Level 1 or baseline within 14 days	Re-test blood routine in accordance with clinical practice and use of interleukin-11 for active treatment is permitted if necessary. Close follow-up is required subsequently.
Level 4: platelet count is <25×10^9/L	Permanent termination of the treatment	Re-test blood routine every until the results return to level 2 or below, active transfusion of apheresis platelets and active treatment of interleukin-11 should be provided.

Bleeding

It is recommended to carry out treatment for bleeding of any sites with reference to **Table 12**, and perform corresponding evaluation and monitoring with reference to Appendix 6.

The investigator is required to report bleeding with CTC AE Grade ≥ 3 as AE of special interest (AESI) in accordance with protocol requirements.

Table 12 Dosage Adjustment Regimen for Bleeding of Any Sites

Classification of AEs	Dosage adjustment regimen	Comments on handling
Level 1	maintain the current dose level	Conduct visits as scheduled

Classification of AEs	Dosage adjustment regimen	Comments on handling
Level 2	Temporary suspension of medication administration is recommended, reduce the dose one level below if recovered to Level 1 or below within 14 days	Active treatment
≥Level 3	Permanent termination of the treatment	Intervention of emergency medicine

Decreased Thyroid Function and Elevated TSH:

If the patient shows clinically significant decreased thyroid function (accompanied by or not accompanied by clinical symptoms) or persistent elevated TSH (accompanied by or not accompanied by T4 reduction), receiving hormone replacement therapy is advised.

Other Toxicities

Treatment for other adverse events should all be patiented to the principles for handling toxic reactions mentioned above and take patient's benefits into full consideration before giving adequate treatment.

9.4.3 Dosage Adjustment of Paclitaxel

Dosage adjustment of paclitaxel shall be determined by the investigator based on the package insert and different tolerance of individuals. See the package insert of paclitaxel injection for more information on temporary suspension of medication and dosage adjustment.

General Rules

The procedures for dosage adjustment and dose reduction of paclitaxel can be found in **Table 13**.

Table 13 Pre-Set Level for Dosage Adjustment of Paclitaxel

Number of dose reduction of paclitaxel	Dosage
Dose level - 0 (current dose)	80mg/m^2, once every week in the first 3 weeks of each cycle
Dose level - 1 (first dose reduction)	60mg/m^2, once every week in the first 3 weeks of each cycle
Dose level - 2 (second dose reduction)	40mg/m^2, once every week in the first 3 weeks of each cycle

If NCI-CTCAE (version 4.03) Grade 4 hematotoxicity or Grade 3 non-hematotoxicity (except for hair loss) is observed, reduce the dose of paclitaxel to one level below. If reduction of the dose to below 40 mg/m^2 is required or occurrence of a paclitaxel-related life-threatening event, permanent termination of administration of paclitaxel is required.

Allergic Reactions of Paclitaxel

Pre-medication of anti-allergy treatment as a routine should be conducted before each administration of paclitaxel. However, paclitaxel may have a fatal allergic reaction with or without pre-medication. Suspension of the treatment is not necessary for mild symptoms. If a

Grade 3-4 allergic reaction occurs, immediately discontinue the use of paclitaxel and provide active symptomatic treatment. Use of paclitaxel in patients with known allergic reaction to paclitaxel is prohibited.

Bone Marrow Suppression

Before each administration of paclitaxel, following hematological criteria (within 72 hours per-dosing) must be met :

- ANC：≥1500/mm^3
- PLT：≥100,000/mm^3

To ensure recovery of bone marrow suspension, paclitaxel may be discontinued for up to 14 days, which means a maximum of 28 days will be acceptable (including 2-week break period). If the patient has not recovered within 28 days, discontinue the treatment of paclitaxel. During the study treatment, preventive use of G-CSF is not suggested, but is highly recommended for patients who have neutropenia of CTCAE Grade 3 or 4 lasting for >5 days or febrile neutropenia.

Hepatoxicity

Available evidence has shown increased toxicity of paclitaxel in patients with a TBiL>2×ULN. These patients need to consider dosage adjustment when being treated with paclitaxel.

Patients who show the early symptoms of liver damage (such as loss of appetite, nausea, vomiting, right upper abdominal discomfort, fatigue, etc.) before discovery of blood biochemical abnormalities should be patiented to blood biochemical test immediately. If the above abnormalities are found, increasing the frequency of monitoring is required.

Other Toxicities

For other Grade ¾ toxicities associated with paclitaxel, temporarily discontinuing the use of paclitaxel is required. If the toxicity is relieved (to ≤CTCAE Grade 1 or baseline), administration of 80% of the current dose of paclitaxel (i.e. reduced the dose to 60 mg/m^2) in the next cycle is permitted. If paclitaxel has to be discontinued for > 2 weeks due to amelioration of toxicity or if CTCAE Grade 3 or 4 toxicity occurs again after the second dose reduction, the patient should not start using paclitaxel again.

9.5 Overdose

Fruquintinib

Overdose of fruquintinib is defined as taking fruquintinib that exceeds the dose specified in the trial protocol in one calendar day. Upon identification of overdose, the investigator should close monitors the change of patient's status and provide symptomatic and supporting treatment when necessary.

If the dose of the study drug exceeds the dose specified in the trial protocol and no corresponding symptom(s) or sign(s) is present, the overdose is not considered as an AE. Overdose accompanied by SAE is considered as a SAE. If an overdose is accompanied by AE and said AE is not a SAE, the overdose is considered as an AE.

Paclitaxel

Overdose of paclitaxel is defined as receiving paclitaxel that exceeds the dose specified in the trial protocol in one calendar day.

No drug used for treating paclitaxel overdose is available at present. The major anticipated complications of overdose include bone marrow suppression, peripheral neurotoxicity and mucositis.

9.6 Drug Interactions

9.6.1 Drugs that Interact with Fruquintinib

Preclinical studies have demonstrated that fruquintinib is metabolized by cytochrome P450 3A4 in hepatocytes, strong inhibitors of CYP3A4 enzymes such as phenytoin, phenobarbital, rifampicin and other drugs (not limited to the aforementioned drugs) and strong inhibitors of CYP3A4 enzymes such as ketoconazole oxazole, itraconazole, fluconazole, indinavir and erythromycin (not limited to the aforementioned drugs) may significantly affect the metabolism of fruquintinib in the body. In vitro studies showed that fruquintinib had an inhibitory effect on P-glycoprotein (P-gp) and breast cancer resistance protein (BCRP). For more details, see the Fruquintinib Investigator's Brochure v10.0, which provides comprehensive information on the study drug. Investigators need to exclude patients known to use CYP3A4 enzyme inducers, CYP3A4 enzyme inhibitors, P-gp substrates and BCRP substrates. If these drugs are used in the study, use with caution and conduct close monitoring on the adverse events. For details about materials, see Appendix 4.

9.6.2 Drugs that Interact with Paclitaxel

According to the package insert of paclitaxel, the possible drug interactions caused by concomitant use of paclitaxel with other drugs have not yet been studied.

9.6.3 Interactions between Fruquintinib and Paclitaxel

The pharmacokinetic data of a Ib/II clinical trial that used the combination of fruquintinib and paclitaxel in patients with advanced gastric cancer (2014-013-00CH3) conducted by Hutchison Whampoa Limited has shown that the combination therapy of fruquintinib and paclitaxel has no significant effect on their pharmacokinetic parameters.

9.7 Prohibited Combination Therapy

At enrollment and during the course of the study, the patients are not allowed to receive other anti-tumor therapies, including traditional Chinese medicine treatment with anti-tumor indications, except for the palliative radiotherapy used to control the pain of bone metastases. Other treatments of investigational drugs must be completed four weeks before randomization of the study, and the Chinese medicine treatment with anti-tumor indications should be completed 1 week before randomization.

9.8 Other Combination Therapies

Investigators are required to follow the following guidelines and carefully use the concomitant drugs to maximize the safety of the patients.

Best supportive care (BSC) was allowed during the study period. Other drugs that may induce or aggravate clinical symptoms cannot be used concomitantly and must be monitored carefully when used.

Preclinical studies have demonstrated that fruquintinib is metabolized by cytochrome P450 3A4 in hepatocytes, strong inhibitors of CYP3A4 enzymes such as phenytoin, phenobarbital, rifampicin and other drugs (not limited to the aforementioned drugs) and strong inhibitors of CYP3A4 enzymes such as ketoconazole oxazole, itraconazole, fluconazole, indinavir, erythromycin, etc. (not limited to the aforementioned drugs) may significantly affect the metabolism of fruquintinib in the body. The investigators are expected to carefully enroll the participants who are known to receive the combined CYP3A4 enzyme inducer and the CYP3A4 enzyme inhibitor, as described in Appendix 4. If these drugs are concomitantly used in the study, use with caution and perform close monitoring on drug exposure and adverse reactions.

Patients are allowed to take low-dose oral anticoagulants (such as warfarin and aspirin) but monitoring the relevant coagulation parameters such as INR, aPTT and platelet count is required, and the use of fruquintinib should be discontinued immediately once signs of bleeding occur. Low molecular weight heparin may be used depending on the treatment needs and patients with bone metastases are allowed to receive bisphosphonates and patients in pain are allowed to receive pain medication.

Patients should avoid taking drugs that cause prolonged QT intervals, such as sotalol, cisapride and clozapine, etc.

The investigator decides whether to use other medications concomitantly to protect the interests and safety of the patients. For restricted use and prohibited drugs, see Appendix 4.

9.9 Treatment Compliance

9.9.1 Fruquintinib Treatment Compliance

The patients are asked to return all the unused fruquintinib or fruquintinib placebo and their vials to the study center at the scheduled visit to evaluate compliance.

The investigator should promptly and accurately record the quantity, date, and actual dose of the study drug distributed to and returned by each patient. Medication compliance will be determined based on the number of drugs distributed to, returned by, and lost by the patient at the end of each treatment cycle and at the time of withdrawal. The patient's report on their own missed/excessive medication/drug loss will also be included for comprehensive judgments.

9.9.2 Paclitaxel Treatment Compliance

Investigators should promptly and accurately record the actual dosage of paclitaxel use in each patient and the date of administration.

Since paclitaxel infusion is completed at the study center, the judgment of its treatment compliance will be based on the actual dose used by the patient at the end of each treatment cycle and at the time of withdrawal, as well as the percentage of usage.

10 Safety

10.1 Parameters and Definition of Safety

10.1.1 Adverse Events

Adverse events (AEs), according to the definition of ICH, refer to any adverse medical events that occur in the patient after administration of the study drug, but are not necessarily causally related to treatment. Therefore, an adverse event may be any of the following conditions:

- Any adverse and unwanted signs (including unusual laboratory findings), symptoms or diseases that are linked to the use of the study drug from the perspective of time, but are not necessarily causally related to them.
- Any new disease or deterioration of the disease (including the deterioration of the nature, frequency or severity of a disease)
- Recurrence of the medical conditions (e.g. headache) that are normal or intermittent at baseline
- Deterioration of any laboratory test result or other clinical test (e.g. electrocardiogram, X-ray) accompanied by clinical symptoms or signs.

10.1.2 Severe Adverse Events (SAEs)

SAEs refer to the adverse events that meet at least one of the following criteria:

- **Leading to Death**
- **Life-threatening**: The investigators believe that patients are directly at risk of death when experiencing adverse events, rather than assuming that if the event becomes more serious, it may lead to death;
- **Leading to hospitalization or prolonged hospitalization**: Adverse events lead to hospitalization, or adverse events occur during the hospital stay and extend the length of hospitalization;
- **Leading to permanent or significant disability or dysfunction**: Adverse events result in substantial damage (loss of function does not include the events that are relatively of less medical importance, such as headache, nausea, vomiting, diarrhea, influenza, and accidental trauma such as ankle sprain, etc.) to the ability of the patient in everyday life;
- **Congenital malformations or birth defects**: There is a congenital malformation or birth defects in a live (or abortion) newborn of a drug-exposed female or the female partner of a drug-exposed male patient;
- **Important medical events**: Important medical events may not immediately be life-threatening, cause death or hospitalization, but may jeopardize the patient or medication, or surgical treatment may be required in order to prevent the occurrence of any of the above results (death, life-threatening, hospitalization, extended hospital stay, permanent or significant disability or loss of function, congenital malformations).

10.1.3 Protocol-Specified Adverse Event of Special Interest (AESI)

Though it is rather rare, the sponsor may conduct a continuous and close monitoring on possible damage to liver function and bleeding cause by the drug (Appendixes 5, 6 and 7).

Following events are AESI specified in the protocol:

- Liver function impairment: The serum transaminase (ALT and/or AST) ≥3×ULN combined with TBiL≥2×ULN are found in the blood sample from a single blood collection.
- Bleeding event: CTCAE Grade ≥ 3.

The above AESI must be reported to the sponsor within 24 hours of occurrence.

10.2 Collection and Evaluation of Safety Parameters

10.2.1 Definition of the Reporting Period of Adverse Events

All SAEs should be reported from signing the ICF to 30 days after the last dose of the study drug or intiating the new anti-tumor therapy (whichever occurs first).

All AEs should be recorded from the first dose of the study drug to 30 days after the last dose of the study drug or intiating the new anti-tumor therapy (whichever occurs first), during which time, AESIs should also be reported.

For the SAEs occur after 30 days after the last dose or intiating the new anti-tumor therapy (whichever occurs first), only SAEs related to study drug confirmed by the investigator will be reported.

Please refer to Section 10.3.1 for the reporting period of death and pregnancy events.

10.2.2 Evaluation of the Severity of Adverse Events

The severity of all AEs will be classified into 5 grades (Grade 1-5) based on the Common Terminology Criteria Adverse Events (CTCAE) v4.03. The AEs that are not listed in NCI CTCAE should be determined by using **Table 14**.

Table 14 Criteria for Determination of the Severity of Adverse Events

Classification of CTCAE	Definitions
1	mild: asymptomatic or mild symptoms; only clinical or diagnosed finding is available; no treatment required;
2	moderate: minimal, local or noninvasive treatment indications; age-related instrumental activities of daily living restricted activity [a];
3	serious or important medical significance, but not immediately life-threatening; hospitalization or prolonged hospitalization indications; disability; restricted self-care daily life activities [b];
4	life-threatening, emergency treatment required;
5	Death

[a] instrumental activities of daily living refer to cooking, buying groceries or clothes, using telephone and financial management, etc.

[b] self-care activities of daily living refer to the bathing, dressing and undressing, eating, using toilet, taking drugs, rather than being bedridden.

Pay attention to the difference between severity and seriousness of adverse events. The severity refers to the intensity of adverse events (e.g. mild, moderate or serious headache), but the event itself may have only relatively less clinical significance (e.g. serious headache) and cannot be determined as a SAE, unless it meets the criteria for SAEs. Therefore, when reporting AEs and SAEs, the severity and seriousness should be evaluated independently.

10.2.3 Relationship between Adverse Event and Drug

The causality relationship between AEs and study drug will be evaluated by the investigator according to the following criteria:

The investigator will be required to answer the following question: Is there any reasonable possibility that the event is caused by the study drug?

The investigator can refer to the following contents for judgment:

- Absence of reasonable possible correlationship

  A "No" indicates that there is no reasonable possibility between the AE and the study drug. Assessment factors include, but are not limited to: no reasonable temporal association; the study drug is unlikely to cause the event based on the mechanism of action; there are alternative causes that are more likely to cause the event.

- Presence of reasonable possible correlationship

  A "Yes" indicates that there is a reasonable possibility between the AE and the study drug. Assessment factors include, but are not limited to: a reasonable temporal association; a positive rechallenge (resolved or recovered after dose interruption, and the event reappears or is aggravated in severity after dose resumption); the event is a known reaction of the study drug; the event is improved or recovered after dose discontinuation or reduction; lack of alternative causes for the event.patient

10.3 Documentation and Evaluation of Safety Parameters

10.3.1 Records of Adverse Events

The investigator is responsible for collecting and recording all AEs during the reporting period of adverse events. When recording AEs, the investigator should use correct medical terminology and avoid colloquial expression and abbreviations. The start date, highest severity grade based on NCI CTCAE and end date of the AE, the correlation between the study drug and the AE as well as the effect of the AE on the study drug, accompanied treatment and recovery status must be recorded.

Diagnosis vs Symptoms and Signs

If a diagnosis is provided, the diagnosis should be recorded on the electronic case report form (eCRF), not the symptoms and signs (e.g. recording liver function impairment, instead of recording elevated transaminase and elevated bilirubin level separately; recording acute pancreatitis, instead of recording abdominal pain, bloating, vomiting and elevated amylase separately). However, if the symptoms and signs cannot be classified as a single diagnosis at the time of reporting, each single event should be recorded as an AE on the eCRF. If a diagnosis is confirmed afterwards, said diagnosis should be recorded on the CRF/eCRF.

Subsequent Adverse Events of other Events

In general, the original event of the subsequent severe adverse events (SAEs) (e.g. induced by other events or clinical complications) of other events should be recorded, unless the severity of the subsequent event is higher or the subsequent event is also a SAE.

However, if the time of occurrence of the subsequent event is different from the time of occurrence of the original event and said subsequent event has clinical significance, the event should be recorded as an independent SAE. Alternatively, if the relationship between the subsequent event and the original event is unclear, the original event and subsequent event should be recorded separately.

Continuous/Intermittent or Single Adverse Event (Frequency of Adverse Event)

Continuous AE refers to adverse events that are present during the entire process without remission. For example, an upper respiratory infection that lasts for 5 consecutive days. This type of AEs should only be recorded as one AE on the CRF. When evaluating the severity, the highest level of severity of the AE during the entire process should be recorded.

Intermittent AE refers to changes or remission of the symptoms, signs or lab test indicators from time to time during the entire process without the occurrence of an AE with clinical significance, for example, continuous nausea and vomiting over several days and relative remission is observed during the period; patients with hypertension show intermittent remission at multiple blood pressure measurements but the progression of hypertension is relatively continuous. This type of AEs should only be recorded as one AE on the CRF/eCRF. When evaluating the severity, the highest level of severity of the AE during the entire process should be recorded.

Single AE refers to AEs that logically occur as a single event or those AEs that occur only once during the study period, for example, the patient has a falling incident during the medication period or the patient vomits once during the study period. This type of AEs should only be recorded as one AE on the CRF/eCRF.

It should be noted that if the abovementioned AE shows a clinically significant recovery but occurs again later and the second event is not a continuation of the first event, the occurrence of the two events should be recorded separately on the CRF/eCFR.

Laboratory Test Abnormality or Vital Sign Abnormality

Not all lab test abnormalities/vital sign abnormalities should be recorded as AEs. The investigator should be responsible for examination of all abnormal lab test results and abnormal vital signs and determine whether to record such abnormalities as AEs after conducting medical judgment. If the baseline is normal, any of the abovementioned abnormalities should have clinical significance, for example, such abnormality should be recorded as an AE if one or more of the following criteria are met:

- Accompanied by clinical symptoms
- Resulting in change to the study drug (e.g. dosage adjustment, suspension or permanent termination)
- Medical intervention or change of concomitant therapy is required (e.g. concomitant medications, additional treatment, and suspension, termination or other changes of treatment)

- The investigator determines that such abnormality has clinical significance

If the clinically significant lab-test abnormality or vital-sign abnormality is the indication of a certain disease or syndrome (e.g. elevated ALT/AST and blood bilirubin levels caused by liver function impairment), record only the diagnosis (that is, liver function impairment). On the contrary, the lab-test abnormality or vital-sign abnormality should be recorded in the AE table of the CRF/eCRF and the test value is higher or lower than the normal range should be specified (e.g. "elevated blood potassium" should be recorded instead of "abnormal blood potassium"). If the lab-test abnormality or vital-sign abnormality has a corresponding standard clinical terminology, the clinical terminology should be recorded on the CRF/eCRF (e.g. blood potassium elevated to 7.0mmol/L should be recorded as "hyperkalemia").

Existing Medical Status

For patient's existing symptoms/signs that are present during the screening period, only those showing increased severity, frequency or nature (except for worsening of the disease under investigation) after entering the trial should be recorded as AEs. Changes of the existing conditions should be represented in the record, for example, "increased frequency of headache".

Hospitalization/Prolonged Hospitalization

Any AE that leads to hospitalization or prolonged hospitalization should be recorded as a SAE, except for the following conditions:

- Hospitalization or prolonged hospitalization (e.g. for medication and efficacy evaluation, etc.) required by the protocol
- Planned hospitalization for treating existing disease before enrollment, for example, having the elective surgery that is planned before participating in the study as scheduled.
- Emergency care and treatment less than 48 consecutive hours.

However, if deterioration of a disease (such as having a surgery or treatment earlier than scheduled) is detected during the study period, deterioration of the disease will be considered as an AE.

Surgery

If the disease to be treated by the surgery is clarified, the disease, not the surgery, should be considered as a SAE (e.g. the patient undergoes inguinal hernia repair, the "inguinal hernia" should be recorded, rather than "inguinal hernia repair"); but if the reason of surgery is not clear, the name of the surgery can be recorded as a SAE (for example, the patient undergoes peritoneal exploration and "abdominal exploration" can be recorded as a SAE).

Overdose

During this study period, investigational drug overdose is defined that any patient who receives a higher dose than the protocol scheduled start dose.

Such overdoses should be notified to sponsor’s study representatives no later than 24 hours of when investigators becomes aware of it. The patient with overdose is intended to be monitored closely, managed with appropriate supportive care and with dose interruption as needed.

- An overdose with associated AEs is recorded as the AE diagnosis/symptoms on the relevant

AE/SAE modules in the CRF and overdose details on the study drug CRF module.

- An overdose with no associated AEs is only reported overdose details on the study drug CRF module.

If an overdose with associated SAEs, the SAE report form should be recorded to sponsor no later than 24 hours, except for the above procedure. Drug overdose itself does not satisfy AE or SAE criteria.

Pregnancy

If a female patient gets pregnant (calculated as the date of the beginning of the last menstrual period) during the study period (from the time of initiating the study drug to 30 days after the last dose), discontinue medication and notify the investigator immediately. The investigator should report such event to the sponsor within 24 hours by filling the pregnancy event form after receiving the notification or result of pregnancy. The investigator should discuss the risk and possible effects of pregnancy on fetus with the patient. Monitoring performed on the patient should continue until the end of pregnancy.

If the female partner of a male patient gets pregnant (calculated as the date of the beginning of the last menstrual period) from the male patient received the first dose of the study drug to 90 days after the last dose, the investigator should also report this event to the sponsor within 24 hours by filling the pregnancy event form after receiving the notification, and discuss the risk and possible effects of pregnancy on fetus with the patient. The monitoring of the female partner will be continued until the end of the pregnancy.

Pregnancy itself is not considered as an AE, but spontaneous abortion should be recorded and reported as a SAE. Induced abortion without abnormal performance may be reported as the result of pregnancy. Any congenital abnormality/birth defect of the newborns of the female patients or female partners of the male patients who have taken the study drug should be recorded and reported as an SAE.

Progressive Disease

If the occurred event is clearly consistent with the anticipated progression model for a tumor, and does not lead to death, the event should not be considered as an AE. Hospitalization resulted from such progressive disease also should not be considered as an SAE. If the symptom cannot be clarified to be induced solely by the tumor or is not consistent with the progression model of the tumor, relevant clinical symptoms can be recorded as AEs. The event will be reported referring to the following “death” section if it leads to death.

Death

All deaths including deaths occurring from the initiating the study drug to 30 days after the last dosing, regardless of the causality with the study drug or caused by progressive disease or initiating the new anti-tumor therapy, will be reported timely to the sponsor as SAEs within 24 hours. Death which occur after 30 days post the last dosing need to be reported as an SAE if they are related to the study drug at the discretion of investigators.

When recording a death event, if an AE that causes the death is present, the AE should be recorded as a single medical concept; “death with unclear cause” should be recorded and an SAE report should be filed promptly before conducting further investigation on the cause of

death if the cause of death is unclear and update the record after the cause of death is confirmed.Fast Report on Serious Adverse Events (SAEs)

All SAEs, regardless of their relevance to the study drug, need to be handled with appropriate medical treatment until the investigator determines the patient has recovered from the event or stabilized.

The investigator should inform the sponsor immediately after being informed of the SAE of patient that occurs between the date of signing the Informed Consent Form and 30 days after the last administration of the drug, no matter whether the event is related to the study drug or not, and should complete the Serious Adverse Effect Report Form provided by the sponsor. The SAE report form should be sent to the sponsor or its agent by fax or e-mail within 24 hours. At the post-treatment stage (30 days after the last administration of the study drug or initiating the new anti-tumor therapy to the end of the trial), only SAEs which have reasonable causality with the study drug evaluated by the investigator need to be reported. The investigator should report the follow-up information within the same reporting deadline after receiving such information. For events that have progressed from non-SAE to SAE, the event and its relevant follow-up information should be reported within 24 hours as described in the preceding paragraph.

10.3.2 Report on Protocol-Specified Adverse Event of Special Interest (AESI)

AESI (liver damage and bleeding) that meets the aforementioned protocol requirements, regardless of whether it is an SAE or not and whether it is related to the study drug or not, needs to be reported to the sponsor.

- AESI that meet the requirements of SAE should be reported by following the procedures of fast report on SAEs:
- For AESI that does not meet the requirements of SAE, only relevant case information needs to be sent to the sponsor or its agent within 24 hours after detection of said event.

10.4 Follow-up for Patients

10.4.1 Follow-up for Adverse Events

The investigator should follow up on all AEs until any of the following conditions occurs:

- amelioration of adverse events or improvement of adverse events to baseline;
- the investigator confirms that further improvement is not expected;
- death of the patient;
- loss to follow-up of the patient;
- the patient starts a new anti-cancer therapy;
- no further collection of clinical or safety data or final shutdown of the database.

10.4.2 Follow-up for Severe Adverse Events

Tracking and follow up on SAE patients conducted by the investigator are both to protect patients' safety (active treatment and tracking disease status), and to collect SAE-related information as much as possible. Investigators should actively take initiative and closely understand disease condition and provide necessary treatment while collecting complete case

information and perform and report follow-ups promptly.

Once the correlation between the SAE and the study drug is completely eliminated, the investigator may stop collecting SAE-related information; when SAE is considered to be (possibly) associated with the study drug, the investigator should continue to follow up until the patient dies, SAE improves or the patient is lost to follow-up and improve the information elements needed for the SAE report.

For certain SAEs, the sponsor may request more information on the event of the case from the investigator.

10.4.3 Follow-up for Adverse Event of Special Interest (AESI) Specified in the Protocol

All AESIs (liver damage and bleeding) that meet the protocol requirements should be followed up as SAE follow-up routine. In addition, said follow-up should be conducted by reference to the "Clinical Evaluation of Liver Function Impairment" and “Clinical Management of Severe or Serious Bleeding Event” provided in Appendix 5 and 6, respectively, so as to obtain more effective information for assessing such events.

10.5 Emergency Unblinding

In case of the occurrence of major safety events, the Principal Investigator of the research center will decide whether to unblind the study according to the emergency and rescue needs. The sponsor should be informed of the unblinding and unblinding should be performed via IWRS. The study will end for the case once the unblinding procedures are performed.

11 Statistical Analysis

More detailed rules for censoring will be presented in the statistical analysis plan (SAP).

11.1 Primary and Secondary Study Variables

11.1.1 Efficacy Indicators

Primary Efficacy Variables:

Overall survival (OS): Refers to the time between the date of randomization and the date of death caused by any reason. For patients who have not yet reported death at the time of the analysis, the date of the last known follow-up for survival will be recorded as the date of censoring.

Progression-free survival (PFS): refers to the time from the date of randomization to the first record of PD or date of death, whichever occurs first. The investigators will evaluate PD based on RECIST v1.1. The imaging results that record the progress of the disease should be obtained as much as possible. The date of last valid tumor evaluation will be recorded as the date of censoring for patients who have not yet show progressive disease or are not dead by the time. The date of randomization will be recorded as the date of censoring for those patients who do not have a tumor evaluation after the baseline period. More detailed rules for censoring will be presented in the statistical analysis plan.

Secondary Variables:

Objective response rate (ORR) of the tumor: is defined as the incidence rate at which best overall response (BOR) is complete response (CR) or partial response (PR) by RECISTv1.1. Patients who do not undergo a tumor evaluation after the baseline period would be considered to have no response.

Disease Control Rate (DCR): defined as the incidence of complete response (CR), partial response (PR), and stable disease (SD) by the standard disease control rate (DCR) according to RECISTv1.1.

The duration of response (DoR) is defined as the time (number of days) from the first recorded objective partial response or complete response, to progressive disease or death (if the patient dies before progressive disease is recorded). The duration of response is only applicable to patients who show complete response or partial response. Use the actual date of completion of the tumor scan for calculation. For the patients who do not show progressive disease or die at the time of analysis, the date of the last valid tumor evaluation will be treated as censored date.

Quality of Life Score: Quality of life score based on QLQ-C30, version 3.0 of the European Cancer Treatment Study Group.

11.1.2 Safety Indicators

The safety of treatment is assessed by AE, lab tests, vital signs, electrocardiogram, echocardiography (especially LVEF) and ECOG PS.

Regardless of whether or not related to the drug, the start/end date, the action taken and the effect of the treatment and the outcome of all adverse events will be reported in the Case Report Form. For all events, the causal relationship with the treatment and the severity will be

confirmed by the investigator.

11.2 Methods of Statistical Analysis

11.2.1 Analysis of Population

Intent-to-test Set (ITT Set): Analyze according to the principle of intentionality analysis, including all patients and are analyzed according to the assigned groups after randomization. The set us the major set of the efficacy endpoint.

Per-Protocol Set (PPS): Patients who have a focused ITT set without a major deviation from the protocol that affects efficacy evaluation. PP population is used for sensitivity analysis of the primary efficacy endpoint (OS) and key secondary efficacy endpoint (PFS).

Safety Analysis Set (SAS): Including all randomized patients who had received at least one dose of the study drug. SAS will be used for the analysis of safety endpoint.

11.2.2 Efficacy Analysis

Primary Efficacy Endpoint: OS and PFS

Primary analysis:

The stratified log-rank test was used to compare the OS and PFS of the paclitaxel+fruquintinib group and the paclitaxel group at the bilateral significance level of 0.05. The factors of stratification are: the primary site of the tumor (gastroesophageal junction vs stomach), ECOG PS score (0 vs 1), and peritoneal metastasis (present vs absent).

The Kaplan-Meier method will be used to estimate the OS and PFS of each treatment group, and a Kaplan-Meier curve will be drawn to provide a visual description of the differences among the treatment groups. The estimate of therapeutic efficacy (OS and PFS) will be expressed by the hazard ratio (HR) and its 95% confidence interval estimated by the stratified Cox proportional hazards model.

The effect of subsequent anti-tumor therapy:

If most patients receive the subsequent anti-tumor therapy, in order to correct the effect of subsequent anti-tumor therapy to the OS, the following or more sensitive analyses will be performed. The purpose of the analyses is to provide supportive evidence for benefit on OS:

- OS data will be censored at the time point of starting subsequent anti-tumor therapy
- Exclude the patients who receive the subsequent anti-tumor therapy

Secondary Efficacy Endpoint: ORR, DCR, DoR and Quality of Life Score.

The Clopper-Pearson method will be used to estimate the ORR and DCR values and their 95% confidence intervals of each treatment group. The Cochran-Mantel-Haenszel (CMH) method will be used to calculate odds ratio and its 95% confidence interval and p value. The stratification factor used in the CMH test was the same as that in the OS analysis. If the number of objective response or disease control does not support CMH test, the stratifying exact test will be performed, and the exact confidence interval of odds ratio will also be calculated.

DoR will only be applicable to patients with objective response (CR/PR). Descriptive analysis will be used for the DoR. The Kaplan−Meier method will be used to estimate the median DoR

of each group and the results will be shown by the distribution curve.

For each visit, descriptive statistics will be applied to the change of QLQ-C30 from the baseline. Mixed Models Repeated Measures (MMRM) were used for the inter-group change of QLQ-C30 from the baseline. Baseline values, visits, treatment and the interaction between treatment and visit will be used as fixed effects, and the patients and interception will be used as random effects. Restricted maximum likelihood (REML) will used to estimate the parameters in the model. Covariance structure will be selected from compound symmetry, toeplitz, spatial and unstructured in accordance with information criteria. The final analytical model will depend on the observed data. The detailed description is provided in a separate statistical analysis plan.

11.2.3 Safety Analysis

Adverse Event (AE)

AEs will be coded by MedDRA. The AEs will be determined and classified in accordance with NCI CTC AE 4.03.

We will summarize treatment emergent adverse events (TEAEs), serious adverse events (SAEs), and incidence of AEs that result in discontinuation of administration of the investigational drug, dosage reduction and death of two groups.

TEAE incidence will be summarized in accordance with SOC and PT. Within each SOC and PT, the TEAE incidence will be further summarized in accordance with the highest degree of severity and the relationship with the investigational drug.

Clinical Laboratory Data

Laboratory data will mainly be summarized by descriptive statistics.

Part of the laboratory variables will be evaluated in accordance with the classification system NCI CTC AE 4.03.

The changes from the baseline in laboratory data will be summarized in a change table.

ECOG PS

The shift table will be used to describe changes before and after treatment.

Electrocardiogram and Echocardiography

The shift table will be used to summarize the changes of each treatment group before and after treatment. The patients with abnormal electrocardiogram and echocardiography will be tabulated and summarized.

Vital Signs

Data is summarized by the descriptive statistics. The number and proportion of patients with defined abnormal values at any time after treatment will be summarized.

11.3 Statistical hypothesis and sample size determination

The primary endpoint is to verify if the efficacy (OS or PFS) of fruquintinib combined with paclitaxel is superior to placebo combined with paclitaxel in patients with advanced gastric or

GEJ adenocarcinoma as the second line therapy.

Null hypothesis (H_0) of OS: OS HR = 1.0 (fruquintinib combined with paclitaxel vs placebo combined with paclitaxel)

Alternative hypothesis (H_1) of OS: OS HR < 1.0 (fruquintinib combined with paclitaxel vs placebo combined with paclitaxel)

The number of primary endpoint OS event required for the evaluation of efficacy is calculated based on the following assumptions:

- The significance level was one-sided 0.025 (which is equal to two-tailed 0.05)
- 80% power of test
- The OS hazard ratio (HR) of treatment group/control group is 0.75, in other words, the median overall survival is extended from 7.0 months to 9.33 months.
- Randomization at an 1:1 ratio

As of the date of protocol version 4.0 amendment, the first interim analysis planned in protocol version 3.0 has been completed, which occurred when 100 patients were enrolled and the last enrolled patient completed the 6-month follow-up. The objective of the interim analys is to early terminate the study if negative result is obtained. The criteria of early termination were provided in the Appendix C in IDMC charter (Version 1.1, issued on 11 Mar 2019).

Based on the results from the first interim analysis and pre-supposed criteria for early termination, IDMC suggested to continue the study, and an additional interim analysis could be added when 50% of the OS events have been observed. The sponsor accepted the suggestion. The second interim analysis will use Lan-DeMets spending function to construct the valid or invalid O'Brien-Fleming type stopping boundary for study termination. Then a total of 401 OS events need to be observed at the final analysis to achieve the 80% power of test. Assuming that the non-homogeneous enrollment will be completed within 29 to 30 months and the last patient need to be followed up for 7 months, a total of 544 patients are required to be enrolled to observe 401 OS events.

After the second interim analysis, Sponsor determined to go on the study with increasing the sample size as suggested by Study Scientific Monitor Committee and Center for Drug Evaluation (CDE). The final analysis will be performed when approximately 700 patients are enrolled and approximately 525 OS events are observed.

Considerations for an increase in OS event count to 525:

- ✓ Fuquinitinib placebo plus paclitaxel (paclitaxel monotherapy), median OS = 8.3 months (Keynote-061)
- ✓ Fuquinitinib plus paclitaxel versus paclitaxel monotherapy, 2-month improvement of OS (HR = 0.806)
- ✓ Final analysis significance level was two-sided 0.049 (accounting for consumption of one

completed interim analysis for OS)

- ✓ Power of final analysis: approximately 70%

The protocol was revised to version 6.0. Based on further communication with the CDE, the sponsor decided to change the single primary endpoint of OS to dual primary endpoints of PFS and OS. The final analysis of PFS and OS will be performed simultaneously.

Graphical method will be used (alpha distribution and recycling) to control type I error for the final analysis of the dual primary endpoints at two-sided 0.049 level (considering the consumption of first completed OS interim analysis).

- ✓ The remaining total two sided significance level 0.049 will be split into two parts, 0.01 for PFS test (two-sided) and 0.039 for OS test (two-sided).
- ✓ If it's statistically significant for PFS analysis, the two-sided level 0.01 of PFS will be recycled and transformed to OS, OS will be test on the two-sided 0.049 level.
- ✓ If it's statistically significant for OS analysis, the two-sided level 0.039 of OS will be recycled and transformed to PFS, PFS will be test on the two-sided 0.049 level.

The sample size for this study was determined by the final analysis of OS. The final analysis will be performed when approximately 700 patients are enrolled and approximately 525 OS events are observed. Based on the number of 656 PFS events and a two-sided 0.01 significance level, approximately 97% power would be provided to detect a statistical difference in PFS between the two groups (assumed HR=0.7).

11.4 Interim Analysis

The interim analysis will be conducted by the independent unblinding statistician and reviewed by the IDMC. Analyses of interim data were performed by unblinded statisticians at the CRO who were independent of the study.

The first interim analysis is planned to be conducted after 100 patients were enrolled and the last enrolled patient completed the 6-month follow-up. As of the date of protocol version 4.0 amendment, the first interim analysis planned in protocol version 3.0 has been complted. IDMC suggested to continue the study, and additional interim analysis could be added when 50% of OS events have been observed. The sponsor accepted the suggestion and scheduled a second interim analysis.

The purpose of the second interim analysis is to early terminate the study by superiority or inferiority results, or continue study by adjusting the sample size, and evaluate the safety profile of fruquintinib combined with paclitaxel. If the results of the interim analysis meet the criteria for early termination by superiority results, the sponsor may accept the suggestion from IDMC and terminate the study. The second interim analysis will occurred when about 201 OS events achieved (approximately 50% of toal OS events).

- If the statistic exceed the efficacy critical value ($z \geq 2.958$, two-tailed $p \leq 0.003$), the study will be terminated by superior efficacy. Under the hypothesis of exponential distribution,

the critical value is equal to HR ≤ 0.659.

- If the statistic exceed the futility critical value (z≤0.564, two-tailed *p*≥0.573), the study will be terminated by futility. Under the hypothesis of exponential distribution, the critical value is equal to HR ≥ 0.924.
- If the results of the interim analysis does not meet the criteria for study termination but fall in the promising zone, the sample size will be increased by CHW method (Cui, Hung& Wang, 1999). In the final analysis, the statistic will be calculated by CHW method. For the reasons of confidentiality, the rules of re-estimation of sample size will be provided in an independent document, and will inform all members of IDMC before IDMC meeting.
- If the results of the interim analysis does not meet the criteria for study termination and the sample size does not need to be adjusted, the final analysis will be conducted when around 401 OS events occurred. The corresponding efficacy critical value: z≥1.969, two-tailed *p* ≤0.049. Under the hypothesis of exponential distribution, the critical value is equal to HR ≤ 0.821.
- If the results of the interim analysis indicate serious safet issues, the study will be early terminated.

O's Brien-Fleming boundary values

OS number of events	Ratio	Boundary value for statistic z		Boundary value for nominal α (two-tailed)		Cumulative α consumption	Cumulative β consumption
		Valid	Invalid	Valid	Invalid		
201	0.501	2.958	0.564	0.003	0.573	0.003	0.070
401	1.0	1.969	1.969	0.049	0.049	0.05	0.20

11.5 Final Analysis

The final analysis will be performed when approximately 700 patients are enrolled and approximately 525 OS events are observed.

12 Data Management

This study uses an electronic data management system.

12.1 Monitoring

The monitors assigned by the Sponsor will contact and visit the investigator on a regular basis. The privacy of the patients will be protected in accordance with the regulatory requirements. The monitors will inspect various records of the study in accordance with the requirements (eCRF and other patient data).

12.2 Audit and Inspection

The representatives authorized by HUTCHMED Limited, monitoring department, ethics committees can inspect the research center, including verification of the original data. The purpose of inspection, through systemic and independent inspection, is to ensure all conducts

and documents related to the study are implemented, all data is recorded and the analyses are compliant with the clinical trial protocol, guidelines of the Good Clinical Practice (GCP) and the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), and other regulatory requirements. The investigator shall inform HUTCHMED Limited when they are informed of the inspection in the study site by the regulatory authority.

12.3 Personnel Training

The principal investigator will keep a record of the research work of the relevant personnel (doctors, nurses and others). The Principal Investigator should ensure that all researchers are properly trained and the information related to the study has been communicated to the researchers.

12.4 Data Entry, Recording and Management

The investigator should ensure that the data recorded on the eCRF is accurate, complete and clear. The monitors of HUTCHMED Limited will check the data during the inspection visits. The monitors will send a question form regarding any data deficiency, anomaly and inconsistency, and make a corresponding record. The database will be locked after all questions are answered.

13 Data Monitoring Committee

An Independent Data Monitoring Committee (IDMC) will be established for this study. The Committee will approve/disapprove risk/benefit assessments in regular data review meetings to decide the safety of the patients. Meanwhile, IDMC will evaluate the safety and efficacy data during the midterm analysis, and provide formal advice on whether to continue or terminate the study in accordance with the established criteria.

The Committee will include at least three independent oncologists and an independent statistician. The data review meetings will be held on a regular basis as described in the IDMC charter. The patient enrollment will continue as the IDMC meetings are held.

After the data is reviewed, IDMC will provide advice on whether to continue, modify or terminate the study. HUTCHMED Limited will make the final decision.

14 Ethical Norms

14.1 Ethical Management of Clinical Research

The clinical study cannot be commenced without the approval of the Ethics Committee (EC).

This study must be carried out in strict accordance with the CFDA Good Clinical Practice and the Declaration of Helsinki.

14.2 Ethics Committee (EC)

Protocol, Informed Consent Forms and other materials provided to the patients should be reviewed and approved by the Ethics Committee before the start of the clinical study. The relevant approval will be provided to the Sponsor.

14.3 Informed Consent Form

The responsibility of the investigator or the personnel designated by the investigator (if permitted by local requirements) is to fully explain the purpose, methods, expected benefits and potential hazards of the study to the patients participating in the study prior to obtaining written Informed Consent Forms from the patients. With regard to the patients unable to provide legal Informed Consent Forms, the investigator must obtain the written Informed Consent Form from their legal guardians. If the patient and his/her legal guardian cannot read, a third party witness must be present during the entire informed consent process. Once the patient and his/her guardian orally agree to participate in this study, the witness has to sign on the Informed Consent Forms to prove that its content is accurately explained and understood. The investigator or the person he/she designates must also state that the patient is free to refuse or withdraw from the study at any time for any reasons. In the case of significant change in the risk/benefit assessment due to new safety information, the Informed Consent Form shall be reviewed and updated if necessary. All patients (including those who have been treated) should be informed of the new information and be provided with a revised Informed Consent Forms for the consent to continue to participate in the study.

14.4 Confidentiality of Patient Information

The investigator must ensure that the patients remain anonymous and the privacy of the patients will not be disclosed to an unauthorized third party. In the eCRF or other documents submitted to the Sponsor, the patients cannot be identified by name but identification code only. The investigators should make a patient enrollment registration form containing the patient code, name and address. The investigator shall strictly preserve certain documents that should not be submitted to the Sponsor, such as patients' Informed Consent Forms.

15 Termination of Study

The Sponsor and investigators are entitled to terminate the study at any time. When the study is terminated, the Sponsor and investigator will ensure that the benefits of the patients are taken into account to the maximum extent possible.

16 Preservation of Research Documents, Case Reports and Records

The original documents should be able to prove the existence of the patients and the completeness of the data collected. The original documents are filed at the research center/hospital where the investigator is located.

The data in the eCRF must be consistent with the original documents. If there are any differences, explanations must be provided. The investigator may need previous medical records, transfer records or current medical records.

The investigator should agree to preserve all study data to 5 or more years after the drug is approved for sale on the market. Once the preservation period of the study data in the research institution ends, please contact the Sponsor for matters related to subsequent preservation.

17 Data Publication and Protection of Trade Secrets

The ownership of all information of this clinical trial belongs to the Sponsor. Without the consent of the Sponsor, the information may not be provided to personnel unrelated to this study.

The data related to this study can only be published after the consent of the Sponsor.

The results of the study may be published or released at scientific conferences. If the investigator has such a plan, the investigator agrees to submit all manuscripts or abstract to the Sponsor before submission. In this way, the Sponsor's patent information can be protected. Moreover, the Sponsor can provide comments to the investigator regarding the information of other studies not accessible by the investigator.

According to standard publishing and ethical norms, the Sponsor usually supports overall publication of multi-center data rather than single-center data.

18 Reference

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7. 艾坦®（Apatinib Mesylate Tablets）Package Insert.
8. Jin Li, et al. Randomized, Double-Blind, Placebo-Controlled Phase III Trial of Apatinib in Patients With Chemotherapy-Refractory Advanced or Metastatic Adenocarcinoma of the Stomach or Gastroesophageal Junction. [J] Journal of clinical oncology. 2016, 34(13):1448-1454.
9. Investigator’s Brochure of Fruquintinib, v10.0.

Appendix 1 ECOG Performance Status Score

The patients will be classified in accordance with the ECOG performance status table. The standards are described as follows:

Grade	ECOG
0	Fully active, able to carry on all pre-disease performance without restriction.
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work.
2	Ambulatory and capable of all self care but unable to carry out any work activities, up and about more than 50% of waking hours.
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.
4	Completely disabled, cannot carry on any self-care, totally confined to bed or chair.
5	Death.

Appendix 2 Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1

Response Evaluation Criteria in Solid Tumors version 1.1

(Response Evaluation Criteria in Solid Tumors RECIST - 1.1)

Because RICIST1.1 has no official Chinese version at present, this version is internal translation version. For more details, please refer to the English version.

Abstract

Background

Assessment of changes in the tumor burden is an important feature of the clinical assessment of cancer therapy. Both tumor shrinkage (objective response) and disease progression are meaningful endpoints in clinical trials. Since RECIST was published in 2000, many investigators, cooperative groups, industry and government authorities have adopted these criteria in the assessment of treatment outcomes. However, a number of questions and issues have arisen which have led to the development of a revised RECIST guideline (version 1.1). The corrections (see the topic of each chapter) have come from the assessment of a large database (more than 6,500 patients), simulation studies and literature reviews.

Highlights of revised RECIST 1.1

Main revisions include:

Number of lesions to be determined: For analysis purpose, based on the database merged from assessments of numerous data, the total number of lesions required to determine the response endpoint of tumor burden has been reduced from a maximum of 10 to a maximum of 5 (from 5 to 2 per organ, maximum).

Assessment of pathological lymph nodes is now incorporated: nodes with a short axis of 15 mm are considered measurable target lesions for assessment. The short axis measurement (of nodes) should be included in the sum of lesions (diameter) when one determines the tumor efficacy. Nodes that shrink to <10mm can be considered normal.

Confirmation of effectiveness is required for clinical trials with response rate as the determination of endpoint. However, this is no longer required in randomized control clinical trial because the control group has become effective means to integrate data. Disease progression is clarified in several aspects: In addition to the previous definition of progression in target disease of 20% increase in sum (diameter), a 5 mm absolute increase in sum diameter is now required as well to guard against over calling PD when the total sum is very small. Furthermore, there is guidance offered on what constitutes 'unequivocal progression' of non-measurable/non-target disease, a source of confusion in the original RECIST guideline. Finally, a section on detection of new lesions, including the interpretation of FDG-PET scan assessment is included. Imaging guidance: the revised RECIST includes a new imaging appendix with updated recommendations on the optimal anatomical assessment of lesions.

Future work:

A key question considered by the RECIST Working Group in developing RECIST 1.1 was whether it was appropriate to move from anatomic unidimensional assessment of tumor burden to either three dimensional anatomical assessment or to functional assessment with PET or MRI.

It was concluded that, at present, there is not sufficient standardization or evidence to abandon anatomical assessment of tumor burden. The only exception to this is in the use of FDG-PET imaging as an adjunct to determination of progression. As is detailed in the final paper in this special issue, the use of these promising newer approaches requires appropriate clinical validation studies.

Keywords: Response criteria; solid tumors; guidelines

3.2.1Background

1.1. History of RECIST criteria

Assessment of changes in the tumor burden is an important feature of the clinical assessment of cancer therapy. Both tumor shrinkage (objective response) and the time to the development of disease progression are important endpoints in cancer clinical trials. The use of tumor regression as the endpoint for phase III trials screening new agents for evidence of anti-tumor effect is supported by years of evidence suggesting that, for many solid tumors, agents which produce tumor shrinkage in a proportion of patients have a reasonable (albeit imperfect) chance of subsequently demonstrating an improvement in overall survival or other time to event measures in randomized phase III studies. At the current time objective response carries with it a body of evidence greater than for any other biomarker supporting its utility as a measure of promising treatment effect in phase II screening trials. Furthermore, at both the phase II and phase III stage of drug development, clinical trials in advanced disease settings are increasingly utilizing time to progression (or progression-free survival) as an endpoint upon which efficacy conclusions are drawn, which is also based on anatomical measurement of tumor size.

However, both of these tumor endpoints, objective response and time to disease progression, are useful only if based on widely accepted and readily applied standard criteria based on anatomical tumor burden. In 1981 the World Health Organization (WHO) first published tumor response criteria, mainly for use in trials where tumor response was the primary endpoint. The WHO criteria introduced the concept of an overall assessment of tumor burden by summing the products of bidimensional lesion measurements and determined response to therapy by evaluation of change from baseline while on treatment. However, in the decades that followed their publication, cooperative groups and pharmaceutical companies that used the WHO criteria often 'modified' them to accommodate new technologies or to address areas that were unclear in the original document. This led to confusion in interpretation of trial results and in fact, the application of varying response criteria was shown to lead to very different conclusions about the efficacy of the same regimen. In response to these problems, an International Working Party was formed in the mid 19th century to standardize and simplify response criteria.

New criteria, known as RECIST (Response Evaluation Criteria in Solid Tumors), were published in 2000. Key features of the original RECIST include definitions of minimum size of measurable lesions, instructions on how many lesions to follow (up to 10; a maximum five per organ site), and the use of unidimensional, rather than bidimensional, measures for overall evaluation of tumor burden. These criteria have subsequently been widely adopted by academic institutions, cooperative groups, and industry for trials where the primary endpoints are objective response or progression. In addition, regulatory authorities accept RECIST as an appropriate guideline for these assessments.

3.2.2 Purpose of this guideline

This guideline describes a standard approach to solid tumor measurement and definitions for objective assessment of change in tumor size for use in adult and pediatric cancer clinical trials.

It is expected these criteria will be useful in all trials where objective response is the primary study endpoint, as well as in trials where assessment of stable disease, tumor progression or time to progression analyses are undertaken, since all of these outcome measures are based on an assessment of anatomical tumor burden and its change on study. There are no assumptions in this paper about the proportion of patients meeting the enrollment criteria for any of these endpoints which will signal that an agent or treatment regimen is active: those definitions are dependent on type of cancer in which a trial is being undertaken and the specific agent(s) under study. Protocols must include appropriate statistical sections which define the efficacy parameters upon which the trial sample size and enrollment criteria are based. In addition to providing definitions and criteria for assessment of tumor response, this guideline also makes recommendations regarding standard reporting of the results of trials that utilize tumor response as an endpoint.

While these guidelines may be applied in malignant brain tumor studies, there are also separate criteria published for response assessment in that setting. This guideline is not intended for use for studies of malignant lymphoma since international guidelines for response assessment in lymphoma are published separately.

Finally, many oncologists in their daily clinical practice follow their patients' malignant disease by means of repeated imaging studies and make decisions about continued therapy on the basis of both objective and symptomatic criteria. It is not intended that these RECIST guidelines play a role in that decision making, except if determined appropriate by the treating oncologist.

3.2.3 Measurability of tumor at baseline

3.1 Definitions

At baseline, tumor lesions/lymph nodes will be categorized measurable or non-measurable as follows:

3.1.1 Measurable lesions

Tumor lesions: Must be accurately measured in at least one dimension (longest diameter in the plane of measurement is to be recorded) with a minimum size of:
• 10 mm by CT scan (CT scan slice thickness no greater than 5 mm; see Appendix III on imaging guidance).
• 10 mm caliper measurement by clinical exam (lesions which cannot be accurately measured with calipers should be recorded as non-measurable).
• 20 mm by chest X-ray.
Malignant lymph nodes: To be considered pathologically enlarged and measurable, a lymph node must be ⩾15 mm in short axis when assessed by CT scan (CT scan slice thickness recommended to be no greater than 5 mm). At baseline and in follow-up, only the short axis will be measured and followed. See also notes below on "Baseline documentation of target and non-target lesions" for information on lymph node measurement.

3.1.2 Unmeasurable lesions

All the other pathologic changes, including small ones (with the longest diameter shorter than 10 mm or pathologic lymph nodes with the shortest axis of 10 mm to less than 15 mm in length) and truly unmeasurable pathologic changes. Pathologic changes that are considered as truly unmeasurable include: pharmacologically confirmed disease of the meninges, ascites, pleural or pericardial effusion, inflammatory mammary gland disease, enlarged skin or lungs, abdomen

lumps or enlarged abdomen involving lymphatic vessels; all of these could not be measured applying reproductive imaging techniques.

3.1.3 Special considerations with measurable lesions

Particular attention needs to be paid to bone, cystic, and previously locally treated pathologic changes:
Bone:

- Bone scan, PET scan, or plain films are considered to be insufficient imaging techniques for measuring pathologic changes involving the bone. Pharmacological techniques, however, may be used to confirm presence or disappearance of pathologic changes involving the bone.
- If the soft tissue partially meets the definition of "measurable" as mentioned above and osteolytic lesions or mixed acute osteolytic lesions with identifiable soft tissues may be evaluated by means of cross-over imaging techniques such as CT or MRI, they may be considered as measurable pathologic changes.
- Osteoblastic lesions are unmeasurable.

Cystic lesions:

- The simple cysts that meet the X-ray definition in the enrollment criteria shall not be considered as malignant (they are not measurable or unmeasurable) as they are defined as simple cysts.
- If the definition of "measurable lesion" as mentioned above is fulfilled, cystic lesions that are expressed through cystic metastasis may be considered as measurable lesions. If non-cystic lesions exist inside the body of the same patient, however, they shall be prioritized to be enrolled as the target lesion.
  Previously locally treated lesions:
- Tumor lesions at the site previously irradiated or locally treated are usually not considered measurable unless it is proven that the lesions remain progressive. It shall be specified in the Protocol that under what circumstances such lesions will be considered measurable.

3.2 Measurement methodology guidelines

3.2.1Measuring the lesion

For clinical assessments, the caliper is used as the measuring tool; all measurements are recorded in the metric system. All baseline assessments have to be completed as close to the start of treatment as possible and may not be earlier than four weeks prior to the start of treatment.

3.2.2Measurement methodology

At baseline and during follow-up, the same determining methods and techniques shall be adopted to describe each reported lesion. Unless it is found during follow-up that imaging is not a suitable technique for detecting the lesion, imaging shall usually be adopted for the evaluation, instead of clinical exams.

Clinical lesions: Only superficial lesions (e.g. subcutaneous nodules) with a diameter of 10 mm or greater as measured with a caliper are considered measurable. For skin impairment cases, it is advised to adopt color pictures for documentation purpose; enclose a scale for the size of the lesion obtained in the picture. Just as described above, when a lesion can be detected clinically and imaging tests may also apply, imaging tests shall be adopted; this is because imaging

evaluations render more objective results and may be used during the final review of a clinical study.

Chest X-ray: Chest CT is prioritized to chest X-ray. This is particularly true when progressive disease is adopted as an important endpoint because a CT scan is more sensitive to an X-ray exam in terms of identifying new lesions. If the borders of a lesion are clearly defined as shown through the X-ray exam and the lesion is surrounded by an inflated lung, however, it is considered measurable. See Appendix 2 for details.

CT, MRI: CT is currently the most effective and most desirably reproductive detection method for evaluating efficacy at a lesion. It is defined in guidelines that when a lesion is to be scanned with CT, the CT slice thickness may not exceed 5 mm (See Appendix 2). When the CT slice thickness exceeds 5 mm, the measurable lesion shall be twice as thick of the slice at minimum. MRI may also be used under certain circumstances (e.g. systemic scan). For more opinions on the use of CT and MRI testing to assess the efficacy for solid tumors, see Appendix 3.

Ultrasonography: An ultrasound scan is not indicated for assessing the size of a lesion and hence shall not be used as a measurement approach. It is impossible for an ultrasound to be completely reproduced between two consecutive observations and the results are dependent on the examiner. The same technique and measurement results may not be guaranteed from one measurement to the next measurement (See Appendix 2 for details). If a new lesion is discovered through an ultrasound scan during a study, it is advised that CT or MRI be used for verification purpose. If the radiation associated with CT is a cause for concern, MRI may be used as an alternative to detect the lesion.

Endoscopy, laparoscopic surgery: It is not advised that the said techniques be applied in the objective tumor evaluation. Nevertheless, they are of merits when a biopsy is used to confirm complete pathologic response or complete response or recurrence following surgical resection.

Tumor markers: Tumor markers cannot be used alone to assess objective response of a tumor. Nevertheless, when tumor markers begin to climb higher than the upper limit of normal, they have to be standardized if they are to be used to determine whether a patient has reached complete response. Because tumor markers are disease specific, descriptions of the measurement techniques shall highlight records of baseline measurements for a specific disease. The special guidelines on CA-125 changes (recurrence of ovary cancer) and PSA changes (recurrence of prostate cancer) are already released. In addition, the international gynecologic oncology group (Intergroup) stipulated the CA125 Progression Criteria to help with the overall objective evaluation of tumor during first-line application in trials of ovary cancer.

Cytology, histology: If required for a study protocol, these techniques may be frequently applied to identify partial response and complete response in individual cases (if the type of tumor is benign remnant disease involving gem cells). If it is known that the exudate is a potential serious adverse consequence during treatment (e.g. certain paclitaxel-based chemotherapy agents or anti-angiogensis agents), in order to separate effective treatment (e.g. stable disease) from progressive disease, even if measurable tumor meets the criteria for being effective or stable, attention needs to be paid to any cytologically-confirmed new exudate that appears or worsens during treatment.

4.7.1 Tumor efficacy assessment

4.1 Assessment of all tumors and measurable lesions

In order to evaluate objective response or possible progression in the future, it is necessary to perform a baseline assessment of the overall tumor load of tumor lesions; the results will serve as reference for subsequent measurements. In clinical protocols where objective response is a primary efficacy endpoint, only patients with measurable lesions at baseline will be enrolled. Measurable lesion is defined as presence of at least one site with measurable lesions. For trials with progressive disease (time to disease progression or extent of progression at a fixed date) is used as the primary efficacy endpoint, it has to be specified in the Protocol enrollment criteria whether only patients with measurable lesions or also those without measurable lesions may be enrolled.

4.2 Baseline records of target and non-target lesions

When there are more than one measurable lesions at baseline assessment, all lesions shall be recorded and measured; there may not be more than five such lesions in total (no more than two in each organ) to be the target lesions for all involved organs (that is, for patients with only one or two organs involved, two or four target lesions at maximum are chosen to be the baseline measurement lesions).

Target lesions are to be chosen based on their size (the longest diameter) and must be capable of representing all involved organs and the measurements have to be of desirable reproductivity. Sometimes, when the largest lesion could not be measured repeatedly, another largest lesion that could be repeatedly measured may be re-selected.

Lymph nodes, due to the fact that they are normal tissues, could be picked up by imaging techniques despite the absence of tumor metastasis and hence particular attention is required. Nodes that are defined as measurable and even pathologic lymph nodes of the target lesion must meet the following criteria: Shortest diameter detected through CT of ⩾ 15 mm. Only the short diameter needs to be detected at baseline. A radiologist usually adopts the short diameter of a node to determine if tumor metastasis presents in the node. The size of a node is usually indicated through 2D data from imaging tests (axial plane for CT and axial plane, sagittal plane, or coronal plane for MRI). The minimum value is the short diameter. An abdominal node of 20 mm × 30 mm, for example, has a short diameter of 20 mm and may be considered as malignant and measurable. In this example, 20 mm is the measurement of the node. A node with a diameter of ⩾10 mm yet <15 mm may not be considered as the target lesion. A node of <10 mm, on the other hand, does not fall within the scope of a pathologic node and does not need to be recorded or observed further.

The sum obtained from the diameters of all target lesions through calculation (including the longest diameters of non-nodal lesions and short diameters of nodal lesions) will be the sum of diameters at baseline and be reported. If diameters of lymph nodes are included, as mentioned above, only short diameters will be included in the calculation. The sum of diameters at baseline will be the reference values of the disease at baseline.

All the other lesions, including pathologic lymph nodes, may be considered as non-target ones and do not require measurements; nevertheless, they shall be recorded during baseline assessment. They may be recorded, for example, as "present", "missing", or under extremely rare circumstances, "definitely progressive". Target lesions that exist extensively may be recorded together with the target organ (e.g. extensively augmented lymph nodes of the pelvis or large scale liver metastasis).

4.3 Efficacy Assessment Criteria

This section defines the criteria for measuring the objective response rate of target tumor lesions.

4.3.1. Efficacy Evaluation of Target Lesions

Complete response (CR): All target lesions have disappeared and any pathologic lymph node (regardless whether it is a target lesion or not) has to have a short axis value of <10 mm.

Partial response (PR): With the overall diameter at baseline as reference, the sum of all target lesion diameters have reduced by 30% at minimum.

Progressive disease (PD): With the sum of smallest lesion diameters as reference (including the sum of lesion diameters at baseline, if it is of the minimum value), the sum of all target lesion diameters has increased at least by 20%. In addition, besides the sum of diameters, which has increased by 20% in relation, the absolute sum has also to have increased by 5 mm at minimum (Note: One or more new lesions may also be considered as progressive disease).

Stable disease (SD): With the sum of smallest lesion diameters during the course of study as reference, the shrinkage of a lesion does not qualify as PR and the increase in the number of lesions does not qualify as PD, either.

4.3.2. Precautions for the Evaluation of Target Lesion Efficacy

When the target lesion is a lymph node:

Usually the actual short axial value measurement (on the same cross section when measured at baseline) shall be recorded, even if all lymph nodes have remitted to be below 10 mm during the study. This means that when a target lesion is a lymph node, even if it reaches the criteria for complete response, the sum of target lesion diameters will not be 0 because a lymph node with the short axial value of <10 mm is defined as a normal lymph node. It may be necessary to separately record target nodal lesions while the case report form or other data collection methods are being designed. In order to determine whether complete response has been reached, the short axial value of <10mm has to be fulfilled by each node. For PR, SD, and PD, the sum of target lesions (diameters) will include the actual short axial values of nodes.

When target lesions are too small to be measured:

For all lesions recorded at baseline in a study (nodal and non-nodal), their actual measurements have to be recorded in subsequent assessments even though they are minimal (e.g. 2 mm).

Sometimes, however, it is possible that a radiologist is unwilling to provide a precise measurement and instead will indicate "it is too small to be measured" in the report due to the fact that the signal of a CT scan is too weak when the lesion or the lymph node is recorded at a threshold value.
Under these circumstances, it is important to record next measurements in the case report form. If a radiologist thinks that a lesion might have disappeared, the measurement may be documented as "0 mm." If a lesion exists yet the signal is too weak, it may be recorded as the default value of 5 mm (this rule, however, does not quite work for lymph nodes as the size of a normal lymph node has a defined value and it is usually surrounded by the adipose tissue, such as the lymph node in the retroperitoneal cavity; nevertheless, if a lymph node does exist and the signal is too weak to make a measurement difficult, it can still be documented as the default value of 5 mm).

The default value of 5 mm comes from the thickness of a CT slice (if the thickness is changed, the default value of 5 mm shall remain the same). The measurements of such lesions (that are too small to be measured) are likely to lack reproductivity. By giving a default value, it prevents

against the determination of false cure or false progression in case of a measurement error. What needs to be re-iterated is that any actual measurement provided by the radiologist, even if it is smaller than 5 mm, shall be recorded.

Lesions that Break Apart or Fuse Together During Treatment

As noted in Appendix 3, when a non-nodal lesion "breaks apart", all the longest diameters of the pieces have to be added together in order to calculate the sum of target lesion diameters. By the same token, when lesions fuse, the inter-diameter between them may be reserved as it helps with the acquisition of the maximum diameter value of each lesion prior to the fusion. If lesions completely fuse with one another and are not separated any more, the vector of the longest diameter under this circumstance is that of the maximum and longest diameter of the post-fused lesion.

4.3.3. Assessment of Non-target Lesions

This section defines the tumor response criteria for non-target lesions. Although some non-target lesions are actually measurable, they do not need to be measured; it is necessary only to perform qualitative assessments at time points specified in the Protocol.

Complete response (CR): All non-target lesions have disappeared and tumor markers have returned to a normal level. All lymph nodes are of a non-pathologic size (with a short diameter of ＜10 mm).

Not complete response/not progressive disease (not CR or not PD): One or more non-target lesions and/or persistent tumor marker levels exceeding normal levels exist.

Progressive disease (PD): Defined progression exists in non-target lesions. Note: The presence of one or more new lesions is also considered progressive disease.

4.3.4. Particular Precautions for the Progression Assessment of Non-target Lesions

The definition of non-target lesion progression is supplemented as follows: When measurable non-target lesions exist in patients, even if a target lesion is determined as stable or partially remitted, to render a specific progression definition on the basis of non-target lesions, it is necessary to fulfill the criterion that the extent of overall progression of non-target lesions has necessitated discontinuation of treatment. The general enlargement in the size of one or more non-target lesions, however, is insufficient to reach the progression criteria; as such, while target lesions are stable or have reached partial response, it is almost very rare to be capable of defining the overall tumor progression only based on changes in non-target lesions.

When non-target lesions of patients are consistently unmeasurable: In some Phase III studies, when it is not specified that the presence of measurable lesions is a requirement in the enrollment criteria, this occurs. The criteria above are still to be referred to during overall assessments. Under the said circumstance, however, there are no measurable data of the lesion available. It is uneasy to evaluate the progression of non-target lesions (according to the definition, it is required that all non-measurable targets are indeed unmeasurable). As such, when the extent of change in non-target lesions results in an increase in the overall disease load is equivalent to the disease progression in target lesions, the definition of specific progression rendered according to non-target lesions requires assessment through an effective test method created. One example is that the increase in tumor load is described to be equivalent to an additional increase of 73% in the volume (which is equivalent to an increase of 20% in the diameter of measurable lesions). Another example is that peritoneal exudates have increased

from "minimal" to "excessive" and lymphatic pathologic changes have progressed from "local" to "extensive" or it is described in the Protocol as "sufficient to change the treatment method." Pleural exudates, for example, have developed from "trace" to "excess" and lymphatic load has spread from the primary site to distal areas or it is possible to be described in the Protocol as "necessitate treatment changes." If specific progression is discovered, progressive disease shall be determined as a whole at the said time point for the patient. It is preferred that there are objective criteria to apply in the assessment of unmeasurable lesions. Caution: The additional criteria have to be reliable.

4.3.5. New Lesions

The appearance of new malignant lesions precautions disease progressions. Therefore, it is very important to evaluate new pathologic changes. There are no substantial criteria for detecting lesions through imaging tests available at present; the discovery of a type of new lesion, however, should be specific. Progression, for example, may not be attributed to different imaging techniques, changes in imaging patterns, or other pathologic changes than tumor (some of the so-called new bone lesions, for example, are only evidence of a cure or a relapse of original lesions). When partial or complete response occurs to baseline lesions of a patient, this is very important. A case of necrosis of liver lesion may be reported as new cystic pathologic change in a CT report, for example, while it is in fact not.

A lesion that is detected during follow-up but was not discovered during baseline exams will be considered as new and indicative of disease progression. For a patient found with lesion in his/her internal organ during baseline exams, for example, if he/she is found with metastasis during a head CT or MRI procedure, the intra-cranial metastatic lesion of the patient will be considered as the basis for progressive disease even if no head exams were done at baseline.

If a new lesion is non-specific because of its shape being too small, for example, further treatment and follow-up evaluation are required in order to confirm if it is a new lesion. If repeated tests confirm that it is a new lesion, the time of disease progression should be that when it was first discovered.

Usually additional tests are required for supplementary confirmation in the FDG-PET assessment of a lesion. It is reasonable to combine results of the FDG-PET exam and the supplementary CT test in order to evaluate the status of progression (particularly when new disease is suspected). When a new lesion can be specified through the FDG-PET exam, the following procedures are to be followed:

When the result of FDG-PET exam at baseline is negative and that during subsequent follow-up is positive, it means that the disease has progressed.

When the FDG-PET exam is not done at baseline and subsequent FDG-PET exam results are positive,

As long as the new lesions indicated through the positive results of the PDG-PET exam during follow-up answer to those found through the CT test, it is evidence of disease progression.

If the new lesions indicated through the positive results of the PDG-PET exam during follow-up are unable to be confirmed through CT test results, additional CT tests are required for confirmation purpose (if they are confirmed, the time to disease progression shall be calculated from the time the abnormality was discovered in prior FDG-PET exams).

If the positive results of FDG-PET exams during follow-up answer to existing lesions found through CT tests and no progression of the lesions is detected through imaging tests, there is no disease progression.

4.4 Optimal Overall Efficacy Evaluation

The optimal overall efficacy assessment deals with optimal efficacy records from the start of trial to the completion of trial. Any necessary criterion needs to be taken into consideration at the same time to facilitate verification. Sometimes efficacy response occurs after treatment is completed; as such, it shall be specified in the Protocol whether or not efficacy evaluation after treatment is completed is to be included in the optimal overall efficacy evaluation. It has to be specified in the Protocol how any new treatment prior to progression affects the optimal efficacy response. The optimal efficacy response of a patient relies mainly on results of target lesions and non-target lesions and expression of new lesions. In addition, the nature of the trial, protocol requirements, and results measurement criteria are to be relied on. Substantially speaking, in a non-randomized trial, efficacy response shall be a paramount goal and confirming a PR or CR response result is a must in order to confirm which is the optimal overall efficacy response.

4.4.1 Time Point-based Response

It is assumed that efficacy response occurs at substantial time points in each protocol. Table 1 summarizes the overall efficacy response at each time point of a patient population with measurable disease at baseline.

If patients have no measurable lesions (no target lesions), refer to Table 2 for their assessment.

4.4.2 Descriptions of Assessment Deficiencies and "Non-evaluable"

If lesion images or measurements may not be obtained at a specific time point, the patient is non-evaluable at the said time point. If only part of lesions may be evaluated during an evaluation, it is usually considered "non-evaluable" at the said time point unless there is evidence to support that the deficient lesion does not affect the efficacy response evaluation at the said time point. This is very likely to occur upon disease progression. Example: When a patient has three lesions with a sum of 50 mm at baseline and only two of the lesions are evaluable later and the sum comes to 80 mm, the patient will be determined as having had disease progression, regardless of the size of impacts from the deficient lesion.

4.4.3 Optimal Overall Response: All Time Points

Once all data of a patient are in place, the optimal overall response may be defined.

When there is no need to confirm complete or partial response in the study, the optimal overall response is assessed as follows: The optimal efficacy response shall be the optimal response at all time points in a trial (if a patient's efficacy is determined as SD for Cycle 1, as PR for Cycle 2, and as PD for the last cycle, the optimal overall response is determined as PR.) If the optimal overall response is determined as SD, the shortest time from baseline as specified in the Protocol has to be fulfilled. If there are no criteria available for reaching the shortest time, even if the optimal overall response is determined as SD, it is not approved; the optimal overall response of the patient will be determined during subsequent evaluations. Example: A patient is determined as SD for Cycle 1 and PD for Cycle 2 but fails to fulfill the required shortest time to SD, his/her optimal overall response shall be determined as PD. If the same patient is lost to follow up after having been determined as SD for Cycle 1 will be considered "non-evaluable".

When it is necessary to confirm complete or partial response in the study, the optimal overall response is assessed as follows: Complete or partial response may only be declared when each patient meets the criteria for partial or complete response specified for the trial and the efficacy is confirmed again at subsequent time points particularly mentioned in the Protocol (which usually occurs in four weeks). Under the said circumstance, see the descriptions in Table 3 for optimal overall response.

4.4.4 Particular Highlights Regarding Efficacy Assessment

When nodal lesions are included in the overall target lesion assessment and the size of the node is reduced to a "normal" one (<10 mm), there will still be a scan report on its size. In order to avoid circumstances from an over-estimated increase in the size of the node, even if a node is normal, its measured result will be recorded. Just as what has been described before, this means that even if for patients whose efficacy is complete response, it will not be recorded as 0 on the CRF.

If efficacy confirmation is required throughout the trial, repeated "unmeasurable" time points will complicate the optimal efficacy assessment. It has to be specified in the trial's analysis plan that data/assessments of these deficiencies, once the efficacy is defined, may be clearly explained. In most trials, for example, it is allowed to adopt a patient's PR-NE-PR response as having been confirmed for efficacy.

If a patient requests discontinued medication due to worsened overall health condition, yet there is no objective evidence to support the claim, it shall be reported as "symptomatic progression". Even if treatment has been terminated, efforts shall be made to assess objective progression. Assessment and description of symptomatic progression is not an objective response: It is the cause of discontinued treatment. The objective response of the patient will be assessed according to the target and non-target lesions shown in Tables 1 to 3.

Early progression, early death, and non-evaluable conditions by definition are exceptional in a study and shall be specified in each protocol (depending on the treatment interval and treatment cycle).

Under certain circumstances, it is relatively difficult to tell local lesions from normal tissues. When the assessment of complete response is based on such definition, we recommend that a biopsy be performed prior to the complete response efficacy assessment of a local lesion. When abnormal results shown through some of the images obtained on the patient's local lesion are considered to indicate lesion fibrosis or formation of scars, FDG-PET is used as an assessment criterion equivalent to a biopsy to confirm the efficacy of complete response. Under the said circumstance, prospective descriptions shall be provided in the Protocol regarding the application of FDG-PET. Meanwhile, reports provided in specialized medical publications on the said circumstance are to be used as support. It is required, however, to be aware of the fact that the restrictive nature of FDG-PET and the biopsy itself (including the distinctiveness and sensitivity of the two) will result in a false positive result during complete response assessment.

Table 1. Time point-based response: patients with target lesions (including or excluding non-target lesions)

Target lesion	Non-target lesion	New lesion	Overall response
CR	CR	No	CR
CR	Not CR/Not PD	No	PR
CR	Non-evaluable	No	PR

PR	Not progressive or not completely evaluable	No	PR
SD	Not progressive or not completely evaluable	No	SD
Not completely evaluable	Not progressive	No	NE
PD	Any condition	Yes or no	PD
Any condition	PD	Yes or no	PD
Any condition	Any condition	Yes	PD
CR= Complete response	PR= Partial response	SD= Stable disease	PD= Progressive disease NE= Non-evaluable

Table 2. Time point-based response - patients with only non-target lesions

Non-target lesion	New lesion	Overall response
CR	No	CR
Not CR or not PD	No	Not CR or not PD
Not completely evaluable	No	Non-evaluable
PD that cannot be defined	Yes or no	PD
Any condition	Yes	PD

Note: For a non-target lesion, "not CR/not PD" means that the efficacy is superior to SD. Due to the fact that it is getting more and more common that SD is adopted as the endpoint for evaluating efficacy, the not CR/not PD efficacy is created in order to deal with the situation where no lesion available for measurement is specified.

For undefined progression findings (such as very small uncertain new lesions or cystic change or necrotic change of original lesions), the treatment may be continued up to the next assessment. If disease progression is confirmed during the next assessment, the date of progression should be that when the suspicious progression was discovered previously.

Table 3 Optimal overall response to be confirmed for CR and PR

Overall response at the first time point	Overall response at subsequent time points	Optimal overall response
CR	CR	CR
CR	PR	SD, PD, or PR[a]
CR	SD	If SD continues for a sufficient period of time, it is SD; otherwise, it should be PD.
CR	PD	If SD continues for a sufficient period of time, it is SD; otherwise, it should be PD.
CR	NE	If SD continues for a sufficient period of time, it is SD; otherwise, it should be NE.
PR	CR	PR
PR	PR	PR

PR	SD	SD
PR	PD	If SD continues for a sufficient period of time, it is SD; otherwise, it should be PD.
PR	NE	If SD continues for a sufficient period of time, it is SD; otherwise, it should be NE.
NE	NE	NE

Note: CR is complete response; PR is partial response; SD is stable disease; PD is progressive disease; NE is non-evaluable. a: If CR truly appears at the first time point, any subsequent disease that appears at subsequent time points, even if the patient's efficacy reaches PR criteria as compared to the baseline, it shall be considered as PD as far as efficacy evaluation is concerned (because disease will reappear after CR). The optimal response is determined by whether SD appears at the shortest treatment interval. Sometimes, however, it is determined as CR during the first evaluation but small lesions seem to appear as indicated through scanned images at subsequent time points, which means that the patient's efficacy at the first time point should actually be PR, instead of CR. Under the said situation, the first CR determination should be revised as PR and the optimal response should be PR.

4.5 Frequency of Tumor Re-evaluation

The frequency of tumor re-evaluation during treatment is determined by the treatment solution and shall be consistent with the type of treatment and the schedule. In a Phase II trial where the treatment benefits are unclear, however, a follow-up visit every six to eight weeks (at the end of a cycle) is reasonable. For special protocols or under special circumstances, the length of the interval may be adjusted. It shall be specified in the Protocol which tissue sites need to be assessed at baseline (which are usually the most likely to be closely related to the metastatic disease of the type of tumor under investigation) and the frequency at which the evaluation should be repeated. Under normal circumstances, the target lesions and non-target lesions shall be evaluated during each assessment. Under certain optional circumstances, the frequency at which non-target lesions are evaluated may be reduced. A bone scan, for example, only needs to be repeated when the target disease is confirmed to have reached CR upon efficacy evaluation or bone lesion progression is suspected.

After the treatment is completed, re-evaluation of the tumor is determined by whether the response rate or the time to emergence of a certain event (progression/death) is adopted as a clinical trial endpoint. If it is time to a certain event (e.g. TTP/DFS/PFS), the routine repeated evaluations as required in the Protocol need to be performed. In a randomized comparative trial, in particular, the scheduled evaluation should be shown in the schedule (e.g. at six to eight weeks during treatment or at three to four months after treatment) and should not be affected by other factors, such as delay in treatment, the dosing interval, and any other time points chosen for disease evaluation that might result in the event of unbalanced treatment arms.

4.6 Efficacy Assessment/ Response Phase Confirmation

4.6.1 Confirmation

For a non-randomized clinical study where efficacy is the primary endpoint, the efficacy of PR and CR needs to be confirmed in order to guarantee that the efficacy is not the result of an evaluation error. This also allows rationalization of the results when historical data are available but the efficacy involved in the historical data of the said trials shall also be confirmed. Under all the other circumstances, however, such as a randomized trial (Phase II or III) or a study with stable disease or progressive disease as its primary endpoint, no efficacy confirmation is required as it is of no merits for the explanation of trial results. Canceling the requirement for

confirming the efficacy, on the other hand, makes site review to prevent against any deviation particularly important. This particularly true in a non-blind experimental study.

With SD, at least one measurement meets the SD criteria specified in the Protocol at the shortest interval after the trial begins (which is generally no less than six to eight weeks).

4.6.2 Overall Response Phase

The overall response phase is the time between measurement of the first compliant CR or PR (whichever occurs first) and the first truly recorded disease relapse or progression (the minimum measurement recorded in the trial is adopted as the reference for progressive disease). The overall complete response is the time from the first measurement of compliant CR to the first truly recorded disease relapse or progression.

4.6.3 Stable Disease Phase

It is the time from the start of treatment to disease progression (in a randomized trial, it begins with randomization) and the minimum sum in the trial is adopted as reference (if the sum at baseline is the smallest, it is used as the reference during PD calculation). The clinical correlation of the stable disease phase varies with the study and the disease. If in a certain trial, the proportion of patients maintaining the shortest stable phase is the endpoint, it shall be specified in the Protocol the minimum interval between two measurements in the definition of SD.

Caution: The response phase, stable phase, and PFS are affected by the frequency at which follow-up visits take place after the baseline evaluation. Defining the standard follow-up frequency is not part of these Guidelines. The follow-up frequency shall take into account many factors, such as the type and stage of disease, the treatment cycle, and the standard guidelines, among others. If it is necessary to compare trials, however, restrictions set by the accuracy of these measured endpoints shall be considered.

4.7 PFS/TTP

4.7.1 Phase II Clinical Trials

These Guidelines mainly focus on the application of objective response in Phase II clinical trials as an endpoint. Under certain circumstances, the response rate may not be the optimal choice for the evaluation of the potential anti-cancer activity of a new drug/new solution. When this happens, PFS/PPF at the borderline time point may be considered as a suitable alternative to provide source signals on the biological activity of the new drug. It is obvious, however, that in a non-controlled trial, these evaluations will be questioned as the seemingly valuable observations may have to do with biological factors such as the screening of patients, instead of the intervening action of the drug. Therefore, Phase II clinical trials with these as the endpoints are better designed to be randomized and controlled. Given the fact that the clinical expression of certain tumors are always consistent (and usually remain undesirable), however, a non-randomized trial also makes sense. Nevertheless, due to the lack of a positive control, efficacy evidence needs to be carefully documented during the assessment of the expected PFS or PPF.

The contents that follow are Phase IIII evaluation endpoints, independent evaluations, and result reports, etc. See the English version for details.

Appendix 3 EORTC QLQ-C30 (V3)

EORTC QLQ-C30 (V3)

We would like to learn about you and your health. Please answer all the questions below in person. There is no "right" or "wrong" about your answer. Simply circle the number that better reflects your condition as required. We will keep your data in strict confidentiality.

Please provide your name initials: _____

Date of birth: _____ YYYY_____MM_____DD

Today's date: _____ YYYY_____MM_____DD

	Not at all	A little	Somewhat	Extremely
1. Is it difficult for you to engage yourself in some strenuous activities such as carrying a very heavy shopping bag or suitcase? such as carrying a very heavy shopping bag or suitcase?	1	2	3	4
2. Is it difficult for you to walk a long distance?	1	2	3	4
3. Is it difficult for you to walk a short distance outdoors?	1	2	3	4
4. Do you need to stay in bed or in a chair during the day?	1	2	3	4
5. Do you need someone to help you when eating, getting dressed, taking a bath, using the bathroom?	1	2	3	4
Over the past one week,	**Not at all**	**A little**	**Somewhat**	**Extremely**
6. Were you restricted at work or during daily activities?	1	2	3	4
7. Were you restricted while engaging in your preferred or leisure activities?	1	2	3	4
8. Were you short of breath?	1	2	3	4
9. Were you in pain?	1	2	3	4
10. Did you need to take a rest?	1	2	3	4
11. Did you have difficulty falling asleep?	1	2	3	4
12. Did you feel weak?	1	2	3	4
13. Did you not feel like eating (lose your appetite)?	1	2	3	4

14. Did you experience nausea?	1	2	3	4
15. Did you vomit?	1	2	3	4
16.				
16. Did you suffer from constipation?	1	2	3	4
17. Did you have diarrhea?	1	2	3	4
18. Did you feel tired?	1	2	3	4
19. Did the pain affect your daily activities?	1	2	3	4
20. Was it difficult for you to stay focused, such as when reading the newspaper or watching TV?	1	2	3	4
21. Did you feel nervous?	1	2	3	4
22. Did you feel worried?	1	2	3	4
23. Did you feel that you had a bad temper?	1	2	3	4
24. Did you feel suppressed (depressed)?	1	2	3	4
25. Did you feel difficulty remembering things?	1	2	3	4
26. Did your physical condition or treatment affect your family life?	1	2	3	4
27. Did your physical condition or treatment affect your social events?	1	2	3	4
28. Did your physical condition or treatment lead to financial difficulties for you?	1	2	3	4

Select and circle between 1 and 7 for the following questions that suits you the best.

29. How do you rate your overall health condition for the past week?

1 2 3 4 5 6 7

Very bad Very good

30. How do you rate your overall quality of life for the past week?

1 2 3 4 5 6 7

Very bad Very good

Appendix 4 Drugs and food whose use is limited or prohibited

Drugs whose use is limited and traditional Chinese medicine and food that may not be used during the trial are as follows:

1. Potent CYP3A4 inhibitors and inducers

The following shows more commonly used potent CYPA34 inhibitors, including without limitation:

Ketoconazole
Itraconazole
Fluconazole
Voriconazole
Nefazodone
Indinavir
Nelfinavir
Saquinavir
Erythromycin
Clarithromycin
Telamycin
Grapefruit
Grapefruit juice

The following shows more commonly used potent CYPA34 inducers, including without limitation:

Rifabutin
Rifampicin
Rifapentine
Carbamazepine
Phenobarbital
Phenytoin
Hypericum perforatum

2. Traditional Chinese medicine: including all traditional Chinese medicine with anti-tumor indications

Regenerative tablets
Brucea javanica oil soft capsules
Zhemu Syrup
Cantharidin
Cinobufotalin
Kang'ai Injection
Kanglaite
Glabrous sarcandra herb for injection
Brucea javanica oil emulsion injection
Aidi injection
AweiHuapiGao
Anti-cancer flat pill
Rehabilitation capsule
Carcinoma Eliminating Injection
Pingxiao capsule
Pingxiao
Shendansanjie Capsule
Kanglixin Capsule
Ankangxin capsule
Bosheng Aining
Zedoary turmeric oil and glucose injection
Kanglixin Capsule
Cidan capsule

3. Food: Avoid consuming grapefruits, citruses, or pomelos or beverages containing the said fruits during the trial.

Appendix 5 Liver impairment clinical evaluation

According to the descriptions under Section 10.1.3, patients confirmed with ALT/AST ⩾ 3 × ULN combined with TBiL ⩾ 2 × ULN are the AESI defined in the Protocol and need to repeat blood biochemistry (including ALT, AST, ALP, TBiL, and DBiL) and the monitoring frequency needs to be increased in order to further describe the trends in changes to biochemistry indicators. In addition, it is necessary for the investigator to exclude other factors that result in abnormal indicators by inquiring about the patient's medical history, performing the physical examination, and related auxiliary tests, adopt corresponding treatment measures, and provide the report on AESI as required in the Protocol. If it meets the potential Hy's Law, the suspected medication must be discontinued.

- Common causes of liver impairment that need to be considered:
  - Acute viral hepatitis
  - Alcoholic or autoimmune hepatitis
  - Non-alcoholic steatohepatitis
  - Hypoxic-ischemic liver disease
  - Disease of biliary tract
  - Congestive right heart failure
  - New or progressive liver metastases

Other relatively uncommon causes also need to be considered.

It is advised that the investigator obtain the following information, evaluate it further, follow up, and complete clinical data:

- ♦ Medical history.
- ✧ Detailed history of current symptoms and diagnoses of complications and history of medical care
- ✧ Prior medical history (viral hepatitis, alcoholic hepatitis, autoimmune disease, disease of biliary tract, and cardiovascular disease, etc.)
- ✧ Concomitant medication history (including non-prescription and prescription drugs, herbal medicine, and dietary supplements), drinking, recreational drugs, and special diet
- ✧ History of exposure to chemicals
- ♦ The following laboratory tests shall be completed:

  ALT, AST, bilirubin (including TBiL and DBiL), ALP, INR

  - ✧ Serum:

    Hepatitis A (Anti-IgM, Anti-IgG), Hepatitis B (HbsAg, Anti-HBs, DNA), Hepatitis C (Anti-HCV and RNA upon a positive result), Hepatitis D (Anti-IgM, Anti-IgG), Hepatitis E (Anti-HEV, Anti-HEV IgM), anti-nuclear antibody, EB virus, cytomegalovirus

- Related auxiliary tests shall be completed:
  - It is advised that one abdominal ultrasound or other clinically applicable imaging tests be performed within 48 hours (to exclude biliary, pancreatic or intra-hepatic causes, such as gall stones or tumor); obtain the liver imaging results (to rule out new or progressive liver metastases) as soon as possible. If the cause of the abnormal liver test result cannot be determined based on the images (e.g. biliary, pancreatic, or intra-hepatic causes), it is advised that puncture pathologic tests be performed with prior consent from the patient.
  - In cases of suspicious cardiovascular causes, it is advised that echocardiography be performed to exclude cardiovascular dysfunction (including right heart failure).
- A gastroenterologist or hepatology specialist is to be invited for a consultation if necessary.

Long-term follow-up: ALT, AST, bilirubin (including TBiL and DBiL), and ALP are to be repeated at least once a week in order to closely monitor the patient's condition till the laboratory ALT/AST and bilirubin abnormalities stabilize or return to normal and then continue with the trial according to the Protocol. Report these data through the electronic case report form.

Appendix 6 Clinical Management of Severe or Serious Bleeding Event

If a bleeding event is assessed as severe (CTCAE Grade ≥ 3) or serious adverse event, fruquintinib therapy should be discontinued or suspended immediately, and appropriate measures should be taken to control bleeding (blood transfusion, radiotherapy, endoscopy, selective surgical intervention and other treatments should be carried out based on CTCAE grading) and necessary symptomatic treatment should given (see dosage adjustment of fruquintinib for bleeding). In addition, the patient must be closely follow-up, and laboratory indicators (INR, aPTT, platelet count, hemoglobin and blood pressure) must be monitored once every two to three days until the results recover to baseline or normal levels. During the preliminary evaluation, it is important to perform medical history gathering and physical examination, including the collection of vital signs and laboratory evaluation, to determine the time, site and severity of bleeding and whether bleeding is still occurring. Clinicians should pay attention to comorbidities and concomitant therapies (such as antiplatelet therapy, thrombocytopenia, or liver disease) that may cause bleeding or affect the treatment and management of bleeding, and manage them appropriately. If necessary, they should consider carrying out consultation with clinicians from other departments.

In addition, the investigator should closely monitor patients who are receiving antiplatelet and/or antithrombotic therapy during study drug treatment. If a CTCAE Grade ≥ 2 bleeding of any site occurs in such patients, the investigator should promptly decide whether to continue or discontinue the use of antiplatelet and/or antithrombotic therapy based on individual risk-benefit assessment results.If severe (CTCAE Grade ≥ 3) or serious bleeding event occurs after taking fruquintinib, the investigator should submit a quick report to the sponsor (see Sections 10.1.3 and 10.3.3).

For all the above management processes, refer to the following **management flowchart**.

Data Collection Recommendations for Severe or Serious Bleeding Event

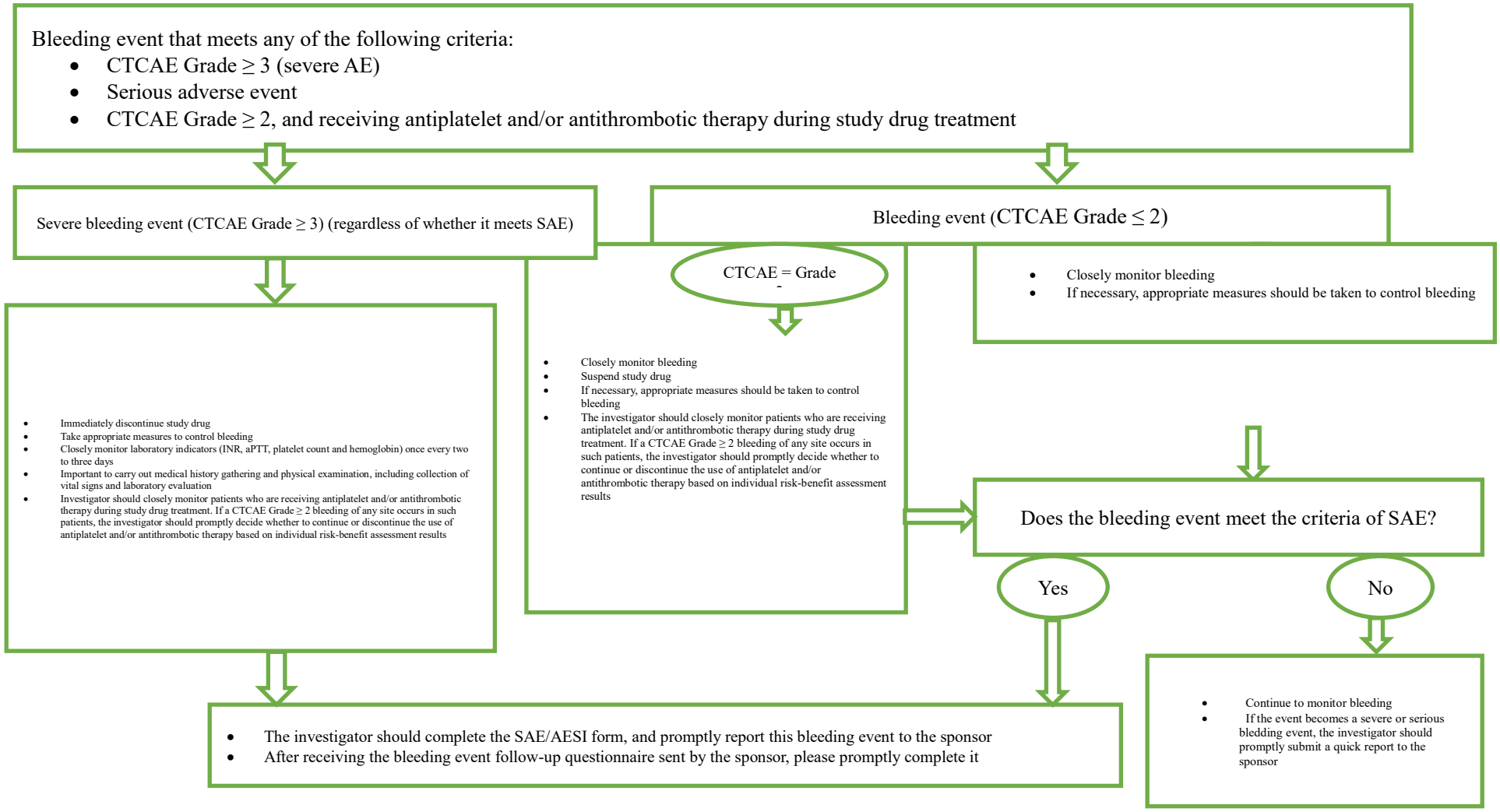
In the event of severe or severe bleeding, the investigator is recommended to obtain the following information for further evaluation and follow-up to complete clinical data. All the data should be recorded in the SAE/AESI report form, and supplemented by a bleeding event follow-up questionnaire:

- Patient's medical history
  - Details of history of present illness and diagnosis of complications and medical history
  - Past medical history
  - Concomitant medications
    - Vitamin K antagonists (such as warfarin)
    - Non-steroidal anti-inflammatory drugs (such as aspirin)
    - Antiplatelet agents (such as clopidogrel/glycoprotein GPIIb/IIIa inhibitor/dipyridamole)
    - Other anticoagulants (such as heparin/thrombolytic/SSRIs)
    - Foods and herbal medicines containing anticoagulant ingredients
    - Immunosuppressive agents
    - Drinking
    - Use of illegal drugs and special diets
  - Family history of bleeding events
- Complete the following laboratory tests:
  - Hematology (hemoglobin, platelets, hematocrit)

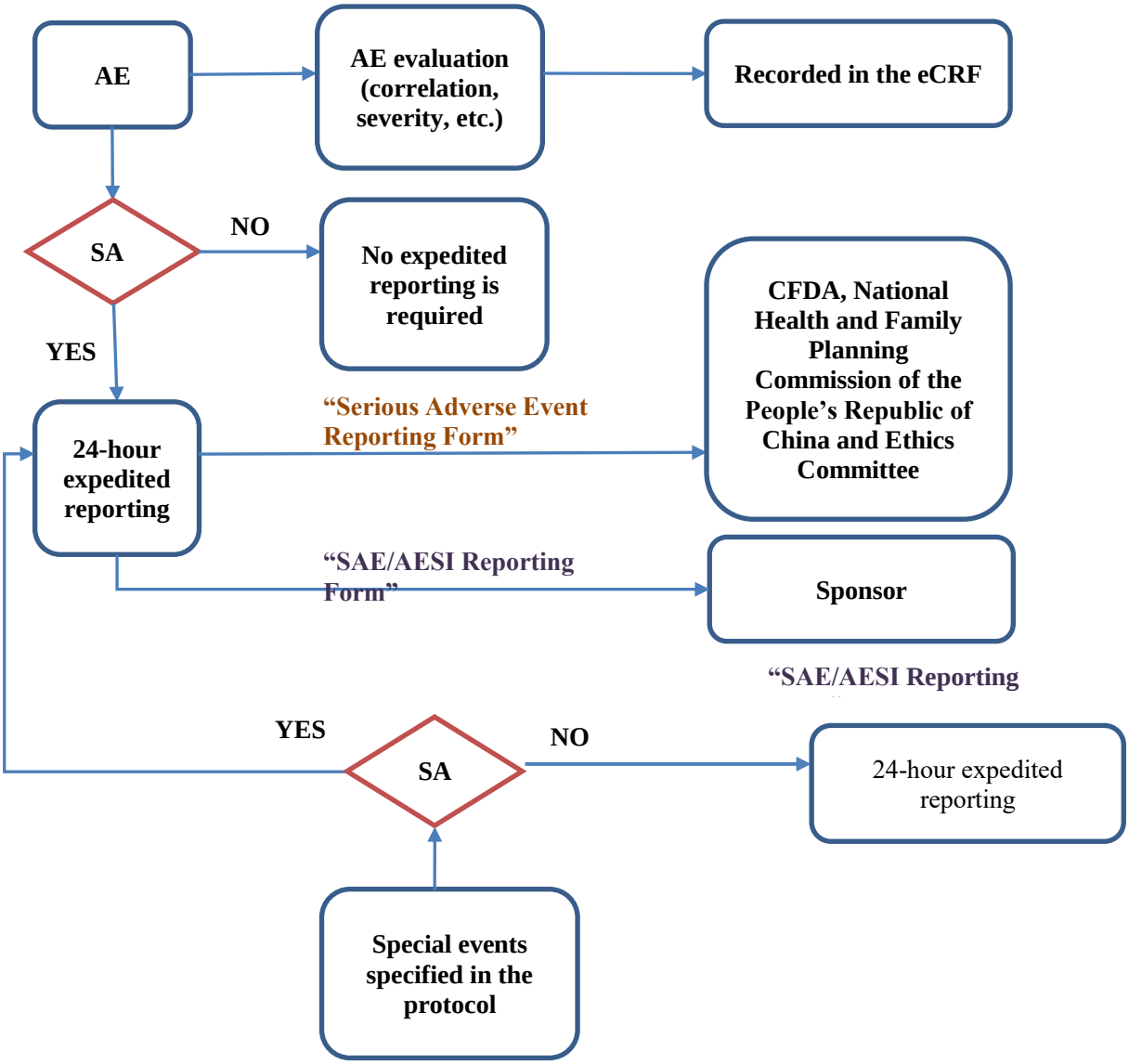
- Clinical biochemistry test (duration of bleeding, PT, aPTT, INR)
- Other tests: stool and urine occult blood tests should be conducted whenever necessary

- Complete appropriate auxiliary examinations:
  - In order to determine the site of bleeding, the patient may need to undergo gastroscopy, bronchoscopy, or clinically needed imaging examination within 48 hours
  - Echocardiography is recommended to rule out cardiovascular dysfunction (such as right heart failure) if cardiovascular disease is suspected

Severe or Serious Bleeding Event Management Flow Chart



Appendix 7 Overview of the safety data reporting procedure



STUDY STATISTICAL ANALYSIS PLAN

2017-013-00CH1

A RANDOMIZED, DOUBLE-BLIND, PHASE III TRIAL TO EVALUATE THE EFFICACY AND SAFETY OF FRUQUINTINIB IN COMBINATION WITH PACLITAXEL FOR THE TREATMENT OF PATIENTS WITH ADVANCED GASTRIC OR GASTROESOPHAGEAL JUNCTION (GEJ) ADENOCARCINOMA

AUTHOR: [redacted]

VERSION NUMBER AND DATE: V3.0,31OCT2022

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STUDY STATISTICAL ANALYSIS PLAN SIGNATURE PAGE

Study Statistical Analysis Plan V3.0 (Dated 31ct2022) for Protocol 2017-013-00CH1 v6.0 (26Aug2021).

	Name	Signature	Date
Author:	[redacted]		
Position:	Biostatistician		
Company:	IQVIA RDS (Shanghai) Co., Ltd		

Upon review of this document, the undersigned approves this version of the Statistical Analysis Plan, authorizing that the content is acceptable for the reporting of this study.

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LIST OF ABBREVIATIONS

Terms and Abbreviations	Definition
AE	Adverse Event
AESI	Adverse Events of Special Interest
ALP	Alkaline Phosphatase
ALT	Alanine Aminotransferase
AST	Aspartate Aminotransferase
CFB	Change from Baseline
CR	Complete Response
eCRF	Electronic Case Report Form
DCR	Disease Control Rate
DoR	Duration of Response
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
ECOG PS	Eastern Cooperative Oncology Group Performance Status
IDMC	Independent Data Monitoring Committee
GEJ	Gastro-Esophageal Junction
ITT	Intention-To-Treat Set
LVEF	Left Ventricular Ejection Fraction
MedDRA	Medical Dictionary for Regulatory Activities
NCI-CTC	National Cancer Institute Common Toxicity Criteria
NCI CTCAE	National Cancer Institute Common Terminology Criteria for Adverse Events
ORR	Objective Response Rate
OS	Overall Survival
PD	Progressive Disease
PFS	Progression Free Survival
PPS	Per-Protocol Set
PR	Partial Response

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QD	Once A Day
RECIST	Response Evaluation Criteria in Solid Tumors
SAE	Serious Adverse Event
SAF	Safety Analysis Set
SAP	Statistical Analysis Plan
SD	Stable Disease
SMQ	Standardised MedDRA Queries
ULN	Upper Limit of Normal
WHO	World Health Organization
WHO-DD	World Health Organization Drug Dictionary

Note: Standard units such as mg and mL as used in the text are not included in the abbreviations list.

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1. INTRODUCTION

This document describes the rules and conventions to be used in the presentation and analysis of efficacy and safety data for Protocol 2017-013-00CH1. It describes the data to be summarized and analysed, including specifics of the statistical analyses to be performed. It will be used for both interim analysis and final analysis.

This study statistical analysis plan (SAP) is based on protocol version 6.0, 26Aug2021(CN).

2. STUDY DESIGN

2.1. GENERAL DESCRIPTION

This is a randomized, double-blind, placebo controlled, multi-center phase III clinical study to evaluate the efficacy and safety of Fruquintinib + Paclitaxel vs. placebo + Paclitaxel for second-line treatment in subjects with advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma. Approximately 700 subjects will be randomly assigned to the following treatment groups in a 1: 1 ratio:

- Treatment group: Fruquintinib+ Paclitaxel
  Subjects will receive Fruquintinib 4 mg/time QD, orally administered for 3 weeks and followed by one-week off, plus Paclitaxel injection 80 mg/m^2 once weekly for 3 continuous weeks followed by one-week off, till disease progression, death, unacceptable toxicity or voluntary withdrawal.

- Control group: Placebo+ Paclitaxel
  Subjects will receive placebo 4 mg/time QD, orally administered for 3 weeks and followed by one-week break, plus Paclitaxel injection 80 mg/m^2 once weekly for 3 continuous weeks followed by one-week off, till disease progression, death, unacceptable toxicity or voluntary withdrawal.

The study is divided into 3 periods: screening period, treatment period and follow-up period, from the start of treatment to death, loss of follow-up, refusal of survival follow-up or end of the entire study.

All subjects will receive fruquintinib (or placebo) plus Paclitaxel injection on a 3 weeks on/1 week off regimen. In a 28-day cycle, Paclitaxel is administered at DAY1, DAY 8 and DAY 15 respectively. Paclitaxel will be administered for at most 6 cycles of treatment and then, if no disease progression occurs, patient will receive fruquintinib (or placebo) monotherapy till disease progression or unacceptable toxicity.

Approximately 700 subjects will be enrolled in this study. The final analysis will be conducted when at least 525 cases of death are observed in this study.

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2.2. SCHEDULE OF EVENTS

Schedule of events can be found in protocol Section 2: Summary of clinical study and Study Flow Chart of the protocol.

3. PLANNED ANALYSES

3.1. INDEPENDENT DATA MONITORING COMMITTEE

Safety monitoring will be conducted by independent data monitoring committee (IDMC). The 1st Independent Data Monitoring Committee (IDMC) data review meeting will occur at approximately 1 month post the first 100 subjects enrolled. The 2nd IDMC data review meeting (the first interim analysis) will occur at approximately 1.5 months post the first 100 subjects completed the 6 months follow-up. The last IDMC data review meeting (the second interim analysis) will occur at about 201 OS events have been observed. The actual timing for the IDMC meetings will be taking into account of recruitment speed and the need of safety monitoring. There will be a total of four IDMC meetings if including the kick-off meeting.

The IDMC will review accumulating data on an ongoing basis. The statistical analyses will consider all available data collected from screening until the cut-off date which will be defined for each IDMC meeting.

For the planned first IDMC data review meeting, the IDMC members will not be charged with the application of formal statistical stopping rules. The recommendations of the IDMC will be based on the members' clinical assessment of the cumulative safety data provided for review.

For the second IDMC data review (the first interim analysis), a futility rule will be defined by sponsor prior to the second data review meeting. Please refer to IDMC charter for details.

For the third IDMC data review meeting (the second interim analysis), IDMC members will review the results and make a recommendation of stopping the study for futility or efficacy or for serious safety concern or continue the study with or without sample size adjustment. Please refer to IDMC charter for details.

In addition to these planned data review meetings, the Sponsor or IDMC can call for an ad hoc meeting, when necessary.

3.2. INTERIM ANALYSIS

The first interim analysis will be conducted once the 100th randomized subject has either been treated for 6 months, died or have prematurely discontinued trial treatment for any reason, whichever comes first. This SAP covers the analyses for efficacy and safety based on the data cut-off for the first interim analysis. The first Interim analysis will be performed using cleaned eCRF data and external data of the first 100 randomized subjects collected until data cut-off date. In addition, SAEs will be summarized based on all subjects recruited into study on or before data cut-off date. All SAE data to be included in the first interim analysis should be cleaned.

There will be two sets of analyses for the first interim analysis, i.e. one for 100 randomized subjects (primary set

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for all data analysis) and another one for all randomized subjects until the cut-off date for SAE only as supplemental analysis.

With the small sample size in first interim analysis, summary to overall survival (OS) and progression free survival (PFS) will be performed with unstratified analysis.

The second interim analysis will be conducted around 201 OS events observed.

For the second interim analysis, overall survival (OS) and progression free survival (PFS) will be performed with stratified analysis. If the interim analysis meets the criteria of early termination of the study, sponsor may accept IDMC's recommendation to early terminate the study. Otherwise, the study will continue with or without sample size adjustment.

3.3. FINAL ANALYSES

The final analysis of OS and PFS will be conducted when approximately 700 subjects enrolled and at least 525 deaths occurred. Statistical analyses of all efficacy and safety data will be performed using cleaned eCRF data and other external data.

3.4. CHANGES TO ANALYSIS FROM PROTOCOL

The following analyses were changed compared with the study protocol:

- The all subjects provided ICF set was added.
- AESI used term 'Haemorrhage' instead of 'bleeding events' in protocol.
- Two different safety analysis sets will be applied for the first interim analysis, i.e. the 100 randomized subjects who have ever received study medication and all randomized subjects who have ever received study medication until the cut-off date. The safety-supplement analysis set was added. Refer to section 3.2 and 4.3 for details.
- Subgroups were added for sensitivity analysis. Refer to Section 4.5 for details.
- Tumor shrinkage: Waterfall plot for best percentage change in target lesions. Refer to Section 14.3.1.4 for details.

4. ANALYSIS SETS AND SUBGROUPS

Agreement and authorization of subjects included/excluded from each analysis set will be conducted at Blinded Data Review Meeting to be held prior to database lock.

4.1. ALL SUBJECTS PROVIDED ICF SET

The all subjects provided ICF set will contain all subjects who provide informed consent for this study.

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4.2. INTENTION-TO-TREAT SET [ITT]

Intention-To-Treat Set (ITT)

The ITT analysis set will be the primary analysis set for the efficacy endpoints. The ITT set will include all randomized subjects. Following the intention-to-treat principle, subject data will be analysed according to randomized treatment group. Demographic data and other baseline characteristics will also be summarized with the analysis set of ITT.

For the first interim analysis, the ITT set will include only the first 100 randomized subjects.

4.3. SAFETY ANALYSIS SET [SAF] AND SAFETY-SUPPLEMENT ANALYSIS SET [SAF-S]

Safety Analysis Set (SAF)

The safety analysis set (SAF) will include all subjects in the ITT set who received at least one administration of study medication (including Fruquintinib/Placebo or Paclitaxel) and subjects will be classified according to the treatment actually received. The SAF set will be used for analysis of safety endpoints.

If there is any doubt whether a subject was treated or not, they will be assumed treated for the purposes of analysis.

If a patient has only received Paclitaxel and no record of Fruquintinib/Placebo, the patient will be included in the control group for safety analyses.

If a patient received both wrong study treatment and correct study treatment, the patient will be classified to the group of Fruquintinib +Paclitaxel for summary as long as this patient treated Fruquintinib. If a patient received wrong study treatment during the whole study period, the patient will be classified to the treatment actually received.

.

For the primary analysis of safety data at the first interim analysis, the SAF set will include only the first 100 randomized subjects who received at least one administration of study medication.

Safety-Supplement Analysis Set (SAF-s)

SAF-s will be used for the first interim analysis only. All subjects until the cut-off date for the first interim analysis will be included in the SAF-s set, as long as they were randomized and received at least one administration of study medication. The SAF-s set will be used supplemental SAE summary. Subjects will be classified according to the treatment actually received.

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4.4. PER-PROTOCOL SET [PPS]

Per-Protocol Set (PPS)

The per-protocol set (PPS) will be used for sensitivity analyses of the primary efficacy endpoint (OS and PFS). The PPS set is a subset of the ITT set, however subjects who have any major protocol deviations that may affect the efficacy evaluation or subjects who randomized but not treated will be excluded from the PPS set. Refer to section 9 for the details.

4.5. SUBGROUPS

Below subgroup analyses of OS and PFS will be performed based on baseline factors as below:

- Age: < 65 years vs. ≥ 65 years (reference level);
- Gender: male vs. female (reference level);
- BMI: <18.5, >=18.5 and <24.0,>=24.0(reference level);
- Baseline Eastern Cooperative Oncology Group Performance Status (ECOG PS): 0 vs 1 (reference level);
- Primary tumor location: GEJ vs stomach (reference level);
- Time since 1st distal metastasis (or local progression) to randomization: <6 months vs. ≥ 6 months (reference level);
- Time since initial diagnosis to randomization: <12 months vs. ≥ 12 months (reference level);
- TNM Stage of disease at the time of initial diagnosis: Stage 0-III and unknown vs. Stage IV (reference level);
- Pathologic Lauren classification: Intestinal type vs. Diffuse type vs. Mixed type or Unknown or not evaluable or Other (reference level);
- Tumor grade at initial diagnosis: Gx/G1/G2 vs. G3/G4/Unknown (reference level);
- Number of metastatic site: Single vs. Multiple (reference level);
- Multiple metastatic site: 2 sites vs. ≥ 3 sites (reference level);
- Metastasis with liver: No vs. Yes (reference level);
- Metastasis with peritoneal: No vs. Yes (reference level);
- Metastasis with lymph node: No vs. Yes (reference level);
- Metastasis with lung: No vs. Yes (reference level);
- Prior gastrectomy: No vs. Yes (reference level);
- Primary therapy class: chemotherapy vs immunotherapy plus chemotherapy (reference level);
- Chemotherapy: doublet vs triplet (reference level).

All above are captured from CRF "Demographics", "ECOG score", "Primary Diagnosis", "Chemotherapy and Medication history related with primary disease", "Surgery history related with primary disease" pages.

In order to assess the impact of patients treated with VEGF/VEGFR as subsequent anti-tumor therapy, OS subgroup analysis based on treated VEGF/VEGFR as subsequent anti-tumor therapy (Yes or No) will be done.

It should be noted that the study is not designed to detect treatment differences with high statistical power within subgroups. If a subgroup is too small, it may be pooled with others. If the number of events in a subgroup is not sufficient (less than 10 in a subgroup), analysis will not be performed.

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For variables with more than two categories, an indicator variable will be defined for each category except for the first category, which defines the reference always.

5. General Considerations

5.1. Reference Start Date and End Date and Study Day

Generally, reference start date is defined as the date of the first administration of study treatment (first administration date) which will be reported on the electronic Case Report Form (eCRF) 'Study Medication - fruquintinib or placebo' and 'Study Medication – Paclitaxel' form. Day 1 is the day of the first dose of study treatment in Cycle 1. However, for tumor assessment and efficacy analysis, reference start date is defined as the date of randomization being recorded on eCRF "Randomization" form.

Reference end date is defined as the day of the last administration of study treatment (fruquintinib/ placebo and Paclitaxel) within the subject's last treatment cycle. In the situation where the reference end date is partial or missing, please refer to section 6.1 for imputation of missing last dose.

Study day will be calculated from the reference start date, and it will be used to show start/stop day of assessments and events relative to the first administration of study treatment.

- If the date of the event is on or after the reference start date then:
  - Study day = (date of event – reference start date) + 1.
- If the date of the event is prior to the reference start date then:
  - Study day = (date of event – reference start date).

In the situation where the event date is partial or missing, the date will appear partial or missing in the listings. Study day, and any corresponding durations will be presented based on the imputations specified in section 6.1: Handling of Missing Data.

In addition to the overall reference start and end date, the start and end of treatment within each cycle will be identified based on the data collected on the eCRF form and will be used when assigning events or assessments to a particular treatment cycle.

5.2. Baseline

Unless otherwise specified, baseline is defined as the last non-missing measurement taken within 21 days prior to the reference start date (including unscheduled assessments). In the case where the last non-missing measurement and the reference start date coincide, that measurement will be considered baseline; but Adverse Events (AEs) and medications commencing on the reference start date will be considered post-baseline.

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5.3. ON TREATMENT PERIOD AND WORST POST BASELINE

For the analysis purpose, on treatment period is defined as the period start on the first administration date to 30(+3) days after last administration date or initiate new anti-tumor therapy (whichever occurs first). For on treatment AE and on treatment death, as the collection is not by visit, time window of 3 days for visit will not be considered.

The worst post baseline is defined as the worst cases during on treatment period including both schedule and unscheduled visit.

5.4. END OF TREATMENT MEASUREMENTS

The latest end of treatment date among the two dates from 'End of Treatment- fruquintinib or placebo' and 'End of Treatment- Paclitaxel' will be defined as the EOT date for identifying subsequent anti-tumor therapy.

5.5. LAST KNOWN ALIVE DATE

The last known alive date will be derived for subjects not known to have died at the analysis cut-off date using the latest date among the following data:

- All assessment dates (e.g. QLQ C-30, laboratory, vital signs assessments, ECG, ECOG, tumor assessments).
- Medication dates including study medication, concomitant medications, anticancer therapies administered after study treatment discontinuation.
- Adverse event start and end dates, and date became serious.
- Last known alive date collected on the eCRF 'Survival Status' form or 'End of Study' form (do not use date of survival follow-up assessment or date of last contact unless status is 'alive') or data cut-off date (if still alive).
- Randomization date.
- Last visit date if subjects are not end of treatment from study treatment regimen.

The last known alive date will be used for censoring of subjects in the analysis of overall survival.

Partially missing date of above data will be imputed as below:

- Missing day only: using the 1st of the month
- Missing day and month: using the 1st January of the year

5.6. RETESTS, UNSCHEDULED VISITS AND EARLY TERMINATION DATA

In general, for by-visit summaries, data recorded at the nominal visit will be presented. Unscheduled measurements will not be included in by-visit summaries, but they will contribute to the best/worst case value where required (e.g. shift table).

In the case of a retest (same visit number assigned), the latest available measurement for that visit will be used for

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by-visit summaries.

For any subject early withdrew from treatment, the last assessments post the end of treatment will be recorded at the "End of Treatment" visit and hence will be summarized.

Listings will include scheduled, unscheduled, retest and early withdrawal data.

5.7. COMMON CALCULATIONS

For quantitative measurements, change from Baseline will be calculated as:

- Change from baseline= Test Value at Visit X – Baseline Value

Percentage change from baseline will be calculated as:

- Percentage change from Baseline= (Test Value at Visit X – Baseline Value) / Baseline Value × 100%

5.8. SOFTWARE VERSION

All analyses will be conducted using SAS version 9.4 or higher version.

5.9. DATA INCLUDED IN THE ANALYSIS

Only data with an assessment date or event start date (e.g. vital sign assessment date or start date of an adverse event) prior to or on the cut-off date will be included in the analysis. (Example: If cut-off date is 28FEB2018 then an AE starting on 13FEB2018 will be included in the analysis, whereas an AE with start date on 01MAR2018 will not be included in the analysis.)
All events with a start date before or on the cut-off date and with an end date after the cut-off date will be reported as "ongoing at the cut-off date''. The same rule will be applied to events starting before or on the cut-off date and not having documented end date. This approach applies, in particular, to adverse event and concomitant medication reports.

6. STATISTICAL CONSIDERATIONS

6.1. HANDLING OF MISSING DATA

In all other cases missing or incomplete dates will not be imputed if they are not indicated otherwise.

Adverse Event Grading

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If the NCI CTC grade of an adverse event is still missing after query by data manager, it will be imputed to grade 3 for AE summary purpose. However, it will be displayed as missing in the listing.

Adverse Event Causality

For any reasonable possibility that the events related to Fruquintinib/Placebo or paclitaxel missing, and it's treatment emergent AE, it will be imputed as related to TEAE. However, it will be displayed as missing in the listing.

Adverse Event Date

In case the start date of an AE is completely missing or the start date is in the same year as reference start date (when only year is recorded), or the start date is in the same month and year (if only the day is missing) as the reference start date, then the start date of the AE will be imputed by the minimum of the reference start date and the AE resolution date. In all other cases the missing start day will be imputed by the 1st and the missing start month will be replaced by January.

In case only the month is missing while year and day are recorded, then the start month of AE will be imputed by the month of reference start date and the AE resolution month which is earlier, the start month of SAE will be imputed by the month of informed consent date and the AE resolution month which is earlier. If the imputed date is later than the date based on which it was imputed, then the imputed month shall be adjusted by imputing with 1 month earlier.

Incomplete stop dates will be imputed by the last day of the month (if only the day is missing), and the missing stop month will be replaced by December (if both day and month are missing) or the latest possible month (if only month is missing), if not resulting in a date later than the date of subject's death. In the latter case the date of death will be used to impute the incomplete stop date. In all other cases the incomplete stop date will not be imputed.

Medication/Procedure Date

There will be no imputation for complete missing start date or stop date.

Impute partial start date as earliest possible date, i.e. first day of month if day unknown or 1st January if day and month are unknown.

Impute stop date as latest possible date, i.e. last day of month if day unknown or 31st December if day and month are unknown, if not resulting in a date later than the date of subject's death. In the latter case the date of last known alive date will be used to impute the incomplete stop date.

Above rules will be applied for prior and concomitant medications/procedures, prior and concurrent oncology therapies.

Duration will not be calculated in case either the start date or end date is completely missing, but it will be calculated based on the imputed start/end date if either the start date or end date is partial missing.

Oncology History Dates

If year is missing, do not impute; if year and month are known but the day is unknown, day will be imputed as 15. If only year is known, month and day would be imputed as July 1st. The imputed date will be compared to inform consent date. If it is later than the inform consent date, the incomplete date will be imputed as the inform consent

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date.

Efficacy Analyses Date
Missing efficacy data will be handled as described in section 14.

Death Date

- If year and month of death date are known but the day is unknown, day will be imputed as 15 (for example, if a patient is reported to die on Dec 2017, the death date will be imputed as 15 Dec 2017).
- If only year are known, death date will be imputed as 1st Jul of the given year.
- If the imputed death date is prior to the last known alive date, then the death date will be imputed as last known alive date+1.
- If death date is totally missing, dates will be imputed as the last known alive date + 1.

if the imputed death date is after cut-off date, then the death date will be imputed as cut-off date.

Anti-Tumor Therapy Date
When a partial new anti-tumor therapy start date is reported, every effort will be made to identify the precedence relationship of starting date of new anti-tumor therapy relative to the reference end date.

Below rules will be used:

- If the date is completely missing, then impute the date as reference end date+1.
- If only the day is missing:
  In case of the year and month are same as reference end date, then impute the day on the 15th of the given month. If the imputed date is before the reference end date, then re-impute the day to the day after reference end date. If the imputed date is after the reference end date, then keep the day as the 15th of the given month. Otherwise, if the year or month is not the same as reference end date, impute on the 15th of the given month.
- If both month and day are missing:
  In case of the year is same as reference end date, then impute as the 1st Jul of the given year. If the imputed date is before the reference end date, then re-impute the date as reference end date+1. If the imputed date is after the reference end date, then keep the date as the 1st Jul of the given year.

Then compare the imputed date to last known alive date or death date. If the imputed date is after last known alive date or death date, then re-imputed to last known alive date or (death date-1).

Concurrent Therapy Date
When a partial concurrent oncology chemotherapy/radiotherapy start date is reported, the start date will be imputed to the earliest possible date on or after randomization. if the start date is completely missing, then impute the date with earlier date among reference end date, disease progression date and end date of the therapy (if known). When a partial concurrent oncology surgery and procedure date, the date will be imputed to the earliest possible date on or after randomization. If the start date is completely missing, then impute the date with earlier date between reference end date and disease progression date.

Last Dose Date of Study Medication

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- If the last dose date in the last cycle of the study medication is partially missing, then impute the date as the first dose date in the last cycle+21. Compare the imputed date with the original last dose date, if it is after the original date based on the known part, then it will be replaced with the last day of the given part of original last dose date.
- If the last dose date in the last cycle of the study medication is completely missing, then impute the date as the first dose date in the last cycle+21.

If the imputed date is after cut-off date or EOT date, then it will be replaced with the cut-off date or EOT date.

6.2. MULTICENTRES

For this study, treatments were assigned to subjects using a central randomization independent of centres.

For the first interim analysis, it is anticipated that subject accrual will be sparse across centres and hence it is not necessary to include centre effect in statistical modelling for the first interim analysis.

For the secondary interim analysis and final analysis of OS and PFS, to evaluate the robustness of primary analysis of OS and PFS, centre as a random effect will be included in the model for sensitivity analyses. If deemed necessary, small centers may be combined with the other centers.

6.3. ADJUSTMENTS FOR COVARIATES AND FACTORS TO BE INCLUDED IN ANALYSES

The following factors are used as stratified factors for randomization.

- Primary site of the tumor (gastroesophageal junction vs stomach)
- ECOG PS score (0 vs 1)
- Peritoneal metastasis (present vs absent)

The comparison of OS and PFS of fruquintinib+Paclitaxel group with placebo+Paclitaxel group by stratified log-rank test, as well as the estimation of Hazard Ratio (HR) by stratified Cox PH model, and the comparison of ORR and DCR of fruquintinib+Paclitaxel group with placebo+Paclitaxel group by Cochran-Mantel-Haenszel test will involve the above three stratification factors.

All stratified efficacy analyses will be performed based on data collected from eCRF. A sensitivity analysis of OS and PFS will be performed based on strata collected through IWRS as stratification factors at randomization. The concordance and discordance between the actual levels and the assigned levels at randomization will be summarized in frequency and percentage by treatment group and also listed by treatment group and subject.

For the first interim analysis based on the first 100 randomized subjects, stratification factors will not be considered in the statistical testing due to the small sample size.

For the second interim analysis and final analysis of OS and PFS, stratification factors will be considered in the statistical testing. If a strata is too small, it may be pooled with others.

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6.4. MULTIPLE COMPARISON/MULTIPLICITY

The overall Type I error rate is two-sided 0.05 for the dual primary analysis of OS and PFS. All the other analyses will be supportive. No formal adjustment for multiplicity will be undertaken for sensitivity or subgroup analyses involving the primary endpoint, analyses of secondary endpoints, or analyses involving exploratory endpoints.

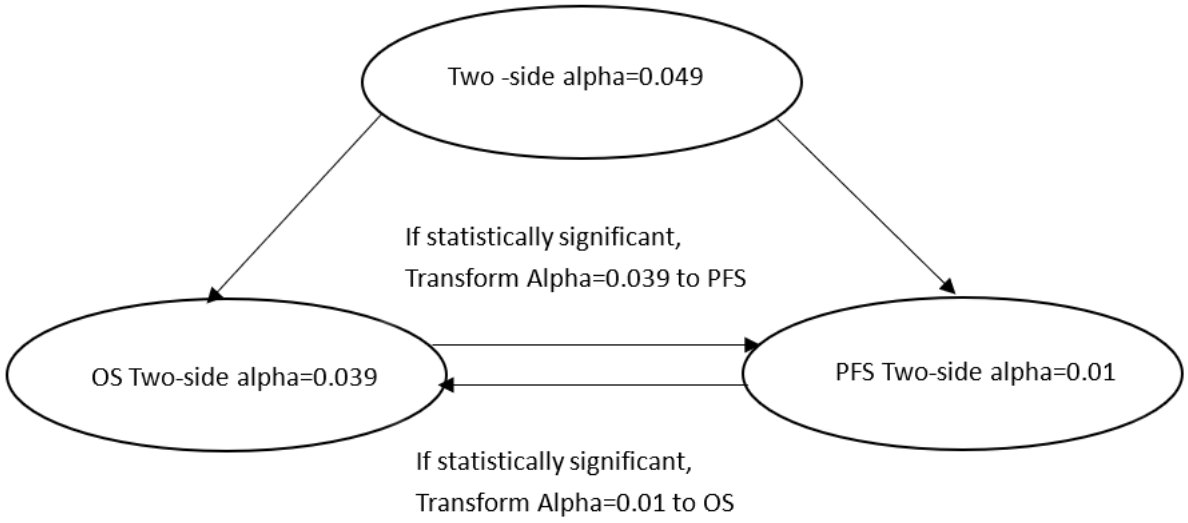
Although the first interim analysis will be performed, there is no intention to stop the trial earlier for efficacy if the superiority of the treatment under study is clearly established. No alpha will be spent. Instead, beta will be spent due to potential early stop for futility, leading to loss of power. This is sponsor's risk rather than subjects' risk.

In the secondary interim analysis, stop for efficacy will be considered. O'Brien-Fleming (Lan-DeMets procedure) method will be used to control type I error. If the secondary interim analysis happened at 201 events, the alpha spending is 0.003 (two sided).

In the final analysis, graphical method will be used (alpha distribution and recycling, Figure 1) to control type I error for the final analysis of the dual primary endpoints at two-side 0.049 level (considering the consumption of OS in the secondary interim analysis).

- ✓ The remaining total two sided significance level 0.049 will be split into two parts, 0.01 for PFS test (two-sided) and 0.039 for OS test (two-sided).
- ✓ If it's statistically significant for PFS analysis, the two-sided level 0.01 of PFS will be recycled and transformed to OS, OS will be tested on the two-sided 0.049 level.
- ✓ If it's statistically significant for OS analysis, the two-sided level 0.039 of OS will be recycled and transformed to PFS, PFS will be tested on the two-sided 0.049 level.

Figure 1: Distribution and recycling of alpha in the final analysis



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7. OUTPUT PRESENTATIONS

Appendix 1 details the conventions for presentation of data in outputs.

The templates provided with this SAP describe the presentations for this study and therefore the format and content of the summary tables, figures and listings to be provided by IQVIA Biostatistics.

Continuous variables will be summarized using descriptive statistics, i.e., number of subjects (n), mean, median, standard deviation (Std), 25th and 75th percentiles (Q1, Q3), minimum and maximum.

Qualitative variables will be summarized by count(s) and percentages (%). Unless otherwise stated, the calculation of percentages will either be based on the total number of subjects in the relevant analysis set with data available or relative to the total number of subjects included in the relevant analysis set. In the case of missing data, a category of "Missing" shall be added to summaries, so that the total percentage would sum up to 100%.

8. DISPOSITION AND WITHDRAWALS

All subjects who provide informed consent will be accounted for in this study.

Subject disposition, withdrawals as well as withdrawal reasons will be presented for the all subjects provided ICF set.

The number/percentage of subjects included/excluded in all subjects provided ICF set, ITT set, SAF set, as well as exclusion reasons will be summarized and listed for all subjects provided ICF set. The break down to each analysis set will be presented at subject level as well.

9. PROTOCOL DEVIATIONS

Apart from the directly programmatic checks from the information collected on the study database, the study team will discuss, identify and classify eligibility criteria or post-entry deviations from Clinical Trial Management System (CTMS) system monitoring reports prior to database lock. The study team will also discuss and identify protocol deviations due to COVID-19.

Subjects who experience any of the following major protocol deviation criteria or who randomized but not treated will not be included in the PPS set:

- Deviation from inclusion criteria 1-3,5-6 and 8
- Deviation from exclusion criteria 1, 2, 4, 5, 7, 18, 23-24, 26-28.
- Subjects who were treated with the incorrect study treatment versus as randomized (case by case, depending on the duration of receiving the incorrect study treatment)
- Subjects who received additional concurrent antineoplastic therapy during the study treatment period

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(case by case discussion)

All subjects with major protocol deviation (except for subjects with incorrect study medication) leading to exclusion from the PPS will be identified and confirmed at data review meeting prior to database lock no matter whether the protocol deviation is related to COVID-19 or not.

All major protocol deviations including those won't lead to exclusion of subjects from the PPS will be summarized by each category for ITT set. Protocol deviations related to COVID-19 will be flagged as well and summarized. One listing to include only major protocol deviations and one listing to include all protocol deviations will be generated.

10. DEMOGRAPHIC AND OTHER BASELINE CHARACTERISTICS

Demographic data and other baseline characteristics will be summarized for the ITT set.

No statistical testing will be carried out for demographic or other baseline characteristics.

The following demographic and other baseline characteristics will be reported for this study:

- Age (years) - Calculated relative to date of informed consent; <65 vs. ≥65
- Gender (male, female)
- Child Bearing Potential (for female subjects only)
- Race (white, black or Afro-American, Asian, American Indian or Alaska Native, Native Hawaiian or other Pacific Islander, Other)
- Baseline ECOG PS (0 vs 1)
- Baseline Weight (Kg)
- Height (cm)
- Baseline BMI (kg/m^2), categorized into <18.5, ≥18.5 and <24, ≥24
- Baseline Body Surface Area (m^2)
- Alcohol use history
- Tobacco use history

10.1. DERIVATIONS

- BMI (kg/ m^2) = Weight (kg)/ Height (m)^2
- Age (Years): Calculated relative to the date of informed consent. Age (Years) is to be calculated using the following SAS® code:
    - Age (Years) = int ((intck('month', <date of birth>, <date of IC>) - (day(<date of IC >) < day(<date of birth>))) / 12)

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11. MEDICAL HISTORY AND ONCOLOGY HISTORY

11.1. MEDICAL HISTORY

All medical history from eCRF page 'Medical History except primary disease' will be coded with Medical Dictionary for Regulatory Activities (MedDRA) central coding dictionary, Version 22.1 (MedDRA v22.1) or higher and summarized by system organ class (SOC) and preferred term (PT) for the ITT. Number of subjects and percentage to the analysis set will be presented in frequency tables, ordered by primary SOC and PT in descending order of the frequency in Fruquintinib + Paclitaxel group. For SOCs or PTs with the same frequency in Fruquintinib + Paclitaxel group, categories will be sorted alphabetically.

11.2. ONCOLOGY HISTORY

Oncology history, collected from eCRF page 'Primary Diagnosis', will be presented for the ITT. The following oncology history information will be summarized for the study:

- Time since initial diagnosis to randomization (months)
- Time since initial diagnosis to randomization (<12 months; ≥12 months)
- Primary Tumor Location (Stomach, GEJ)
- Stage of disease at the time of initial diagnosis (Stage 0, Stage I, Stage II, Stage III, Stage IV, Unknown)
- Pathologic Lauren classification (Intestinal Type, Diffuse Type, Mixed Type, Or Unknown or not Evaluable Type, other)
- Histological or Cytological Diagnosis (Adenocarcinoma, Other)
- Tumor Grade at initial diagnosis (Gx Grade Cannot Be Assessed, G1 Well-Differentiated, G2 Moderately Differentiated, G3 Poorly Differentiated, G4 Undifferentiated, Unknown)
- Metastasis (Yes, No)
- Metastatic site (Single, Multiple)
- Number of Metastasis sites
- Metastasis with liver(Yes, No)
- Metastasis with peritoneal(Yes, No)Metastasis Sites (Peritoneal, Pancreas, Spleen, Liver, Transverse Colon, Esophagus, Duodenum, Lung, Bone, Brain, Meninges, Skin, Ovary, Lymph Node, Other)
- Prior Gastrectomy(Yes, No)
- Time since 1st distal metastasis (or local progression) to randomization (months)
- Time since 1st distal metastasis (or local progression) to randomization (<6 months; ≥6 months)
- Time since Most Recent Progression to Randomization (months)

11.2.1. DERIVATIONS

- Time since initial diagnosis to randomization (months) = (Date of randomization – Date of primary diagnosis+1)/ 30.4375
- Time since 1st distal metastasis (or local progression) to randomization (months) = (Date of randomization

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– Date of 1st distal metastatic (or local progression) +1) / 30.4375

- Time since most recent progression to randomization (months) = (Date of randomization – Date of most recent progression+1) / 30.4375

12. PRIOR AND CONCOMITANT MEDICATIONS/PROCEDURES/SURGERIES

All prior and concomitant medications and procedures data will be analysed with the SAF analysis set.

12.1. PRIOR AND CONCOMITANT MEDICATIONS/PROCEDURES/SURGERIES

Prior and Concomitant Medications, collected from eCRF page 'Concomitant Medications ', will be coded using World Health Organization Drug Dictionary (WHO-DD) version 01Sep2018 or higher. The number and proportion of subjects with prior and concomitant medications will be summarized by ATC level 2 text and PT terms for each treatment group. The output will be tabulated in the order of descending frequency of ATC level 2 text followed by descending frequency of PT in Fruquintinib + Paclitaxel group. If multiple ATCs/PTs have the same frequency, they will be sorted alphabetically. Prior or concomitant medications will be listed.

Prior and Concomitant Procedures/Surgeries from eCRF page 'Concomitant Procedures/Surgery' will be coded with Medical Dictionary for Regulatory Activities (MedDRA), version 22.1 or higher. All prior and concomitant procedures and surgeries will be summarized and listed by SOC and PT.

Concomitant medications are defined as medications other than study medications which are taken at any time on-treatment, i.e. on or after the first administration date of study medication and no later than 30 (+3) days following the last administration date on treatment of study medication or initiate new anti-tumor therapy (whichever occurs first)

- See section 6.1 for handling of missing data, in the case where it is not possible to define a medication as prior or concomitant, the medication will be classified by the worst case; i.e. concomitant.
- 'Prior' medications are medications which started and stopped prior to the first administration date of study medication.
- 'Concomitant' medications or procedures are those which:
    - Started prior to the first administration date of study medication and ended on or after the first administration date on treatment;
    - Started on or after the first administration date of study medication but no later than 30 (+3) days following last administration date on treatment or initiate new anti-tumor therapy (whichever occurs first)

Any concurrent oncology therapy is not allowed and will be reported as protocol deviation. The concurrent oncology therapy will be listed.

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12.2. PRIOR ONCOLOGY TREATMENT (RELATED WITH PRIMARY DISEASE)

Prior Oncology treatments related with primary disease are collected from the eCRF page 'Radiotherapy related with primary disease', 'Chemotherapy and Medication history related with primary disease', 'Surgery history related with primary disease' and 'Meaningful Operation within 4 weeks'.

The following variables will be summarized.

- Radiotherapy
  - Radiotherapy Performed (Yes, No)
  - Total Radiotherapy Dose
  - Treatment Purpose
  - Best response (Complete Response, Partial Response, Progressive Disease, Stable Disease, Not evaluable, Unknown)
- Medication
  - Any anti-tumor medication
  - Protocol defined first line therapy failure
    Protocol defined first line failure is defined as disease progression (that occurs during treatment or within 4 weeks after final administration of drugs (including a single first-line drug for maintenance treatment) at the end of treatment). Neoadjuvant or adjuvant therapy is permitted, in which paclitaxel-containing drug can be included. But if relapse or metastasis occurs during the neoadjuvant or adjuvant therapy period or within 6 months after completion of the above treatment, that adjuvant therapy is considered as the failure of the first-line systemic chemotherapy for progressive disease.
  - Protocol defined first line Therapy class (Chemotherapy: Doublet, Triplet, Targeted therapy: Her 2 antibody, Other, Immunotherapy
  - Protocol defined first line therapy primary therapy class (Chemotherapy only, Immunotherapy plus chemotherapy)
  - Most recently best response (Complete Response, Partial Response, Progressive Disease, Stable Disease, Not evaluable, Unknown, Not applicable)
- Surgery/Procedure and meaningful operation
  - Surgery/Procedure Performed (Yes, No)
  - Name of Surgery (Total Gastrectomy, Subtotal Gastrectomy, Abdominal Exploration, Other)
  - Surgery Purpose (Diagnostic, Curative, First-Line Therapy, Palliative, Other)
  - Meaningful Operation Within 4 Weeks to Randomization (Yes, No)
  - Operation name (Endoscopy, Biopsy, Non-Invasive Operation, Other)

Besides above summarized information will be included in the listing, below information will also be included in the listing.
Treatment objective of medication, all therapy class, regimen number, treatment failure or not, medication name, start or stop date and so on that information collected in the CRF.

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Prior oncology medication treatment by ATC2 and PT will be summarized.

12.3. SUBSEQUENT ANTI-TUMOR THERAPY

Subsequent anti-tumor therapies are collected from eCRF pages under the folder of 'Survival Follow-Up' **,** including 'Radiotherapy', 'Systemic Anti-tumor Medical Treatment' and 'Surgery and Procedures'.

Subjects with any subsequent anti-tumor therapy, type of anti-tumor therapy(radiotherapy, systemic anti-tumor medical treatment and surgery and procedures), time from randomization until initiation of subsequent anti-tumor therapy (months), time from last dose of study treatment until initiation of subsequent anti-tumor therapy, time of initiation of subsequent anti-tumor therapy relative to progression, number of regimens, therapy class of systemic anti-tumor medical treatment (chemotherapy, targeted therapy, immunotherapy, hormonal therapy, traditional Chinese medicine and other). . An accompanying listing will be produced.

Subsequent Anti-Tumor Medication will be summarized by Specific Type of medication (chemotherapy, Anti-VEGF/VEGFR inhibitor, EGFR inhibitor and HER-2, PD-1/PD-L1, traditional Chinese medicine, other) and PT. The type of medication will be mapped by medical.

Subsequent Surgery/Procedures will be summarized by SOC and PT.

12.3.1. DERIVATIONS

- Time from randomization until initiation of subsequent anti-tumor therapy (months) = (Date of first subsequent anti-tumor therapy– Date of randomization +1) / 30.4375
- Time from last dose of study treatment until initiation of first subsequent anti-tumor therapy (months) = (Date of first subsequent anti-tumor therapy- Date of last study drug administration date+1) / 30.4375

13. STUDY MEDICATION EXPOSURE AND COMPLIANCE

The extent of exposure to the study treatment regimen will be analysed with the SAF.

Study medication exposure will be derived according to data collected on eCRF 'Study Medication - fruquintinib or placebo' page and 'Study Medication – Paclitaxel' page. Fruquintinib or placebo is given 4mg QD and it is assumed that for each cycle the subject should take medication in the first 3 weeks followed by 1 week off treatment.

Paclitaxel will be administrated concomitantly with fruquintinib or placebo at the center. Subjects will receive Paclitaxel 80 mg/ m^2 on Day 1, 8 and 15 in a 28 day per treatment cycle. The allowed time window for injection is +3 days (except for C1D1). For each cycle, the first injection of Paclitaxel should be performed on the same day of the administration to the first dose of fruquintinib in the cycle. If there is any delay to the injection of Paclitaxel in the cycle, the subsequent injection of Paclitaxel in the same cycle will be delayed accordingly. However, the last or

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the 3rd injection of Paclitaxel in the cycle must be completed no later than Day 22. Otherwise, the last administration will be interrupted. The dose of Paclitaxel will be decided based on the baseline height (Ht) and weight (Wt), and in each cycle of the study treatment period, the dose level will be based on weight (Wt) assessed in the cycle. Paclitaxel injection will be given until disease progression or the presence of intolerable toxicity, or **6 treatment cycles** are completed. If there is no disease progression or intolerable toxicity after the 6 treatment cycles of Paclitaxel, monotherapy of fruquintinib (or placebo) may be given until disease progression or intolerability toxicity.

For study treatment regimen, the following variables will be summarized:

- Number of initialized cycles and the corresponding frequency distribution
- Total duration of exposure (months)

For 'Fruquintinib or Placebo' medication, the following variables will be summarized by treatment group:

- Number of initialized cycles and its frequency distribution
- Total duration of exposure (months)
- Actual duration of exposure (months)
- Cumulative dose (mg)
- Average daily dose (mg/day)
- Dose intensity (mg/day)
- Relative dose intensity (RDI), and category by <50%; 50% -< 70%; 70% - < 90%; 90 - < 100%; 100 - < 110%; > =110%
- Number and percentage of subjects experiencing drug interruption or dose reduction with reason as AE, Non compliance or other.
- Number and percentage of subjects with drug interruption with reason as AE, Non compliance or other.
- Number and percentage of subjects with any dose reduction with reason as AE, Non compliance, other.
- Number of subjects with dose reduction from 4mg to 3mg by investigator
  Number of subjects with dose reduction from 3mg to 2mg by investigator

For 'Paclitaxel' medication, the following variables will be summarized:

- Number of initialized cycles and its frequency distribution
- Number of Paclitaxel injections planned
- Number of Paclitaxel injections received
- Total duration of exposure (months)
- Cumulative dose (mg)
- Dose intensity (mg/m^2/ week)
- Relative dose intensity (RDI), and category by <50%; 50% -< 70%; 70% - < 90%; 90 - < 100%; 100 - < 110%; > =110%
- Number and percentage of subjects experiencing drug interruption/delay, or dose reduction with reason as AE, Non compliance or other.
- Number and percentage of subjects with drug interruption/delay with reason as AE, non compliance, other.
- Number and percentage of subjects with any dose reduction with reason as AE, non compliance, other.
  Number of subjects with dose reduction from 80 mg/m^2 to 60 mg/ m^2 by investigator
- Number of subjects with dose reduction from 60 mg/m^2 to 40 mg/ m^2 by investigator

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In addition, the study medication exposure on treatment by cycle will be presented in listing.

13.1. DERIVATIONS

If stop date of study drug administration is missing, please refer to section 6.1 for imputation of missing last dose date of study drug.
Total duration of exposure on study and on treatment will be derived following different rules as below.

- Exposure to study regimen
  - Number of initialized cycles is the total number of cycles that a subject has received any study fruquintinib/placebo or Paclitaxel injection.
  - Total duration of Exposure (months) = (Last date of last cycle on study – first date of cycle 1 + 1)/30.4375. Last date of last cycle on study is defined as earliest date among following three dates, i.e. the last administration date +7 (which is reported in eCRF "Study Medication – fruquintinib or placebo" form) or the last administration date + 14 (which is "Study Medication – Paclitaxel" form), death date and date of cut-off.

- Exposure to fruquintinib/placebo
  - Number of initialized cycles is the total number of cycles that a subject has received at least one dose of study fruquintinib/placebo.
  - Total duration of Exposure (months) = (Last date of last cycle on study – first date of cycle 1 + 1)/30.4375. Last date of last cycle on study is defined as the earliest date among three dates, i.e. last administration date + 7 days (which is reported in eCRF "Study Medication - fruquintinib or placebo" form), death date and date of cut-off.
  - Actual duration of exposure (months) = sum of days with non-zero dosing/30.4375. Dose interruption will be excluded from actual duration of exposure.
  - Cumulative dose (mg) = total dose of the subject actually received during study treatment period. For subjects who did not receive any dose of fruquintinib/placebo, the cumulative dose will be set to zero.
  - Average daily dose (mg/day) = Cumulative dose (mg) /Actual duration of Exposure(months)*30.4375. For subjects who did not receive any dose of fruquintinib/placebo, the average daily dose will be zero.
  - Dose intensity (mg/day) = Cumulative dose (mg) / Total duration of Exposure(months)*30.4375. For subjects who do not receive any dose of fruquintinib/placebo, the dose intensity will be zero.
  - Relative dose intensity = Dose intensity (mg/day) / planned dose intensity (mg/day). The planned dose intensity for Fruquintinib or matching placebo is calculated as Planned total dose (mg)/Total duration of exposure(days), in which Planned total dose (mg) is (Total number of cycles - 1)* (4mg/day*21 + 0mg/day*7) + minimum ((last date of last cycle – the first date of the last cycle+1), 21 days)*4mg/day .
  - Dose interruption is collected from eCRF 'study medication – fruquintimib or placebo' page of 'Was the dose adjusted' with 'Dose interrupted' selected.
  - Dose reduction is collected from eCRF 'study medication – fruquintimib or placebo' page of 'Was the dose adjusted' with 'Dose reduced' selected. A subject will experience at most twice reduction medication, from 4mg/day to 3 mg/day and then 3mg/day to 2 mg/day.

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- Exposure to Paclitaxel
  - Number of initialized cycles is the total number of cycles that a subject has received Paclitaxel injection.
  - Number of Palitaxel injection planned = 3*number of largest initialized cycles
  - Number of Palitaxel injection received = total number of Palitaxel injections of the subject have received from eCRF 'study medication- Paclitaxel' page
  - Total duration of Exposure (months) = (last date of Paclitaxel on study – date of first administration of Paclitaxel on study +1)/30.4375. Last date of Paclitaxel on study is defined as earliest date among three dates, i.e. the last administration date, which is reported in eCRF "Study Medication – Paclitaxel" form + 14 days, death date and date of cut-off.
  - Cumulative dose (mg) = sum of the total dosage that the subject actually received during each exposure period. For subjects who do not receive any injection of Paclitaxel, the cumulative dose will be set to zero.
    Dose intensity (mg/m^2/day) = the average of all dose levels (mg/m^2) administered on each scheduled day prior to end of treatment of Paclitaxel. When dose is interrupted, the dose level is 0 mg/m^2. Dose Level on each scheduled day is collected by eCRF form "Study Medication – Paclitaxel". For subjects who do not receive any study drug, the DI is set to zero.
  - Relative dose intensity (%) = Dose intensity (mg/m^2/day)/ planned dose intensity (mg/m^2/day)* 100%. The planned dose intensity of Paclitaxel is 80mg/m^2/day.
  - Dose interruption is collected from eCRF 'study medication- Paclitaxel' page of 'Was the dose adjusted' with 'Dose interrupted' selected and will sum up all interruptions for a subject.
  - Dose reduction is collected from eCRF 'study medication-Paclitaxel' page of 'Was the dose adjusted' with 'Dose reduced' selected. A subject will experience at most twice reduction medication, from 80mg/m^2 to 60mg/m^2 and then from 60mg/m^2 to 40mg/m^2.

13.2. STUDY MEDICATION COMPLIANCE

Compliance to study medication will be assessed using Relative Dose Intensity (RDI) as described in section 13.1.

14. EFFICACY OUTCOMES

Tumors should be evaluated according to the standard of Response Evaluation Criteria in Solid Tumors (RECIST) Version 1.1. Either computed tomography (CT) or magnetic resonance imaging (MRI) evaluation method can be used, but it should be consistently used during the entire study period for a given subject. After randomization, tumor assessments in both treatment groups are to be performed every 4 weeks in the first 3 cycles and every 8 weeks thereafter till the presence of disease progression within a time window of (-5, +3 days) around the scheduled visits, while unscheduled scans may be performed when disease progression is suspected.
The dual primary endpoint of the study are OS and PFS, PFS will be evaluated base on investigator assessment according to the RECIST 1.1 criteria .

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14.1. DUAL PRIMARY ENDPOINT ANALYSES

OS and PFS are co-primary endpoints. The alpha spending of co-primary endpoints refer to section 6.4. If OS is statistically significantly longer for the treatment group than for the control group comparing with pre-defined two-sided significance level at the secondary interim or at final analysis, or if PFS is statistically significantly longer for the treatment group than for the control group comparing with pre-defined two-sided significance level at final analysis, the result can be declared positive.

14.1.1. **OS**

The primary efficacy analysis of Overall Survival (OS) will be performed on the ITT set. All data required for the calculation of time to event will be taken from the eCRF.

OS is defined as the time from randomization to the date of death caused by any reason. For subjects who have not yet reported death at the time of the analysis, the date of the last known alive will be the date of censoring. Please refer to section 5.5 for how to determine last known alive date.

- OS (in months) = (Date of death – Date of randomization +1) / 30.4375

Censoring reasons are as follows:

- Alive
- Withdrawal of consent
- Lost to follow-up: data recorded in CRF prior to the analysis cut-off

Table 1. The date of event / censoring for OS

Survival Status		**Source**	**Censoring**	**Date of event/ censoring**
Died	Before cut-off	Death eCRF	Event	Date of death
	After cut-off	Death eCRF	Censored	Date of cut-off
Alive or Lost to Follow-up/ Withdrew Consent	After cut-off	Follow-Up eCRF	Censored	Date of cut-off
	Otherwise	Section 5.5	Censored	Last date known to be alive

14.1.1.1. OS

Frequency (number and percentage) of subjects with an event (death) and censoring reasons will be presented by treatment group. The OS time or censoring time and the reasons for censoring will also be presented in a subject listing.

The stratified log-rank test will be performed to compare OS between fruquintinib+Paclitaxel group and placebo+Paclitaxel group. Stratification factors included into the analysis are the primary site of the tumor (gastroesophageal junction vs stomach), ECOG PS score (0 vs 1), and peritoneal metastasis (present vs absent) collected from eCRF.

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The median and quartiles for overall survival and survival rate at the 3rd, 6th, 9th, 12th and 18th months, together with the corresponding two-sided 95% confidence interval will be estimated for each treatment arm using Kaplan-Meier method. A Kaplan-Meier plot will be presented to provide a visual description of the differences among the treatment groups. Furthermore, the treatment effect will be characterised by the "treatment/control" HR estimated using a stratified Cox proportional hazards model. The estimated HR and corresponding two-sided 95% confidence interval (CI) will be presented.

The SAS procedure LIFETEST will be used for log-rank test and Kaplan-Meier estimation, and the SAS procedure PHREG will be used for fitting the Cox proportional hazards model analysis. A STRATA statement will be used to implement the stratified analyses in both procedures.

```
proc lifetest data=adtte&n outsurv=ci method=km timelist=(3 6 9 12 18)
      atrisk conftype=loglog;
      time aval * CNSR(1,2,3,4,5,6,7);
      strata &stratvar/group= trtpn test=(logrank);
      ods output ProductLimitEstimates=mmaxgrp&n Quartiles=medgrp&n;
run;
```

The SAS procedure PHREG will be used for Cox proportional hazards model analysis. Efron method will be used to handle ties. Breslow is SAS default; however, it does not do well when data are heavily tied. Efron is preferred over Breslow [Allison, Paul D. 2010].

```
proc phreg data=adtte1;
      class trtpn (ref="2");
      model aval*CNSR(1,2,3,4,5,6,7)=trtpn/rl alpha=0.05 ties=EFRON;
      strata   &stratvar;
      hazardratio trtpn / diff=all;
Run;
```

For the planned first interim analysis of the first 100 subjects, stratification factors will not be considered in the analysis due to the small sample size. Unstratified log-rank test and unstratified Cox proportional hazards model will be the primary analysis method for OS at first interim analysis.

14.1.1.2. Subgroup Analysis of OS

To assess consistency of treatment effect in OS in different subsets, subgroup analyses of OS will be performed. For each subgroup level, the number of subjects and events at the time of analysis, median OS from Kaplan-Meier [KM] estimate), hazard ratios and 95% CI from an unstratified Cox proportional hazards model (ties=Efron) with treatment as the only covariate will be presented for the ITT set. Forest plot will be presented with hazard ratios and 95% CIs of treatment for the overall treatment effect and for each subgroup level. Also, the p-value for treatment by subgroup type interaction test (Likelihood Ratio test) will be provided together. For the list of subgroups, please see Section 4.5.

If there are too few events (less than 10) available for a meaningful analysis of a particular subgroup, the relationship between that subgroup and OS will not be formally analysed. In this case, only the event number and total number of subjects in the subgroup for each treatment arm will be provided.

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Furthermore, interaction of treatment by each subgroup factor will tested using a Cox proportional hazards model (ties=Efron) that contains the treatment term, factor and treatment-by-factor interaction term. P value for the interaction item will be presented in the summary table and forest plot. Significance level for interaction test is set to two-sided 0.10.

No adjustment to the significance level for testing will be made since all these subgroup analyses will be considered exploratory and may only be supportive of the primary analysis of OS.

14.1.1.3. Sensitivity Analyses of OS

The following sensitivity analyses for OS will be performed on the ITT set, except the second one, i.e. b.

a. To take centre effect into account, centre as random effect will be added to the Cox proportional hazard model as described in section 14.1.1.1.

b. To assess the robustness of the OS, the primary analyses as described in section 14.1.1.1, i.e. the stratified log-rank test, Kaplan-Meier estimates, and hazard ratio obtained from Cox proportional hazard model will be repeated for **PPS set**, if numbers of subjects in the ITT set and the PPS set differ by more than 10%.

c. The primary efficacy analysis as described in section 14.1.1.1 will be repeated for the ITT set, except that strata data used in the analysis are strata collected through **IWRS as stratification factors**.

d. As a sensitivity analysis to assess the impact of stratification, the two treatment groups will be compared using **unstratified** log-rank test. The hazard ratio together with associated 95% confidence interval obtained using the unstratified Cox regression model will also be presented. **Note: This is actually the primary analysis of OS for the first interim analysis.**

e. As an exploratory analysis, a stratified Cox regression model with actual strata collected through eCRF will be fitted to evaluate the effect of other baseline demographic or disease characteristics on the estimated hazard ratio. This model will include treatment and a set of **potential prognostics/predictive factors** as described in section 4.5. Potential prognostics/predictive factors included in the model will be assessed for co-linearity and a stepwise selection process (Significance levels for entering or removing an explanatory variable are set to 5%) will be applied to identify a final subset of prognostics/predictive factors in the model. Note: Treatment is always included in the model during the stepwise selection process.

f. An additional exploratory analysis of treatment by factor interactions using the stratified Cox regression model will be conducted, using the factors identified in the above model (see e) . This will be done by adding interaction terms of each factor with treatment to the model. Stepwise selection will be performed on the saturated model, whereby (using a 10% level throughout) the least significant interaction terms are removed one-by-one and any newly significant interactions re-included until a final model is reached where all included interactions are significant and all excluded interactions are non-significant. Throughout this process all main effects will be included in the model regardless of whether the corresponding interaction term is still present. This approach will identify the factors that independently alter the treatment effect and prevent identification of multiple correlated interactions. The hazard ratios, 95% CI of the HR, and p values for treatment from the final multivariate model will be presented. For any covariate where the

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treatment by covariate interaction is significant in the final multivariate model, the hazard ratio for treatment within each level of the covariate will also be shown.

g. As an exploratory analysis to assess the impact of subsequent anti-tumor therapy to OS, OS for patients with any anti-tumor therapy will be censored at the first initialized date of subsequent anti-tumor therapies. The analysis will be performed with the method in the primary analysis to overall survival, which is described in section 14.1.1.1.

h. As an exploratory analysis to assess the impact of subsequent anti-tumor therapy to OS, all of patients with any subsequent anti-tumor therapy will be removed from ITT. The analysis will be performed with the method in the primary analysis to overall survival, which is described in section 14.1.1.1

i. The assumption of proportional hazards will be tested firstly by examining plots of complementary log-log (event times) versus log (time) and, if these raise concerns, by adding a treatment*log(time) interaction in the Cox model. If a lack of proportionality is evident, one sensitivity analysis with a stratified piece-wise Cox model will be used to calculate piece-wise HR in different periods. In such circumstances, the HR from the primary analysis can still be meaningfully interpreted as an average HR over time unless there is extensive crossing of the two curves.

Note: the above sensitivity analyses won't be performed at the first interim analysis due to small sample size.

14.1.1.4. Duration of Follow-up for OS

Duration of follow-up refers to the time interval between randomization date and last date known to be alive for subjects who have not yet been reported to have died by the time of analysis. In order to assess duration of follow-up for OS, Kaplan Meier estimates will be calculated in the same way as in the analysis described above for primary OS, while using different censoring rule which reverse censoring indicator of primary analysis of OS instead.

14.1.2. **PFS**

PFS time is defined as the time (in months) from randomization date to either the first observation of PD (as assessed by Investigator) or death due to any cause, whichever occurred earlier. Subjects who have not progressed or died at the time of analysis will be censored at the time of the latest date of assessment from their last evaluable RECIST assessment.

If subject progresses or dies after two or more consecutive scheduled missed/non-evaluable tumor assessments, the subject will be censored at the time of the latest evaluable RECIST assessment just **before the consecutively missed/non-evaluable tumor assessment(s).** According to study protocol, Tumor RECIST assessment will be performed every 4 weeks (-5 days or +3 days) in the first 3 cycles followed by every 8 weeks (-5 days or +3 days) thereafter. This means the scheduled tumor assessments will be performed on C2D1, C3D1, C4D1, C6D1, C8D1, C10D1, … and so on. Depending on the latest evaluable RECIST assessment visit prior to the PD event (disease progression or death), the two consecutive tumor assessment visit window may varies. The following is the rule:

- If the latest evaluable RECIST assessment visit prior to PD event **falls before C3D1 nominal visit**, then visit window is 62 days [2*(4 weeks+ 3 days window)]

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- If the latest evaluable RECIST assessment visit prior to PD event **falls between C3D1 nominal visit (inclusive) and C4D1 nominal visit (exclusive)**, then visit window is 90 days [(4 weeks+ 3 days window+ 8 weeks + 3 days window)]
- If the latest evaluable RECIST assessment visit prior to PD event **falls on or after C4D1 nominal visit**, then visit window is 118 days [2* (8 weeks+ 3 days window)]

If a subject has no evaluable tumor assessment visits after randomization or does not have evaluable baseline tumor assessment data the subject will be censored on the date of randomization unless they die within two consecutive tumor assessments since the randomization.

If there is any new anti-tumor therapy initiated prior to documented disease progression or death on study, the subject will be censored at date of last evaluable tumor assessment on or prior to the date of new anti-tumor therapy.

The above censoring rules for primary analysis of PFS are summarized in Table 3. Another set of censoring rules for sensitivity analysis are also presented in Table 3. The only difference from the primary analysis is that it considers initiation of new anticancer treatment as PD event for subjects without any documented PD or death prior to new anticancer treatment.

The PFS time will always be derived based on scan/assessment dates not RECIST evaluation dates. If PD is documented between scheduled visits, the actual date of documented progression will be used as an uncensored value in the analysis of PFS.

RECIST assessments/scans contributing towards a particular visit may be performed on different dates. The following rules will be applied:

- Date of progression will be determined based on the **earliest** of the dates of the component that triggered the progression.
- When censoring a subject for PFS the subject will be censored at the **latest** of the dates contributing to a particular overall visit assessment.

PFS (in months) will be calculated as (first event date – randomization date +1) /30.4375 when a subject is deemed to experience a PFS event or calculated as (censored date - randomization date +1) /30.4375 when a subject is deemed to be censored, according to the rules specified in Table 3.

Event date is the earliest date of the following events that occurred ⩽ two consecutive tumor assessment visit window s from last evaluable tumor assessment or randomization date if no evaluable post-baseline tumor assessment at all:

- Observed PD
- Death

Censoring rules:

- Data cut-off will be applied to raw data first.
- Censored on the randomization date, if patient didn't have baseline or post-baseline assessment and without any PD or death.

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- Censored on the date of last evaluable tumor assessment or randomization date, whatever occurs later, if:
  - PD or Death (without previously documented PD) is observed after two or more consecutive missing/non-evaluable tumor assessments , or
  - Initiation of new anticancer treatment without PD/death observed, or
  - Lost to follow-up / withdraw consent without PD/death observed, or
  - No PD/death observed up to the cut-off date (administrative censoring)

Table 3. An overview of event/censoring rules for PFS:

Status		**Censoring**	**Date of event / censoring**
Progressed or died	PD occurred at or before next scheduled tumor assessment	Event	Date of PD
	PD after two or more consecutive scheduled missed/non-evaluable tumor assessments	Censored	Date of last evaluable tumor assessment before the date of progression or randomization date, whatever occurs later
	Death (without previously documented PD is observed) at or before next scheduled tumor assessment	Event	Date of death
	Death (without previously documented PD is observed) after two or more consecutive scheduled missed/non-evaluable tumor assessments	Censored	Date of last evaluable tumor assessment or randomization date, whatever occurs later
	Initiation of new anticancer treatment	Censored	Date of last evaluable tumor assessment or randomization date, whatever occurs later AND before or on new anticancer treatment
		Event (apply for sensitivity analysis)	Date of new anticancer treatment, however if the new anticancer treatment is initialized after two or more consecutive scheduled missed/non-evaluable tumor assessments, then subject will be censored at the date of last known tumor assessment or randomization date (whatever occurs later)
Neither progressed nor died	Initiation of new anticancer treatment* without PD/death observed	Censored	Date of last evaluable tumor assessment or randomization date, whatever occurs later and before or on new anticancer treatment
	Lost to follow-up / withdraw consent without PD/death	Censored	Date of last evaluable tumor assessment or randomization date, whatever occurs

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	observed		later
	No PD/death observed up to the cut-off date (administrative censoring)	Censored	Date of last evaluable tumor assessment or randomization date, whatever occurs later
	No baseline or evaluable post-baseline assessment	Censored	Date of randomization date

*New anticancer treatment includes therapies collected from eCRF pages under the folder of "Survival Follow-Up", including "Radiotherapy", "Systemic Anti-tumor Medical Treatment" and "Surgery and Procedures", as well as concomitant anticancer therapy collected from eCRF page "Concomitant Medications" which will be identified by checking if ATC2 is antineoplastic agents.

Note: When there were (consecutively) missed/non-evaluable tumor assessment, date of last evaluable tumor assessment refers to the date of tumor assessment that occurred just before the (consecutively) missed/non-evaluable tumor assessment(s). Note: 'Evaluable' is defined if overall response isn't equal to "Not Evaluable" or missing.

14.1.2.1. PFS analysis

The primary analysis of PFS will be based on ITT set. The primary comparison of PFS between the two groups is to be performed using stratified log-rank test; specifically, treatment efficacy expressed by the HR and associated two-sided 95% CI will be estimated from a stratified Cox regression model. Stratification factors are the same as that for the primary analysis of OS. The median and quartiles for PFS and the PFS rates at the 1st, 3rd and 6th months, together with their two-sided 95% confidence intervals will be estimated for each treatment group using the Kaplan-Meier method, and the Kaplan-Meier curves for PFS will be presented.

However, for the planned first interim analysis of the first 100 subjects, stratification factors will not be considered in the analysis due to the small sample size. For the second interim analysis and final analysis, stratification factors will be considered in the analysis.

Progression events (death or progression) will be summarized by treatment groups.

The status of subjects who are censored in the primary analysis of PFS will be tabulated for each treatment group using the following categories:

- Censored RECIST progression: RECIST progression event after two or more consecutive scheduled missed/non-evaluable tumor assessments
- Censored death: Death (without previously documented PD is observed) occurred after two or more consecutive scheduled missed/non-evaluable tumor assessments
- No baseline tumor assessment and no death/ disease progression observed
- No evaluable post-baseline tumor assessment and no death/ disease progression observed
- Initiation of new anti-cancer therapy before the observed disease progression or death
- Progression free at time of analysis: Subjects known to be alive without RECIST progression at data cut-off date. This excludes subjects alive without evaluable baseline RECIST assessment or evaluable post-baseline RECIST assessment.
- Lost to follow-up without PD/ death observed.
- Withdrawn consent without PD/ death observed.

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14.1.2.2. Subgroup Analysis of PFS

Subgroup analyses of PFS will be performed in the same way as OS. All assignments of censoring and progression will be same as in the primary method of derivation for PFS.

14.1.2.3. Sensitivity Analysis of PFS

The following sensitivity analyses for PFS will be performed on the ITT set, except the second one, i.e. b.

a. To take into account centre effect, centre as random effect will be added to the stratified Cox proportional hazard model as described in in section 14.1.2.1.

b. To assess the robustness of the PFS endpoint, the analysis as described in section 14.1.2.1, i.e. the stratified log-rank test, Kaplan-Meier estimates, and hazard ratio obtained from Cox proportional hazard model will be repeated for the **PPS set**, if numbers of subjects in the ITT set and the PPS set differ by more than 10%.

c. The primary efficacy analysis as described in section 14.1.2.1 will be repeated for the ITT set, except that strata data used in the analysis are strata collected through **IWRS as stratification factors**.

d. As a sensitivity analysis to assess the impact of stratification, the two treatment groups will be compared using unstratified log-rank test with same censoring rule as primary analysis of PFS. The hazard ratio together with associated two-sided 95% confidence interval obtained using the **unstratified** Cox regression model will also be presented. Note: This is actually the primary analysis of PFS at the first interim.

e. As an exploratory analysis, a stratified Cox regression model with actual strata collected through eCRF will be fitted to evaluate the effect of other baseline demographic or disease characteristics on the estimated hazard ratio. This model will include treatment and a set of **potential prognostics/predictive factors** as described in section 4.5. Potential prognostics/predictive factors included in the model will be assessed for co-linearity and a stepwise selection process (Significance levels for entering or removing an explanatory variable are set to 5%) will be applied to identify a final subset of prognostics/predictive factors in the model. Note: Treatment is always included in the model during the stepwise selection process.

f. An additional exploratory analysis of treatment by factor interactions using the stratified Cox regression model will be conducted, using the factors identified in the above model（see e）. This will be done by adding interaction terms of each factor with treatment to the model. Stepwise selection will be performed on the saturated model, whereby (using a 10% level throughout) the least significant interaction terms are removed one-by-one and any newly significant interactions re-included until a final model is reached where all included interactions are significant and all excluded interactions are non-significant. Throughout this process all main effects will be included in the model regardless of whether the corresponding interaction term is still present. This approach will identify the factors that independently alter the treatment effect and prevent identification of multiple correlated interactions. The hazard ratios, 95% CI of the HR, and p values for treatment from the final multivariate model will be presented. For any covariate where the treatment by covariate interaction is significant in the final multivariate model, the hazard ratio for treatment within each level of the covariate will also be shown.

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g. The primary analysis as described in section 14.1.2.1 will repeated for PFS endpoint using a different set of censoring rule as described in Table 3 for sensitivity analysis. Note: For the first interim analysis, unstratified analysis using a different set of censoring rule will be performed. For the second interim analysis and final analysis, stratified analysis using a different set of censoring rule will be performed.

Note: the above sensitivity analyses except (g) won't be performed at the first interim analysis due to small sample size.

14.1.2.4. Duration of Follow-up for PFS

In order to assess duration of follow-up for progression-free survival, Kaplan Meier estimates will be calculated in the same way as in the analysis described above for primary PFS, while using different censoring rule which reverse censoring indicator of primary analysis of PFS instead.

14.2. SECONDARY EFFICACY ENDPOINTS ANALYSES

The secondary efficacy variables include Objective Response Rate (ORR), Disease Control Rate (DCR), Duration of Response (DoR), tumor shrinkage and quality-of-life score (EORTC QLQ-C30).

14.2.1. SECONDARY EFFICACY VARIABLES & DERIVATIONS

14.2.1.1. Objective Response Rate (ORR)

Objective Response Rate (ORR) is defined as the proportion of subjects whose best overall response (BOR) is either a CR or PR per RECIST 1.1 criteria for the ITT set.

BOR is determined by the best response designation recorded between the date of randomization and the date of objectively documented progression or the date of subsequent anti-cancer therapy, whichever occurs first. For subjects without documented progression or subsequent anticancer therapy, all available response designations will contribute to the BOR determination. For subjects without any post-baseline tumor assessment, their BOR will be assigned as Not Assessable. For subjects who die in the absence of progression and have no evidence overall visit response (i.e. not evaluable for overall response) prior to death, PD will be assumed to the BOR to this patient if patients died within two consecutive tumor assessment visit window from last evaluable assessment or randomization, CR or PR after initializing any further anti-tumor therapy will not be included into consideration.

Note: Duration between randomization and the visit at which SD is to be claimed must be at a minimum of 51 days. If SD is less than 51 days, then BOR will be assigned as Not Assessable.

14.2.1.2. Disease Control Rate (DCR)

Disease Control Rate (DCR) is defined as the proportion of subjects whose best overall response (BOR) from baseline is either a CR, PR or SD per RECIST 1.1 criteria for the ITT set.

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The best response of stable disease (SD) can be assigned if SD criteria was met at least once after the randomization at a minimum interval of 51 days (8 weeks -5 days).

14.2.1.3. Duration of Response (DoR)

Duration of Response (DoR) is defined as the time (month) from the first time that the objective response reaches CR or PR, whichever comes first, until the occurrence of PD or death (whichever comes first). The DoR is just applied to subjects whose BOR is CR or PR. The calculation is performed based on the actual completion date of tumor scan. The end of response should coincide with the date of progression or death from any cause used for the PFS event. If a subject does not progress following a response, then their DoR will use PFS censoring time.

14.2.1.4. Tumor shrinkage

Tumor shrinkage will be summarized as the best percent change from baseline in target lesions (sum of longest diameter for non-nodal lesions and short axis for nodal lesions). It will be derived for subjects with post baseline target lesion assessment as:

Percent change from baseline (%)= 100% $*$ (Sum of target lesion diameters -sum of target lesion diameters at baseline)/ sum of target lesion diameters at baseline

The best change in target lesions from baseline will be derived across all the post-baseline assessments until documented disease progression and initial of new anticancer treatment as the minimum value of percentage change from baseline.

Waterfall plot for the best relative change of the sum of the longest diameter will be provided.

14.2.1.5. Quality-of-Life Score (EORTC QLQ-C30)

The European Organization for Research and Treatment of Cancer Quality of Life Questionnaire Core 30 (EORTC QLQ-C30, Version 3.0) questionnaire consists of 30 multiple choice questions that can be classified into following categories:

- Functional scales: physical, role, cognitive, emotional, social
- Symptom scales: fatigue, nausea and vomiting, pain
- A global health status scale
- Single items: dyspnea, loss of appetite, insomnia, constipation, diarrhea, financial difficulties.

First the average of the items that contribute to the scale is estimated; this is the raw score; then a linear transformation is used to standardize the raw score, so that scores range from 0 to 100; a higher score represents a higher ("better") level of functioning, or a higher ("worse") level of symptoms. Refer to Appendix 3 for details on scoring.

14.2.2. MISSING DATA METHODS FOR SECONDARY EFFICACY VARIABLES

When no imaging/measurement is done at all at a particular time point, overall response for the subject will be considered as not evaluable (NE) at that time point. If only a subset of lesion measurements is made at an

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assessment, overall response for the subject is also considered NE at that time point, unless a convincing argument can be made that the contribution of the individual missing lesion(s) would not change the assigned time point response. See Appendix 2 in protocol for specification.

All the secondary efficacy variables will be analysed based available data, therefore no imputation method will be implemented, unless other specified.

14.2.3. Analysis of Secondary Efficacy Variables

14.2.3.1. ORR and DCR

The analysis of ORR and DCR will be based on the ITT set. Actual levels of strata collected through eCRF will be used for the stratified analyses, where applicable.

The exact 95% CI for the estimated ORR and DCR will be calculated for each treatment group using the Clopper-Pearson method. The Cochran-Mantel-Haenszel (CMH) test, with strata as specified above, will be used to compare the ORR and DCR between treatment groups. The results from the stratified CMH analysis will be expressed in terms of the odds ratio and its associated CI and p-value. If the number of objective response or disease control is not sufficient to utilize CMH test, the stratified exact test will be performed instead. Exact confidence limits for the common odds ratio will be presented.

PROC FREQ with BINOMIAL option in EXACT statement in SAS will be used to calculate the CI of the ORR and DCR for each treatment group. PROC FREQ with CMH option in TABLE statement will be used for CMH test. PROC FREQ with COMOR option in EXACT statement will be used to execute stratified exact test.

For the first interim analysis, stratification factors will not be considered. The treatment difference in ORR and DCR will be tested using Chi-Square test or Fisher's exact test. The 95% confidence interval for ORR will be calculated using asymptotic or exact method, where appropriate.

For the second interim analysis and final analysis of OS and PFS, stratification factors will be considered.

PROC FREQ with 'OR FISHER' option in EXACT statement will be used to execute Fisher's exact test and exact confidence interval for ORR.

14.2.3.2. DoR

Duration of response only applies for subjects who have responded to the treatment, i.e. CR or PR. Descriptive analysis will be adopted for DoR. For each treatment group, results will be presented by Kaplan-Meier estimates.

14.2.3.3. EORTC QLQ-C30

Analysis of EORTC QLQ-C30 is not applicable for the first interim analysis.

Each scale and item will be summarized by visit using summary statistics descriptively. Summary statistics (mean [and standard deviation], median, Q1, Q3, range) of standardized scores will be reported for the scales of the

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European EORTC QLQ-C30 questionnaire. These summary statistics will be presented for observed and change from baseline scores. A listing of all items scores values will be produced for each patient.

Missing data will be handled in accordance with the scoring manual. Namely, for multi-item scales, if at least half of the items from the scale have been answered, assume that the missing items have values equal to the average of those items which are present for that respondent.

Mixed Model Repeated Measures (MMRM) via SAS® MIXED procedure will be used for the inter-group change of QLQ-C30 from the baseline, assuming missing at random (MAR). Baseline values, visits, treatment and the interaction between treatment and visit will be used as fixed effects, and the subjects and interception will be used as random effects. Restricted maximum likelihood (REML) will used to estimate the parameters in the model. Covariance structure will be selected from compound symmetry, toeplitz, spatial and unstructured in accordance with information criteria. The final analytical model will depend on the observed data.

MMRM analysis will be making use of all data from baseline up to the xx cycles. The analysis will be to compare treatment effect at each visit and the average treatment effect from the point of randomisation until xx cycles unless there is excessive missing data (defined as >75% missing data). The number of cycles will be specified in the footnote to the table.

14.3. EXPLORATORY EFFICACY

Analyses of biomarkers related to the antitumor activity of Fruquintinib will be described via a separate document.

15. SAFETY OUTCOMES

All outputs for safety outcomes will be based on the Safety Analysis Set. Safety endpoints include AEs, laboratory results (hematology, blood chemistry, coagulation, urinalysis, 24 hours urine protein, stool for occult blood and thyroid function test), vital signs (blood pressure, heart rate, respiratory rate, and temperature, weight, BSA, BMI), 12 Lead ECG, Echocardiography and ECOG PS.

In general, the safety summary tables will only include “on-treatment” events/assessment, i.e. those collected on or after the first date of study treatment and collected no later than 30 (+3) days after the date of last study treatment administration or initiation of subsequent anti-tumor therapy, whichever comes first.

Unless otherwise specified with the relevant section, no statistical comparisons will be performed between these two treatment groups for safety data.

15.1. ADVERSE EVENTS

All AE verbatim descriptions will be coded using the Medical Dictionary for Regulatory Activities (MedDRA), version

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22.1 or higher.

Treatment-emergent adverse events (TEAEs) are defined as AEs that started or worsened in severity on or after the first administration date of study medication and no later than 30 days after the last administration date on study of study medication or initiate new anti-tumor therapy (whichever occurs first). An exception is that study medication related SAE collected later than 30 days after the last administration date on study or after initialte new anti-tumor therapy (whichever occurs first) will be treated as TEAEs. This definition is aligned with protocol AE/SAE data collection procedure.

If the start date is coincident to the reference start date, a query "Prior/ post 1st dose" will be used to identify TEAEs (only when 'Post 1st dose' will be considered as TEAE, if 'Prior' then not TEAE). When there is no definite answer to the query, the AE will be considered as a TEAE. See section 6.1 for handling of partial dates for AEs. In the case where it is not possible to classify an AE as treatment-emergent or not, the AE will be classified by the worst case; i.e. treatment-emergent.

An overall summary of number of subjects and number of events within each of the categories described in the sub-section below, will be provided as specified in the templates.

Listings will include all AEs. TEAEs (Y/N) will be flagged in the listings.

15.1.1. ALL TEAEs

Incidence of TEAEs will be presented by SOC and then PT and broken down further by maximum NCI CTC AE grade. Related TEAE will be presented by SOC/PT by maximum CTCAE grade. Number (n) and percentage (%) will be presented with the percentage calculated relative to the total number of subjects in the SAF set.

15.1.2. SEVERITY

The severity of all TEAEs shall be graded according to 5 grades (Grade 1 to Grade 5) in accordance with the national cancer institute common terminology criteria for adverse event (NCI CTC AE) V4.03 as defined in the following document:
http://evs.nci.nih.gov/ftp1/CTCAE/CTCAE_4.03_2010-06-14_QuickReference_5x7.pdf

TEAEs classified as grade 3 or higher will be summarized. Incidence rates (frequencies and percentages) by SOC and PT will be presented.

15.1.3. RELATIONSHIP WITH STUDY MEDICATION

Relationship, as indicated by the Investigator and collected in 'Adverse Events' eCRF page by indicating relationship with Fruquintinib/Placebo or Paclitaxel by 'Yes' or 'No'. Relationship would be summarized as "Related" or "Unrelated" in the overall summary of TEAE and by relationship with Fruquintinib/Placebo or Paclitaxel. Separated summary of fruquintinib/placebo related TEAE and paclitaxel related TEAE and any drug related TEAE by SOC and PT and maximum CTCAE grade will be prepared.

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15.1.4. Serious TEAEs

Serious adverse events are those events with a response of "Yes" for the item "Is the adverse event serious?" on the AE form. Collection of serious adverse event (SAE) will be from the date of signing the Informed Consent Form to 30(+3) days after last administration or start of new anti-tumor treatment and, afterwards, only the SAEs related to the investigational drug will be collected. All SAEs will be listed. Only treatment emergent SAEs will be tabulated for SAF set.

Summaries of incidence rates (frequencies and percentages) of serious treatment emergent AEs by SOC/PT by maximum CTCAE grade will be prepared. The SOC and PT should be sorted by descending frequency of the number of subjects in Fruquintinib + Paclitaxel group.

15.1.5. TEAEs Leading to Drug Discontinuation

TEAEs leading to permanent discontinuation of study medication will be identified as those TEAE with a response of "Drug withdrawn" to the item "Action taken with the study drug'' for -Fruquintinib/Placebo or Paclitaxel on the "Adverse Event" form of eCRF.

For TEAEs leading to drug discontinuation, summaries of incidence rates (frequencies and percentages) by SOC/PT by maximum NCI-CTCAE will be prepared.

For TEAE leading to drug discontinuation, summaries of any TEAE leading to fruquintinib/placebo discontinuation and any TEAE leading to paclitaxel discontinuation will be prepared. The same summary for fruquintinib/placebo related TEAE and paclitaxel related TEAE leading to discontinuation will be prepared.

15.1.6. TEAEs Leading to Drug Interruption/Delay or Dose Reduction

TEAEs leading to drug interruption/delay or dose reduction will be identified as those records with a response of "Drug interrupted" or "Dose delayed" or "Dose reduced" to the item "Action taken with the study drug" on the AE/SAE form of eCRF.

For TEAEs leading to drug interruption/delay or dose reduction, summaries of incidence rates (frequencies and percentages) by SOC/PT by maximum NCI-CTCAE will be prepared.

For TEAE leading to drug interruption/delay or dose reduction, summaries of any TEAE leading to fruquintinib/placebo interruption or dose reduction and any TEAE leading to paclitaxel interruption/delay or dose reduction will be prepared. The same summary for fruquintinib/placebo related TEAE and paclitaxel related TEAE will be prepared.

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15.1.7. TEAEs WITH CTCAE GRADE >=3

The incidence rates of TEAEs/drug related TEAEs with CTCAE grade at "Grade 3" "Grade 4" or "Grade 5" will be summarized by SOC/PT by severity as above.

15.1.8. TEAEs LEADING TO DEATH

TEAEs leading to Death are those events with 'Death' response recorded as 'Fatal' on the Adverse Events page of eCRF. Or collected on Death page of eCRF with 'Adverse Event' recorded as primary cause of death. The TEAE leading to death and drug related TEAE leading to death will be summarized by SOC/PT.

TEAE leading to death will be summarized by any TEAE leading to death, any drug related TEAE leading to death, any fruquintinib/placebo related TEAE leading to death, and any paclitaxel related TEAE leading to death.

15.1.9. ADVERSE EVENTS OF SPECIAL INTEREST (AESIs) OF FRUQUINTINIB

The adverse events of special interest (AESI) of fruquintinib will be identified by PT terms included in Standardised MedDRA Queries (SMQ) or SOC or HLGT current version from Appendix 4 which provided by Drug safety based on available MedDRA version prior to each planned analysis. Incidence rates (frequencies and percentages) will be presented by treatment group. A total of 11 AESI category will be summarized for this study, which are 'haemorrhages', 'hepatic function abnormal', 'infections', 'hypertension', 'dermatological toxicity', 'proteinuria', 'thyroid dysfunction', 'embolic and thrombotic events,', 'gastrointestinal perforation', 'Left ventricular ejection fraction decreased and 'Aneurysms and artery dissections'. Treatment Emergent Adverse Events of Special Interest will be summarized by AESI category, Preferred Term and Maximum NCI-CTCAE . Under each AESI category, any drug related, fruquintinib/placebo related, paclitaxel related, leading to any drug discontinuation, leading to Fruquintinib/placebo discontinuation, leading to paclitaxel discontinuation, leading to any drug interruption/delay/reduction, leading to Fruquintinib/placebo drug interruption/reduction, leading to paclitaxel interruption/delay/reduction, leading to any drug interruption/delay, leading to Fruquintinib/placebo drug interruption, leading to paclitaxel interruption/delay, leading to any drug reduction, leading to Fruquintinib/placebo drug reduction, leading to paclitaxel reduction, leading to death, meeting SAE criteria will be summarized.

Time to onset of first AESI is defined as time interval from date of randomization to the earliest onset date among AEs within the same AESI category.

Besides the programmed 11 AESIs, the AESIs collected from eCRF page 'Adverse Event' will also be listed.

15.1.9.1. Derivations

- Time to onset of first AESI (months) = (Date of earliest onset date among AEs within the same AESI category – Date of randomization +1) / 30.4375

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15.2. DEATHS

If any subjects die during the study, as recorded on the 'Death' form or 'Adervse Events' page with 'Fatal' outcome selected, the information will be presented in a summary table and a data listing for overall and on treatment period (from first administration date to 30 days after last administration date on study) and follow up period. Subject died on treatment period (from the first dose to the 30 days after last dose), will be flagged in the listing.

15.3. LABORATORY EVALUATIONS

Results from the local laboratory will be included in the reporting of this study for hematology, blood chemistry, coagulation, urinalysis, 24 hours urine protein, Fecal Occult Blood and thyroid function test.

The laboratory assessments to be included in the outputs are in Protocol Table1 study follow chart.

Data recorded by the laboratory will be converted to the International System of Units (SI) and all presentations will use SI units. Quantitative laboratory measurements reported as "< X", i.e. below the lower limit of quantification (LLQ), or "> X", i.e. above the upper limit of quantification (ULQ), will be converted to X for the purpose of quantitative summaries, but will be presented as recorded, i.e. as "< X" or "> X" in the listings.

Baseline is defined as the last non-missing evaluation (including post-randomization evaluation) prior to the first dose of study medication.

The following summaries will be provided for laboratory data:

- Summary of quantitative lab test results and changes from baseline by scheduled visit (for hematology, blood chemistry, coagulation, thyroid function, urine analysis (Red Blood Cell Count and White Blood Cell Count), 24-hour Urine Protein.
  When summarizing quantitative lab test outputs, CTCAE terms won't be used for reporting.
- Summary of qualitative lab test results and changes from baseline by scheduled visit (Urine Analysis (only Urine Protein, Urine Glucose, Red Blood Cell, White Blood Cell) and Fecal Occult Blood.
  When summarizing qualitative lab test outputs, CTCAE term won't be used for reporting.
- Shift table from baseline to worst post-baseline CTCAE grade for CTC gradable parameters using CTCAE terms
- Shift table from baseline to worst post-baseline value according to abnormality assessed by investigator

All rest lab data will be listed by subject, by lab category, lab test and visit.

15.3.1. CTC GRADING FOR LABORATORY DATA

To identify laboratory values of potential clinical importance, laboratory results will be graded according to NCI CTCAE 4.03, where applicable. All grading will be based on lab values (being direct or some derived, corrected values) only, regardless of its interventional or symptomatic consequences.

Some modifications to the grading system will be applied:

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- Grade 5 refers to fatal outcomes, which cannot be determined solely by lab values, therefore will not appear in the grading system.
- A further category denoted Grade 0 would include all other laboratory values except missing values.
- Missing results shall be graded as missing.

For some specific parameters with CTC grading in both high and low direction (e.g., calcium, glucose, magnesium, potassium, sodium), CTC in high and low directions will be presented separately, i.e. hyper for higher values of concern and hypo for lower values of concern. Section 15.3.2 lists all CTC gradable parameters for this study including the CTC terms.

15.3.2. CTC GRADABLE PARAMETERS

In this study, the laboratory parameters will be tabulated using the shift from baseline in the CTC grade if gradable. The summary of the baseline grade will be summarized versus the worst CTC grade. The worst grade per subject defined as the highest CTC grade among the on-treatment evaluations. Only non-missing baseline and post-baseline data will be summarized, The by CTC tables will use the CTC terms listed below in the bracelet.

- Hematology:
  - Leukocytes / White Blood Cells Count (WBC) (Leukocytosis & White blood cell decreased)
  - Hemoglobin (Hemoglobin increased & Anemia)
  - Lymphocytes (Lymphocyte count increased & Lymphocyte count decreased)
  - Platelets (Platelet count decreased)
  - Neutrophils (Neutrophil count decreased)
- Blood biochemistry:
  - Fasting Glucose (Hypoglycemia& Hyperglycemia)
  - Calcium (Hypocalcaemia& Hypercalcemia)
  - Potassium (Hypokalemia & Hyperkalemia)
  - Sodium (Hyponatremia & Hypernatremia)
  - Magnesium (Hypomagnesemia & Hypermagnesemia)
  - Cholesterol (Cholesterol high)
  - Phosphate (Hyperphosphatemia)
  - Alkaline phosphatase /ALP (Alkaline phosphatase increased)
  - Albumin/ALB (Hypoalbuminemia)
  - Alanine Aminotransferase /ALT (Alanine Aminotransferase increased)
  - Amylase (Serum amylase increased)
  - Aspartate Aminotransferase (Aspartate Aminotransferase increased)
  - Total Bilirubin (Blood bilirubin increased)
  - Creatinine (Creatinine increased)
  - Uric Acid (Hyperuricemia)
  - Triglycerides (Hypertriglyceridemia)
- Coagulation:
  - Activated Partial Thromboplastin Time (Activated partial thromboplastin time prolonged)

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15.3.3. Laboratory Reference Ranges High/Low Criteria and Investigator's Assessment

Quantitative laboratory measurements will be compared with the relevant laboratory reference ranges in SI units and categorized as:

- Low: Below the lower limit of the laboratory reference range.
- Normal: Within the laboratory reference range (upper and lower limit included).
- High: Above the upper limit of the laboratory reference range.

For urine analysis, Red Blood Cell Count and White Blood Cell Count will be collected as Quantitative.

And for urine protein, urine glucose and Fecal Occult Blood will be collected as 'Negative' 'Trace' '+' "++' '+++' '++++' and be summarized for by visit summary table.

All laboratory measurements will also be evaluated as normal, abnormal not clinically significant (ANCS), abnormal clinically significant (ACS) by investigator's assessment.

15.3.4. Potential Drug-Induced Liver Damage

Special events of potential drug-induced liver damage are defined in Protocol Section 10.1.3 and summarized below. Events will be identified via programming and summarized by categories and liver metastasis or not (yes/no) at baseline for SAF.

- AST/ALT $\geqslant$ 3 x ULN and $\leqslant$ 5 x ULN
- AST/ALT > 5 x ULN
- Total bilirubin $\geqslant$ 2 x ULN
- AST /ALT $\geqslant$ 3 x ULN and total bilirubin $\geqslant$ 2 x ULN
- AST/ALT $\geqslant$ 3 x ULN, total bilirubin $\geqslant$ 2 x ULN and ALP < 2 x ULN

15.4. ECG Evaluations

Results from the ECG will be reported for SAF for this study.
Left Ventricular Ejection Fraction (LVEF) results at screening will be present in ECG listing.

The following ECG parameters will be reported:

- Heart rate (beat/min)
- PR Interval (msec)
- RR interval (msec)
- QRS Interval (msec)
- QT Interval (msec)

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- QTc Interval (msec)
- QTcB Interval (msec) = QT/(60/heart rate)$^{1/2}$ (QTcB, The corrected QT interval by Bazett)
- QTcF Interval (msec) = QT/(60/heart rate)$^{1/3}$ (QTcF, The corrected QT interval by Fredericia)
- Overall assessment of ECG (Investigator's judgment):
  - Normal
  - Abnormal, NCS
  - Abnormal, CS

The following summaries will be provided for ECG data:

- Actual and change from baseline by visit (for Heart Rate (beat/min), PR Interval (msec), RR Interval (msec), QRS Interval (msec), QT Interval (msec), QTc Interval (msec)), QTcF Interval (msec), QTcB Interval (msec).
- Table and listing of subjects meeting markedly abnormal criteria (worst post-baseline QTcB and QTcF values and change from baseline)
- Summary of abnormality assessed by investigator by scheduled visit and worst post-baseline result

15.4.1. ECG Markedly Abnormal Criteria

Markedly abnormal quantitative ECG measurements will be identified in accordance with the following predefined markedly abnormal criteria for worst post-baseline result:

- Absolute values for QTcB and QTcF interval will be classified as:
  - > 450 msec
  - > 480 msec
  - > 500 msec
- Change from baseline for QTcB and QTcF interval will be classified as:
  - > 30 msec increase from baseline
  - > 60 msec increase from baseline
  - > 90 msec increase from baseline
  - > 30 msec decrease from baseline
  - > 60 msec decrease from baseline
  - > 90 msec decrease from baseline

15.5. Vital Signs

The following vital signs measurements will be reported and summarized in tables for SAF:

- Systolic blood pressure (mmHg)
- Diastolic blood pressure (mmHg)
- Pulse rate (bpm)
- Respiratory Rate (breath/min)
- Temperature (^{o}C)
- Weight (kg)
- BSA (m^2)

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- BMI (kg/ m^2)

The following summaries will be provided for vital signs data:

- Actual and change from baseline by visit
- Shift table of worst post-baseline category from baseline by investigator's assessment

For Vital sign related listings, the all measurements will be presented.

15.5.1. Vital Signs Specific Derivations

For Oral Body Temperature (°C), conversions to oral route will be made as follows:

- Axillary: +0.5°C
- Oral: No Adjustment
- Forehead: +0.5°C
- Ear: -0.5°C

SBD, DBP will be summarized in table for SAF using the categories defined in Table 6.

Table 6. Vital Sign Categories

Parameter	Unit	Shift	Baseline categories	Change from baseline categories
SBP	mmHg	Increase and decrease (absolute change)	<140 / ⩾140	0 / >0 - ⩽20 / >20 - ⩽40 / >40
DBP	mmHg	Increase and decrease (absolute change)	<90 / ⩾90	0 / >0 - ⩽ 20 / >20 - ⩽40 / >40
Weight	Kg	Increase and decrease (percentage change)		0 / >0 - ⩽ 5 / >5- ⩽10 / >10

The categories will be displayed in vital sign listing for change from baseline category.

15.6. Physical Examination

The physical examinations results, Normal/Abnormal, NCS/Abnormal, CS, of each tested location will be presented in a listing for the SAF.

15.7. Other Safety Assessments

Pregnancy test result, collected from eCRF page 'Pregnancy Test' will be presented in listings for SAF.

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15.7.1. Eastern Cooperative Oncology Group (ECOG) Performance Scores

Number and percentage of subjects for each category of the ECOG performance status at each assessment time will be provided by treatment group. Negative change scores indicate an improvement and positive scores indicate a decline in performance. ECOG will also be summarized using shift table from baseline to post-baseline score for each treatment group by visit.

The ECOG score for each schedule visit will be listed.

15.7.2. Left Ventricular Ejection Fraction (LVEF)

The shift table of LVEF from baseline to the worst post-baseline level according to the abnormality status (normal, abnormal not clinically significant, and abnormal clinically significant) evaluated by the investigator will be summarized for the patients who have baseline value and at least one post-baseline value. The baseline LVEF will be summarized.

16. References

Clinical Study Protocol 2017-013-00CH1_CN Protocol Version 4.0.
Clinical Study Protocol 2017-013-00CH1_CN Protocol Version 5.0.
Clinical Study Protocol 2017-013-00CH1_CN Protocol Version 6.0.
Annotated Study Book for Study Design: 2017-013-00CH1 v70, dated 22Apr2022 (eCRF).
https://support.sas.com/documentation/cdl/en/statug/63033/HTML/default/viewer.htm#statug_lifetest_sect004.htm
http://www.biostat.jhsph.edu/~ejohnson/regression/topic3.pdf
Common Terminology Certieria for Adverse Events Version 4.03, dated 14Jun2010.

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APPENDIX 1. PROGRAMMING CONVENTIONS FOR OUTPUTS

IQVIA OUTPUT CONVENTIONS

Outputs will be presented according to the IQVIA's general guidelines and template for outputs conventions.

GENERAL RULES

- The following conventions will be applied for reporting descriptive statistics of all continuous data (except PK values):
- Mean, Median, Q1, Q3, Minimum, and Maximum will have the one more digit than RAW/SDTM data for non-derived data (eg weight).
- SD will be presented with one digit more than mean.
- Statistics on derived data (eg treatment exposure time in days) will be rounded to reasonable number of digits. Maximal digits should be available in ADaM datasets.

Qualitative variables will be summarized by counts and percentages. Unless otherwise stated, the calculation of proportions will be based on the number of subjects of the analysis set of interest. Therefore, counts of missing observations will be included in the denominator and presented as a separate category.

In case the analysis refers only to certain visits, percentages will be based on the number of subjects of the analysis set of interest, unless otherwise specified. Total of missing and non-missing observations at each time-point will reflect the population still in the trial at that time. For example, if a subject is still in the trial at the time-point but with missing data, it should be counted in the number of missing observations.

DATES & TIMES

Depending on data available, dates and times will take the form DDMMMYYYY and HH:MM:SS.

SPELLING FORMAT

English US

PRESENTATION OF TREATMENT GROUPS

For outputs, treatment groups will be represented as follows and in that order:

Treatment Group	For Tables, Listings and Graphs
Fruquintinib 4mg orally, QD Plus Paclitaxel Infusion	Fruquintinib + Paclitaxel
Placebo 4mg orally, QD Plus Paclitaxel infusion	Placebo + Paclitaxel
Screen Failure	Screen Failures

PRESENTATION OF VISITS

For outputs, visits will be represented as follows and in that order:

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System Name (Default)	Visit Name	Visit Number
SCREEN	Screening	2
C1D1	C1D1	3
C1D8	C1D8	4
C1D15	C1D15	5
C2D1	C2D1	6
C2D8	C2D8	7
C2D15	C2D15	8
C3D1	C3D1	9
C3D8	C3D8	10
C3D15	C3D15	11
C4D1	C4D1	12
C4D8	C3D8	13
C4D15	C4D15	14
C5D1	C5D1	15
C5D8	C5D8	16
C5D15	C5D15	17
…	….	…
EOT	End of Treatment	25
FU	Observation period after EOT	26

Day 8 is only scheduled for Paclitaxel injection without any safety assessment, so it won't be included in by-visit safety summary tables.

LISTINGS

All listings will be ordered by the following (unless otherwise indicated in the template):

- randomized treatment group, first by active dose and then placebo,
- centre-subject ID,
- date (where applicable),
- For listings where non-randomized subjects are included, these will appear in a category after the randomized treatment groups labelled "Screen Failure".

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APPENDIX 2. PARTIAL DATE CONVENTIONS

Imputed dates will NOT be presented in the listings.

ALGORITHM FOR TREATMENT-EMERGENCE OF ADVERSE EVENTS:

START DATE	STOP DATE	ACTION
Known	Known/partial/missing	If start date < study med start date, then not TEAE If start date > study med start date, and 'Prior/Post the first dose' is missing in eCRF then TEAE If start date = study med start date, and 'Prior to the first dose' is ticked in eCRF then not TEAE If start date = study med start date, and 'Post the first dose' is ticked in eCRF then TEAE If start date = study med start date, and both 'Prior to the first dose' and 'Post the first dose' are missing in eCRF then TEAE
Partial, but known components show that it cannot be on or after study med start date	Known/partial/missing	Not TEAE
Partial, could be on or after study med start date	Known	If stop date < study med start date, then not TEAE If stop date >= study med start date, then TEAE
	Partial	Impute stop date as latest possible date (i.e. last day of month if day unknown or 31st December if day and month are unknown), then: If stop date < study med start date, then not TEAE If stop date >= study med start date, then TEAE
	Missing	Assumed TEAE
Missing	Known	If stop date < study med start date, then not TEAE If stop date >= study med start date, then TEAE
	Partial	Impute stop date as latest possible date (i.e. last

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START DATE	STOP DATE	ACTION
		day of month if day unknown or 31st December if day and month are unknown), then: If stop date < study med start date, then not TEAE If stop date >= study med start date, then TEAE
	Missing	Assumed TEAE

ALGORITHM FOR PRIOR / CONCOMITANT MEDICATIONS:

START DATE	STOP DATE	ACTION
Known	Known	If start date & stop date < study med start date, assign as prior If study med start date <= start date <= last study med +33 assign as concomitant If start date < study med start date and ended after study med start, assign as concomitant
	Partial	Impute stop date as latest possible date (i.e. last day of month if day unknown or 31st December if day and month are unknown), then: Follow rule 1
	Missing	If stop date is missing could never be assumed a prior medication If start date <= end of treatment, assign as concomitant If start date > end of treatment, assign as post treatment
Partial	Known	Impute start date as earliest possible date (i.e. first day of month if day unknown or 1st January if day and month are unknown), then: Follow rule 1
	Partial	Impute start date as earliest possible date (i.e. first day of month if day unknown or 1st January if day and month are unknown) and impute stop date as latest possible date (i.e. last day of month if day unknown or 31st December if day and month are unknown), then: Follow rule 1
	Missing	If stop date is missing could never be assumed a prior medication Impute start date as earliest possible date (i.e. first day of month if day unknown or 1st January if day and month are unknown), then: If start date <= end of treatment, assign as concomitant If start date > end of treatment, assign as post treatment

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START DATE	STOP DATE	ACTION
Missing	Known	If stop date < study med start date, assign as prior If stop date >= study med start date, assign as concomitant
	Partial	Impute stop date as latest possible date (i.e. last day of month if day unknown or 31st December if day and month are unknown), then: If stop date < study med start date, assign as prior If stop date >= study med start date, assign as concomitant
	Missing	Assign as concomitant

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APPENDIX 3. SCORING THE EORTC QLQ-C30 (VERSION 3.0)

	Scale	Number of items	Item range*	**Version 3.0** Item numbers	Function scales
Global health status / QoL					
Global health status/QoL (revised)†	QL2	2	6	29, 30	
Functional scales					
Physical functioning (revised)†	PF2	5	3	1 to 5	F
Role functioning (revised)†	RF2	2	3	6, 7	F
Emotional functioning	EF	4	3	21 to 24	F
Cognitive functioning	CF	2	3	20, 25	F
Social functioning	SF	2	3	26, 27	F
Symptom scales / items					
Fatigue	FA	3	3	10, 12, 18	
Nausea and vomiting	NV	2	3	14, 15	
Pain	PA	2	3	9, 19	
Dyspnoea	DY	1	3	8	
Insomnia	SL	1	3	11	
Appetite loss	AP	1	3	13	
Constipation	CO	1	3	16	
Diarrhoea	DI	1	3	17	
Financial difficulties	FI	1	3	28	

* Item range is the difference between the possible maximum and the minimum response to individual items; most items take values from 1 to 4, giving range = 3.
† (revised) scales are those that have been changed since version 1.0, and their short names are indicated in this manual by a suffix "2" – for example, PF2.

For all scales, the RawScore, RS, is the mean of the component items:

$$RawScore = RS = (I_1 + I_2 + \cdots + I_n)/n$$

Then for functional scales:

$$Score = \left\{1 - \frac{(RS - 1)}{range}\right\} * 100$$

and for Symptom scales/items and Global health status /QoL:

$$Score = \{(\mathrm{RS} - 1)/\mathrm{range}\} * 100$$

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APPENDIX 4. ADVERSE EVENTS OF SPECIAL INTEREST (AESI)

The AESIs are defined by the following definition based on MedDRA version 25.0 terminology as below. For each time analysis, the SMQ and PT terms based on latest version of MedDRA used in the study will be used.

Hemorrhages is defined by SMQ

- 20000038 Haemorrhages (Narrow) (Including two sub-SMQs narrow search)

Hepatic function abnormal is defined by SMQ

- 20000006 drug related hepatic disorders-comprehensive search (includes four sub-SMQs narrow search)

Infections

- 10021881 For preferred terms (PTs) under MeDRA System Organ Class of “infections and infestations”.

Hypertension is defined by the SMQ

- 20000147 Hypertension (Narrow)

Dermatological toxicity

- 10040785 For PTs under MedDRA SOC of “skin and subcutaneous tissue disorders”

Proteinuria is defined by the SMQ

- 20000220 Proteinuria (Narrow)

Thyroid dysfunction is defined by the SMQ

- 20000159 Thyroid dysfunction (Broad+Narrow) (Including two sub-SMQs search)

Embolic and thrombotic events, arterial and venous is defined by the SMQ

- 20000081 Embolic and thrombotic events (Narrow) (Includes all three sub-SMQs narrow search)

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Gastrointestinal perforation is defined by the SMQ

- 20000107 Gastrointestinal perforation (Narrow)

Left ventricular ejection fraction decreased is defined by the SMQ

- 20000004 cardiac failure (Narrow)

Aneurysms and artery dissections is defined by the HLGT

- 10002363 Aneurysms and artery dissections

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Hutchison MediPharma Limited
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Independent Data Monitoring Committee (IDMC) Charter

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IDMC CHARTER TABLE
DATA SAFETY MONITORING BOARD ROSTER: IDMC MEMBERS

IDMC CHAIRPERSON

[redacted]

IDMC MEMBERS

[redacted]

[redacted]

[redacted]

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DATA SAFETY MONITORING BOARD ROSTER: IDMC CONTACTS AND CONTRIBUTORS

SPONSOR (PRIMARY)

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

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IQVIA CLINICAL PROJECT MANAGER

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IQVIA IDMC COORDINATOR

[redacted]

IQVIA UNBLINDED BIOSTATISTICIAN

[redacted]

IQVIA BLINDED BIOSTATISTICIAN

[redacted]

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IQVIA DATA MANAGEMENT

[redacted]

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APPROVING OFFICIALS:

Name	Title	Signature/Agreed	Date
Prof. [redacted]	Chairperson IDMC	_____	_____
Prof. [redacted]	Member IDMC	_____	_____
Prof. [redacted]	Member IDMC	_____	_____
Prof. [redacted]	Member IDMC	_____	_____
[redacted]	CSO, Execute Vice President Hutchison China Medi Tech Limited	_____	_____
[redacted]	CMO, Senior Vice President Hutchison MediPharma Limited	_____	_____
[redacted]	IDMC Coordinator IQVIA CEVA	_____	_____

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ABBREVIATIONS:

Abbreviation	Definition
CEVA	Clinical Event Validation and Adjudication Services at IQVIA
CI	Coordinating Investigator
GCP	Good Clinical Practice
ICH	International Conference on Harmonization
IDMC	Independent Data Monitoring Committee
IRB	Institutional Review Board
IWRS	Interactive Web Response System
KOM	Kick-off Meeting
SAE	Serious adverse event
SAP	Statistical analysis plan
SMR	Senior Management Representative
SUSAR	Suspected unexpected serious adverse reaction

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1.0 Introduction: Protocol

Protocol 2017-013-00CH1: A Randomized, Double-blind, Phase III trial to evaluate the efficacy and safety of Fruquintinib in combination with Paclitaxel for the treatment of patients with advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma.

Please refer to the study protocol for additional study details.

IQVIA has been contracted by the Sponsor of this trial, Hutchison MediPharma Limited, to provide IDMC coordination services.

2.0 SCOPE OF IDMC CHARTER

The objectives of the 2017-013-00CH1 IDMC Charter are to:
1) Identify the experts of the Data Safety Monitoring Board (IDMC) appointed by the sponsor of this trial,
2) Outline the specific purposes and functions/scope of the IDMC
3) Outline the roles and responsibilities and operational procedures of the IDMC, and
4) Describe the procedures for data abstraction and data delivery conventions to and from the IDMC members for review purposes.

3.0 COMPOSITION IDMC MEMBERSHIP AND SCOPE

The IDMC is composed of four (4) members: one Chairperson, and three (3) additional individuals. The IDMC membership includes three (3) therapeutic experts in oncology, as well as an independent Biostatistician. Prof. Lu Shun will serve as Chairperson of the IDMC. Additional IDMC members are named on the IDMC roster.

The IDMC is an expert advisory group of experts who function as a review committee independent from the sponsor. IDMC members meet without the sponsor or study team members to review and discuss provided data. The IDMC scope is to review the study data to ensure the safety and efficacy of subjects enrolled in this study, including the first IDMC safety data review meeting when 100 patients randomized, the first interim analysis when 100 patients randomized and the last randomized subject completed the 6 month follow up, and the second interim analysis when 50% plan overall survival event (about 201 deaths) observed.

Through signature of this IDMC charter, IDMC members:
1) Confirm that they are not (and will not become) involved as investigators in the 2017-013-00CH1 protocol,
2) Verify that they do not have a serious conflict of interest that would bias their review of trial data (e.g. IDMC members do not have a financial interest that could be substantially affected by the outcome of the study, strong views on the relative merits of the study compound, or relationships with individuals in trial leadership positions that could be considered reasonably likely to affect their objectivity),
3) Agree to communicate to the IQVIA IDMC Coordinator any and all potential minor conflicts of interest, and
4) Agree to notify the IQVIA IDMC Coordinator, should they identify an emerging potential conflict of interest (minor or major) during the course of their service on the IDMC.

All IDMC members are expected to serve until the study is completed. Should it be necessary for

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a member to resign, the member should follow any guidance contained in their contract andshould submit the effective date of resignation in writing to the IQVIA IDMC Coordinator, who will notify Hutchison MediPharma Limited and the IDMC Chairperson. In the event a member resigns, Hutchison MediPharma Limited, in consultation with the IDMC Chairperson, will initiate the process to identify a replacement member. Should the IDMC Chair resign, the IDMC Coordinator in consultation with Hutchison MediPharma Limited will initiate the process to identify a replacement.

The IDMC may, with prior approval from Hutchison MediPharma Limited, contact and involve outside expert consultants who may provide additional, relevant insight or expertise to the IDMC, regarding any specific issues that may arise (once they have entered into confidentiality disclosure agreements).

In order to maintain the independence of the IDMC, IDMC contacts and consultants may ***not*** attend closed sessions of IDMC Data Review Meetings during which deliberations with respect to IDMC recommendations or voting procedures take place. Outside expert consultants should not be exposed to confidential, unblinded data unless absolutely necessary for the consultant's input to be meaningful.

4.0 IDMC RESPONSIBILITIES

The 2017-013-00CH1 IDMC is an independent expert advisory group commissioned and charged with the responsibility of evaluating cumulative safety and efficacy of the subjects enrolled and to be enrolled in the 2017-013-00CH1 study by reviewing the available clinical data at scheduled time points, as described in this IDMC Charter, as well as on an ad hoc basis, as needed.

For 2017-013-00CH1, the IDMC will play the role of data and safety monitoring board and review the results from an interim analysis for stopped because of serious safety issues or futility performed by the IQVIA Unblinded Biostatistician. The IDMC will review the results of the interim analysis and will make appropriate recommendations based on those results. Per the study protocol, an interim futility analysis will be conducted by the time when 100 patients randomized and the last randomized patient completed the 6-month follow-up.

At the data and safety monitoring meeting held on Oct 16, 2018, all the experts suggested to continue study without modifications after evaluating the safety profiles. At FRUTIGA Scientific Steering Committee meeting held on Feb 16, 2019, and the first interim analysis meeting held on Mar 28, 2019, all the experts suggested to add one more interim analysis to provide more precise estimate of the treatment effect since the first interim analysis was performed based on a very small sample which included only the first 100 patients enrolled into this study. Based on the above considerations, Hutchison decided to add one more interim analysis when there are about 201 OS events achieved (approximately 50% of total OS events). IDMC will additionally review unblinded results from the secondary interim analysis (IA) and will make appropriate recommendations based on those results.

After the review of each Data Report has been completed from secondary IA, the IDMC Chairperson will provide the official IDMC recommendation to Hutchison MediPharma Limited regarding the appropriateness of continuing the study without any change, or continuing study with sample size increase or early stop study because of futility or early stop enrollment because of efficacy or enhance safety monitor from a safety

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perspective, as well as any other recommendations relevant to study conduct and/or patient safety.

The operating procedures of the IDMC are based on and are in compliance with the Food and Drug Administration's "Guidance for Clinical Trial Sponsors [on the] Establishment and Operation of Clinical Trial Data Monitoring Committees" (March 2006, OMB Control Number: 0910-0581) and with the European Medicines Agency's "Guideline on Data Monitoring Committees" (January 2006, EMEA/CHMP/EWP/5872/03 Corr).

Specifically, the IDMC members are authorized and charged to perform the following functions:

- Provide timely review and approval and operate in accordance with the specifications outlined in this IDMC Charter.
- Review the provided 2017-013-00CH1 study documents that are approved by sponsor for IDMC review such as protocol, the Investigator Brochure (latest version), safety overview of Fruquintinib.
- Provide timely response to requested review of IDMC documentation and meeting availability requests.
- Attend the IDMC Kick-Off meeting to receive IDMC process training, protocol orientation and IDMC operational training.
- Monitor the safety of patients enrolled and to be enrolled in 2017-013-00CH1, through review of accumulating clinical data from the ongoing clinical trial based on protocol and IDMC recommendation.
- Review and evaluate the content of all Data Reports received.
- Ensure that blinded IQVIA representatives (specifically including the IQVIA CEVA IDMC Coordinator and the IQVIA Blinded Biostatistician), and blinded Hutchison MediPharma Limited representatives (specifically including the staff who directly participate in the study conduct, analysis or reporting) are not exposed to the confidential, unblinded data made available to the IDMC. The IQVIA Unblinded Biostatistician has the same access to the data that is reviewed by the IDMC; therefore, it is not necessary for the IDMC to take steps to maintain this individual in a blinded position.
- Participate in IDMC discussions and vote on IDMC recommendations.
- Submit timely payment invoices as required in member contracts.

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Throughout the trial, the IDMC Chairperson will serve in a leadership role and will be authorized and charged with the following additional responsibilities:

- Provide leadership at all IDMC Data Review meetings; ensure that all relevant data have been reviewed by the IDMC members and that any issues have been addressed.
- Ensure that blinded individuals (i.e. IDMC contacts and IDMC consultants) are not inappropriately exposed to confidential and/or unblinded data.
- Ensure that only the members of the IDMC, Unblinded Biostatistician and invited consultants are present for data discussions. The IDMC Chairperson may elect to exclude the Unblinded Biostatistician and external consultants for a portion of the Closed session to allow for discussion by the IDMC only when IDMC deliberations are made, IDMC recommendations are discussed and IDMC voting procedures are conducted.
- Generate confidential, written minutes of all executive sessions of any IDMC meetings (using Appendix B) and maintain these minutes as confidential to IDMC members,

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5.0 SPONSOR RESPONSIBILITIES

The Sponsor, Hutchison MediPharma Limited, will have the following responsibilities with respect to the 2017-013-00CH1 IDMC:

- Assign a sponsor representative to serve as a primary sponsor contact for the IDMC Coordinator and the IDMC for IDMC issues.
- Provide timely response to IDMC Lead for member compilation and meeting logistics decisions as required.
- Provide final approval of the IDMC Chairperson and members to serve on the IDMC.
- Ensure relevant clinical or other data available to Hutchison MediPharma Limited on the safety of Fruquintinib capsules (HMPL-013) is provided for communication to the IDMC.
- Collaborate with IQVIA to ensure that IDMC members are informed of trial progress and issues on a regular basis via study updates to the IDMC.
- Ensure timely response to IDMC requests for additional information.
- In preparation for data review meetings, collaborate with IQVIA to prepare a general summary of the status of the trial and any relevant clinical issues.
- Provide Sponsor representation at all open and final sessions of IDMC meetings, as needed.
- Provide review and approval, in conjunction with the IDMC Chairperson, of minutes of open and final sessions of IDMC meetings.

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- Arrange for fair and reasonable compensation (consistent with any contractual guidelines) to IDMC members for their data monitoring activities (i.e., data monitoring services provided and any committee related travel costs, such as transportation, lodging, and meals). NOTE: Payments and reimbursements to be coordinated and facilitated by the IQVIA CEVA IDMC Coordinator, on the behalf of Hutchison MediPharma Limited.

- Provide a primary contact representative to receive and take action based on recommendations from the IDMC.

- Maintain ultimate responsibility for safe study conduct, according to ICH Good Clinical Practices (GCP) Guideline.

- Communicates decision to implement or not to implement the IDMC recommendation(s) to the IDMC Chairperson.

6.0 IQVIA BIOSTATISTICS RESPONSIBILITIES

IQVIA will provide a Blinded Biostatistician in support of the IDMC process. The responsibilities of the IQVIA Blinded Biostatistician are as follows:

- Provide approval for and operate in accordance with the specifications outlined in this IDMC Charter.

- Work with the study team to determine the data that are necessary for the IDMC Data Reports. Author and finalize the IDMC Statistical Analysis Unblinding Plan and provide copy to the IQVIA Unblinded Biostatistician.

- Provide all programs and blinded data to the IQVIA Unblinded Biostatistician for use in the IDMC Data Report.

- Maintain an archive of electronic copies of datasets and SAS programs used to generate the output for the IDMC blinded Data Reports.

IQVIA will also provide a IQVIA Unblinded Biostatistician in support of the IDMC process. The responsibilities of the IQVIA Unblinded Biostatistician are as follows:

- Provide approval for and operate in accordance with the specifications outlined in this IDMC Charter.

- Discuss data needs, reports format with IDMC members.

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- Assist in establishing timelines for each data review meeting to determine data cut-offs and required data transfer dates to ensure timely delivery of data reports to the IDMC.

- Generate, summarize and provide the final version of each IDMC fully unblinded Data Report.

- Distribute each final fully unblinded IDMC Data Reports directly to all four [4] IDMC members in a secure electronic email format or SharePoint at least 3 business days prior to the IDMC data review meeting.

- Confirm to the IDMC Coordinator that Data Reports have been submitted and received by the IDMC members.

- Attend the Closed Sessions of the IDMC to provide consultation regarding the information presented in the IDMC Data Reports, as requested by the IDMC members.

- Receive and obtain response to statistical and data content inquiries and commentary from the IDMC members.

- Maintain records related to the production of the fully unblinded version of each IDMC Data Report, as well as an exact file copy of each final, fully unblinded version of the Data Report. Transfer this file to Hutchison MediPharma Limited after the study database is locked and the treatment groups for the entire study are unblinded.

The IQVIA Unblinded Biostatistician will remain independent from the project teams at IQVIA Inc. and Hutchison MediPharma Limited who are responsible for oversight of the study.

7.0 IQVIA IDMC COORDINATOR RESPONSIBILITIES

IQVIA will provide a CEVA Lead for the 2017-013-00CH1 study to serve as IDMC Coordinator. The IDMC Coordinator will provide full administrative, logistical and coordinating support to the IDMC members.

The IQVIA IDMC Coordinator will have the following responsibilities:

- Provide approval for and operate in accordance with the specifications outlined in this IDMC Charter.

- Assist IQVIA Contract Negotiations group with preparation of IDMC contracts using scope and payment terms approved by the sponsor.

- Serve as the primary, central point of contact for the IDMC members and as the main liaison between the 2017-013-00CH1 study operations teams and the IDMC members.

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- Ensure the IDMC receives copy of the latest study protocol and final Statistical Analysis Plan.
- Coordinate the implementation of the schedule for preparation and distribution of Data Reports to IDMC members.
- In preparation for data review meetings, collaborate with Hutchison MediPharma Limited and IQVIA Clinical Project Management and/or Medical Monitor to ensure that the IDMC members are provided a general summary of the status of the trial and any relevant clinical issues.
- Obtain IDMC recommendation letters or forms and distribute them, as described in this IDMC Charter.
- Coordinate arrangements for all data review meetings and any IDMC ad hoc meetings, as outlined in this charter.
- Distribute meeting logistics, agenda and any supplemental data prior to the IDMC meetings.
- Draft open and final session meeting minutes, as specified in the Charter.
- Maintain a central file of all key IDMC-related correspondences. Provide this file to Hutchison MediPharma Limited according to the project filing plan.
- Process IDMC member invoices and expense reports on behalf of Hutchison MediPharma Limited.

8.0 IDMC MEMBER TRAINING

All IDMC members will receive 2017-013-00CH1 IDMC process training. To this end, an IDMC Kickoff Meeting (KOM) will be held. The objectives of the IDMC Kickoff meeting will be to orient the IDMC members to the relevant study protocol, and to familiarize them with study materials such as the Investigator Brochure and the contents of the IDMC Charter, review of the proposed data report formats, as well as to provide them with operational instructions/training. The KOM will be scheduled to occur prior to the anticipated first patient enrollment if at all possible. In the event the meeting cannot be held by that time point, provisions will be made to inform the IDMC of any potential adverse safety information that occurs prior to the initiation of the IDMC process.

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9.0 ONGOING COMMUNICATIONS & NOTIFICATIONS TO IDMC

The IQVIA IDMC Coordinator will ensure that the IDMC Chairperson is provided copies of all study Safety SUSAR/Alert Letters and any applicable compound cross-reporting Safety/SUSAR/Alert Letters related to the 2017-013-00CH1 study when they are distributed.

The IDMC Coordinator will ensure that the IDMC Chairperson is provided a cumulative SAE Listing from the safety database for the 2017-013-00CH1 study on a monthly basis.

The IDMC Chairperson will determine whether further distribution of this material to the remaining IDMC members is necessary.

Process of producing IDMC report will be detail recorded in Unblinding Plan. IQVIA Blinded Biostatistician team will provide all relevant documents, blinded data and programs to IQVIA Unblinded Biostatistician team. IQVIA Unblinded Biostatistician team will receive random code list from IWRS, the data/program from blinded IQVIA Biostatistician team, and then only change variables of treatment groups (no further programming).

The IQVIA Unblinded Biostatistician disseminates the reports to the IDMC members for deliberation and acts as a liaison between the IDMC and IQVIA blinded Biostatistian regarding any requests for additional analyses. After deliberation, the IDMC Chairperson will provide verbal or electronic recommendation to the Hutchison Senior Management Representative (SMR). Ultimately, the SMR has the responsibility to accept or reject the recommendation on behalf of the sponsor.

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10.0 IDMC DATA REPORTS

10.1 Data Submission

IDMC members will receive all IDMC Data Reports directly from the IQVIA Unblinded Biostatistician via a secure electronic email format or SharePoint.

Whenever possible, IDMC Data Reports will be provided to the IDMC members at least 3 business days prior to the scheduled data review meetings.

10.2 Data Source and Content

Data included in each IDMC Data Report will be cumulative-to-date at the time of the established data cut-off. The cut-off date for the data included in the Data Reports, as well as the current enrollment figures, will be stated in a cover letter.

Please reference the *IDMC Statistical Analysis Plan (*separate document), which contains specifications for the IDMC Data Reports, including the data sources used to compile the reports.

The IDMC may request additional information on individual patients and/or events, as needed.

10.3 Degree of Blinding Applied to IDMC Data Reports

Data Reports for review by the IDMC will be presented in a full unblinding fashion.

10.4 IDMC Data Reports Timelines

When an IDMC Data Review Meeting has been scheduled, the sequence of events leading to preparation of the data for the IDMC meeting will be triggered,updated the timeline for 3rd IDMC meeting :

Activity	Responsible	timeline
Data cut-off		Fri 6/3/2020
Data entry by site staff and data cleaning by I QVIA Data Management	Clinical	Fri 8/5/2020
SDV	Clinical	Fri 8/5/2020
Data Transfer， DBL	DM	Mon 11/5/2020
IQVIA Biostatistician develops tables, listings and figures and validate and transfers to IQVIA Unblinded Biostatistician	Blinded Bios	Wed 20/5/2020
IQIVA Unblinded Biostatistician prepares unblinded DSMB data reports and submits final data report to DSMB	Unblinded Bios	Wed 27/5/2020
IDMC's review of data report prior to the DSMB meeting	IDMC members	Mon 6/1/2020
IDMC Data Review Meeting	ALL	3/6 2020

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11.0 IDMC MEETINGS

11.1 IDMC Data Review Meeting Format & Frequency

It is anticipated that IDMC data review meetings will be scheduled via face-to-face/ teleconference at predefined intervals during the remaining portion of the study timeline, resulting in a minimum of 3 planned data review meetings.

Besides the kick-off meeting, the first IDMC safety data review meeting occurred at approximately 1 months post the first 100 patients enrolled. The second IDMC efficacy and safety data review meeting occurred at approximately 1.5 months after the first 100 subjects completed 6 months treatment (data cut-off date). The third IDMC efficacy and safety data review meeting will occur at approximately 2 months after about 201 OS events achieved (approximately 50% of total OS events).

In addition to these planned data review meetings, the IDMC will have the ability to hold ad hoc meetings, should they be deemed necessary (e.g. due to a need identified by the IDMC or Hutchison MediPharma Limited).

11.2 IDMC Meeting Agendas

With input from the Hutchison MediPharma Limited, the IQVIA CEVA IDMC Coordinator will establish the agenda for each planned data review meeting and for any ad hoc IDMC meetings.

11.3 IDMC Attendance

At least three (3) IDMC members including the chairperson, the IDMC statistician and at least one IDMC clinician must be in attendance, in order for each IDMC data review meeting to be convened, and in order for voting procedures to be conducted.

11.4 IDMC Meeting Structure

IDMC data review meetings will consist of a minimum of two sessions: an Open session and a Closed session. If it is necessary for the IQVIA Unblinded Biostatistician (a non-IDMC member) to attend the closed session of an IDMC meeting, then the IDMC members should have the opportunity to proceed to an Executive session, once this individual departs the meeting.

It is recommended that there be a Final Session upon conclusion of the Closed Session where the IDMC returns to an Open Session with non-IDMC members to provide final IDMC recommendations and IDMC comments following Closed Session reviews.

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11.4.1 Open Session(s)

During the initial Open Session attendance is not restricted. Therefore, the IDMC members, as well as the IQVIA CEVA IDMC Coordinator, the Sponsor, the IQVIA Unblinded Biostatistician, CPM, Medical Monitor and any other IDMC contacts and consultants may be present. During open sessions, the IDMC members will receive a project update and will have an opportunity to discuss the progress of the trial with Hutchison MediPharma Limited and IQVIA. Unblinded participants should take care not to provide comment on any unblinded data received by the IDMC only during Open Session discussions.

11.4.2 Closed Session

During this session, attendance is restricted to the IDMC members. The IDMC has the option to have the Unblinded Biostatistician (or an outside, expert consultant) in attendance. During closed sessions, the IDMC members will confidentially review the IDMC Data Report, deliberate, and conduct voting procedures.

Should the IQVIA Unblinded Biostatistician (or an outside, expert consultant) attend a closed session of an IDMC meeting, then the IDMC members can elect to hold an **executive session** during which only members of the IDMC are present and can commence when the Unblinded Biostatistician or outside consultant leaves the session, if desired by the IDMC. This executive session could then be held if the IDMC desires confidential deliberation and voting procedures.

11.5 IDMC Voting Procedures

In general, each IDMC member's vote carries equal weight during IDMC voting procedures.
In case the voting score is 2:2, the chairperson can cast a deciding vote.

11.6 IDMC Voting Procedures: Recommendation to Sponsor

After review and discussion of each Data Report, the IDMC members will vote to determine the final IDMC recommendation within one of the following options:

- The study may continue without modifications;
- The study may continue with sample size increase
- The study should be stopped because of futility
-
- The study may continue follow up and stop enrolment because of efficacy
- The study should enhance safety monitor, because of safety issues from interim analysis result
- Other suggestions or changes

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NOTE: It is recommended that the IDMC members conduct an unblinded review of trial data before recommending any change in the conduct of the study, including amendment, pausing enrollment or termination.

As part of the recommendation to the Sponsor, the IDMC may also make comments and suggestions that might enhance study performance, as deemed appropriate.

For the planned the first IDMC data review meeting, the IDMC members will not be charged with the application of formal statistical stopping rules. The recommendations of the IDMC will be based on the members' clinical assessment of the cumulative safety data provided for review.

For the second IDMC data review (the first interim analysis), IDMC will make a recommendation on whether to terminate the trial due to futility based on the predefined boundaries as shown in Appendix C. The stopping boundaries are primarily determined based on affordable risk in incorrectly stopping the trial at interim when the drug is actually effective. The other operating characteristics including posterior probability, predictive power, conditional power, probability of stopping the trial when drug is ineffective, and overall statistical power are also displayed in the table. As indicated by Appendix C, the stopping boundaries change with the number of events that occurred at interim.

Whether to stop the trial for futility will consider both OS and PFS. [redacted]

[redacted]

[redacted]	[redacted]	
	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]

[redacted]

For the third IDMC data review (the secondary interim analysis), IDMC will make a recommendation on whether to terminate the trial due to futility, or whether to early stop enrollment because of efficacy, or whether to continue study with sample size increase based on the predefined boundaries as shown in Appendix D. The stopping boundaries are generated base on Lan-DeMets procedure with an O'Brien-Fleming stopping rule. Sample size increase criteria are generated base on Cui, Huang and Wang (1999) (CHW) method with condition power. Details information have been clarified in Appendix D.

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Regarding voting, it is required that at least three (3) IDMC members must participate in this voting procedure. The IDMC Chairperson will document the outcome of the vote. From review of the votes, the IDMC Chairperson will assess whether a unanimous opinion has been achieved.

If a unanimous vote is not achieved, majority vote will determine the final recommendation of the IDMC, and the IDMC Chairperson will be responsible for assembling and presenting the majority and dissenting opinions to Hutchison MediPharma Limited, for all recommendations considered.

In the event of a tie vote arising from an even number of IDMC members voting, the IDMC Chairperson can cast a deciding vote.
If deemed necessary, the IDMC may elect to postpone the determination of an IDMC recommendation, pending external consultation(s) and/or receipt of additional data and a subsequent closed (and/or executive) IDMC Data Review Meeting session.

11.7 IDMC Meeting Minutes

IDMC Data Review meeting minutes will be divided by session and will reflect attendance, as well as whether each individual attended in person or via teleconference.

The IQVIA CEVA IDMC Coordinator will produce draft minutes of all open and final sessions of IDMC meetings within 3 business days of a data review meeting. The IQVIA CEVA IDMC Coordinator will provide the draft minutes to Hutchison MediPharma Limited and then the IDMC Chairperson for review and approval, before wider distribution. Once approved by the Hutchison MediPharma Limited and the IDMC Chairperson, the IQVIA CEVA IDMC Coordinator will distribute the final minutes to all attendees of the open and final sessions.

Since all details of IDMC deliberations must be kept strictly confidential among members of the IDMC, minutes of the closed and executive portions of the IDMC Data Review meetings must remain confidential until after the study database is locked and the treatment groups for the entire study are unblinded. The IDMC Chairperson (or a designee selected among the IDMC members) will review and approve minutes of closed sessions and executive portions of IDMC meeting. The IQVIA Unblinded biostatistician will draft all closed session of IDMC meetings minutes within 2 business days. These minutes will be distributed only to IDMC members, until after the study database is locked and the treatment groups for the entire study are unblinded.

The IDMC Chairperson will file all minutes from all sessions, centrally. After the study database is locked, the IDMC Chairperson will forward the central file of all closed IDMC meeting minutes for all sessions to Hutchison Medi Pharma Ltd.

12.0 IDMC Communication of Recommendation

Once the IDMC recommendation is finalized by IDMC vote, the IDMC Chairperson will provide communication of the IDMC recommendation to Hutchison MediPharma Limited, as follows:

- Initial, verbal communication of the IDMC recommendation will typically occur during the final open session of IDMC Data Review meetings.
- Formal, written communication of the IDMC recommendation will take the form of an IDMC

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Recommendation Letter. IDMC recommendation letters should be written by the IDMC Chairperson using attached IDMC Recommendation template (Appendix A), and should be provided to the IQVIA CEVA IDMC Coordinator within 5 business day after the IDMC recommendation is finalized.

Communications of IDMC recommendations will reflect the final recommendation of the IDMC members. In the event that a unanimous decision cannot be reached, majority and dissenting opinions will be summarized and presented by the IDMC Chairperson.

IQVIA CEVA IDMC Coordinator will collect the IDMC Recommendation Letter and distribute it to Hutchison Senior Management Representative.

Requests for copies of the IDMC Recommendation Letters or/and sponsor's final decision by investigators or Institutional Review Boards (IRBs) should be handled by the IQVIA Clinical group.

Hutchison MediPharma Limited will retain responsibility for submitting the IDMC Recommendation Letters or/and sponsor's final decision to relevant regulatory authorities.

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13.0 IMPLEMENTATION OF IDMC RECOMMENDATIONS

The recommendations provided to the Sponsor by the IDMC should be treated as such. The ultimate responsibility to implement or take action based upon the recommendations of the IDMC will be made by Hutchison MediPharma Limited. Hutchison MediPharma Limited will notify the IDMC Chairperson via CEVA IDMC Coordinator in writing of actions taken in response to a given IDMC recommendation, or for cases in which Sponsor action other than to continue the study without modification was recommended.

14.0 IDMC DOCUMENT HANDLING & RECORDS RETENTION

14.1 Confidentiality

The IDMC and IQVIA unblinded statistician will maintain a strictly confidential relationship to the 2017-013-00CH1 study data. The IDMC and IQVIA unblinded statistician will only reveal specific details and information associated with IDMC data review to appropriate parties, as specified by this IDMC Charter.

With specific reference to the above instruction, IDMC members and IQVIA unblinded statistician should take care to maintain the blind of the Sponsor and agents of the Sponsor (e.g., employees of Hutchison MediPharma Limited, and IQVIA project team) at all times for the duration of the trial, until after the study database is locked and the treatment groups for the entire study are unblinded.

14.2 Data Handling

The IDMC Chairperson must retain copies of all data reviewed by the IDMC in a central file.

Following each IDMC Data Review Meeting, the IDMC members other than Chairperson may choose to retain or safely destroy/shred their reference copies of the data reviewed.

Due to its confidential nature, IDMC members should take appropriate measures to properly house and/or destroy data pertaining to IDMC review.

14.3 Records Retention

The IDMC Chairperson should maintain a copy of the original IDMC file (i.e., copies of all data reviewed by the IDMC and copies of final minutes of all sessions of any IDMC meetings) until informed that the documents are adequately filed in the study files and the IDMC Chairperson's files can be destroyed.

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List of investigators and participating sites

Principal investigator	Participating site
Ruihua Xu, Feng Wang	Sun Yat-sen University Cancer Center
Lin Shen	Peking University Cancer Hospital and Institute
Weijian Guo	Fudan University Shanghai Cancer Center
Tianshu Liu	Zhongshan Hospital Fudan University
Jin Li	Tongji University Shanghai East Hospital
Shukui Qin	Chinese People's Liberation Army Cancer Center of Nanjing Bayi Hospital*
Yuxian Bai	Harbin Medical University Cancer Hospital
Zhendong Chen	The Second Affiliated Hospital of Anhui Medical University
Jufeng Wang	Henan Cancer Hospital
Yueyin Pan	Anhui Provincial Hospital
Yongqian Shu	The First Affiliated Hospital of Nanjing Medical University (Jiangsu Province Hospital)
Fuyou Zhao	The First Affiliated Hospital of Bengbu Medical College
Ying Cheng	Jilin Cancer Hospital
Feng Ye	The First Affiliated Hospital of Xiamen University
Kangsheng Gu	The First Affiliated Hospital of Anhui Medical University
Tao Zhang	Union Hospital, Tongji Medical College, Huazhong University of Science and Technology
Hongming Pan	Sir Run Run Shaw Hospital, Zhejiang University School of Medicine
Haijun Zhong	Zhejiang Cancer Hospital
Fuxiang Zhou	Zhongnan Hospital of Wuhan University
Yanru Qin	The First Affiliated Hospital of Zhengzhou University
Lei Yang	Nantong Tumor Hospital
Weidong Mao	Jiangyin People's Hospital
Qiu Li	West China Hospital, Sichuan University
Wenxiang Dai	The First Affiliated Hospital of University of South China
Wei Li	The First Bethune Hospital of Jilin University
Shubin Wang	Peking University Shenzhen Hospital
Yong Tang	Xinjiang Medical University Cancer Hospital
Dong Ma	Guangdong Provincial People’s Hospital
Xianli Yin	Hunan Cancer Hospital
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Wenwei Hu	The First People’s Hospital of Changzhou
Donghui Chen	Shanghai General Hospital, Shanghai Jiao Tong University School of Medicine
Guoxin Li	Nanfang Hospital of Southern Medical University

*This is the former institution for Prof S Qin during the conduction of FRUTIGA study.